ANSWERS TO QUESTIONS ON HARDY SPACES AND HANKEL OPERATORS

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Answer. Open means no informed opinion. Expected yes or Expected no gives a belief with reasons, without a proof. Answered gives the proposed answer; Partial names what remains open.

Evidence. Heuristic means reasons or numerics; Sketch means an argument by Søvik, with or without AI assistance; arXiv means a posted preprint or a public institutional report; Published means a peer-reviewed result. Sketch arguments have not been through arXiv or peer review and should be checked by the reader. Conditional on names an unrefereed result or standing hypothesis used by the argument.

AI assistance was used in adapting and checking the arguments and the exposition.

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Part 1. Replacing factorization: point evaluations

1. Embeddings and boundary values

1.1. The Blaschke condition below two

Do the zero sets of functions in $\mathscr{H}^p$ for $p < 2$ satisfy the Blaschke condition in the half-plane $\operatorname{Re} s > 1/2$?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 2.3.

Answer. No. One function belongs to every $\mathscr{H}^p$, $p < 2$, and has a non-Blaschke zero set in a bounded window.

Answer: Answered: no. **Evidence:** Sketch.

Proof. Let $A = 64$, $a_k = A^k$, $N_k = \lceil \exp(\exp(A^{8k})) \rceil$, $L_k = \log N_k$, and

$$F(s) = \sum_{k \geq 1} \frac{a_k}{\sqrt{N_k}} P_{N_k}(s), \qquad P_N(s) = \sum_{n \leq N} n^{-s}.$$

Harper's low-moment estimate gives $\|P_N\|_{\mathscr{H}^p} \leq C_p \sqrt{N}/(\log\log N)^{1/4}$ for each fixed $p < 2$. Hence the kth block has norm at most $C_p A^{-k}$; summability (or the p-triangle inequality for $p < 1$) gives $F \in \mathscr{H}^p$ for every $p < 2$. The positive coefficient series converges absolutely for every $\operatorname{Re} s > 1/2$, by comparison with $N^{1/2-\sigma}$. Thus the Hardy limit is the displayed analytic function.

Uniform sum-integral comparison in $1/2 \leq \operatorname{Re} s \leq 3/4$, $|\operatorname{Im} s| \leq 2$, gives

$$P_N(s) = \frac{N^{1-s}}{1-s} + O(1).$$

Consequently, with $z = s - 1/2$, the function $G(z) = (1/2 - z)F(1/2 + z)$ equals $\sum_j a_j e^{-L_j z} + H(z)$, where H is uniformly bounded in the strip. Put $R = A^4$, $\Delta_k = L_{k+1} - L_k$. The two consecutive terms $B_k(z) = a_k e^{-L_k z} + a_{k+1} e^{-L_{k+1} z}$ have simple zeros

$$z_{k\ell} = \frac{\log A + (2\ell + 1)\pi i}{\Delta_k}.$$

On the circle $|z - z_{k\ell}| = 1/\Delta_k$, write $z = z_{k\ell} + u/\Delta_k$, $|u| = 1$. Then $|B_k(z)| = a_k |e^{-L_k z}||1 - e^{-u}| > a_k/8$, since $|1 - e^{-u}| \geq 3 - e > 1/4$ and $|e^{-L_k z}| > 1/2$.

Earlier blocks contribute at most $a_k/(A-1) = a_k/63$ on these circles; later blocks contribute less than $a_k/100$ by the supergeometric growth $L_{k+1} \geq R L_k$; the bounded error is less than $a_k/32$ for large k. Thus $|G - B_k| < |B_k|$ on each circle. Rouché's theorem gives exactly one zero of G in each disk. There are at least $\Delta_k/(4\pi)$ such disks with $|\operatorname{Im} z_{k\ell}| \leq 1/2$, and the bands for different k are disjoint. Each zero has $\operatorname{Re} z \geq (\log A - 1)/\Delta_k$ and $1 + |\rho|^2 < 3$, so the contribution from each band to the Blaschke sum is at least $(\log A - 1)/(12\pi)$. Summing over k diverges. □

1.2. Local Nevanlinna membership below two

Are elements of $\mathscr{H}^p$ for $p < 2$ locally in the Nevanlinna class?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, Problem 2.4.

Answer. No. The same function as above has local logarithmic means diverging along a sequence of vertical lines.

Answer: Answered: no. **Evidence:** Sketch.

Proof. Use F and N_k, L_k from the preceding entry. The same Harper estimate proves $F \in \mathscr{H}^p$ for every $p < 2$, and the coefficient series converges absolutely in the half-plane. For fixed T, sum-integral comparison uniformly for $|t| \leq T$ gives

$$P_N(\sigma + it) = \frac{N^{1-\sigma-it}}{1-\sigma-it} + O_T(1).$$

At $s_k = 1/2 + L_k^{-1} + it$, the kth main numerator in $(1-s_k)F(s_k)$ has modulus a_k/e. Earlier terms sum to at most $a_k/63$; later terms sum to less than $a_k/100$ since $L_{k+m}/L_k \geq R^m$ and $R = A^4$. The accumulated sum-integral error is bounded independently of k. Since $|1 - s_k| \leq (1/4 + T^2)^{1/2}$, it follows for all large k that

$$\inf_{|t|\leq T} |F(1/2 + L_k^{-1} + it)| \geq c_T A^k.$$

Thus the integral of $\log^+ |F|$ over every fixed interval of positive length tends to infinity along L_k^{-1}. If F were locally in the classical Nevanlinna class, a positive harmonic majorant on a rectangle adjoining a smaller boundary interval would dominate these integrals: the bottom harmonic-measure density at a fixed interior point has a uniform positive lower bound on that interval. This contradicts divergence. □

1.3. **Small-exponent Carleson classes for Dirichlet series**

Below $p = 2$, Carleson measures don't depend on p.

Søvik, *Where Nehari Survives*, Chapter 3, question B4.

Answer. No. For some fixed $q \in (1, 2)$ there is a finite positive measure with $A_q(\mu) < \infty$ and $A_1(\mu) = \infty$.

Answer: Answered: no. **Evidence:** Sketch.

Proof outline. For $r > 0$, let $A_r(\mu)$ be the least constant in $\int |f|^r \, \mathrm{d}\mu \leq A_r(\mu)\|f\|_{\mathscr{H}^r}^r$ over Dirichlet polynomials, and let $\mathcal{A}_r$ denote the analogous torus embedding constant. Put

$$a(x) = \frac{1}{2}\left(\prod_j (1 - x_j^2)^{-1} + \prod_j (1 + x_j^2)^{-1}\right).$$

The two-point Szegő Gram matrix gives $A_2((\delta_x + \delta_{-x})/2) = a(x)$. For vectors with small coordinates and squared length close to 3, a Gaussian limit of the two reproducing kernels shows that a positive polynomial test has H^1 embedding constant at least $\rho a(x)$ for some fixed $\rho > 1$. Tensorized analytic dilation then gives a fixed $q \in (1, 2)$, $\kappa > 1$, and $\theta = \sqrt{q/2}$ such that $A_q(\sigma_x) \leq a(x/\theta)$ and $A_1(\sigma_x) \geq \kappa a(x/\theta)$.

For each m, choose a height $s_m > 1/2$ and m disjoint prime blocks with squared energy near 3. Their sign configurations give a probability measure ν_m on the square-summable polydisc with $\mathcal{A}_q(\nu_m) \leq D_m$ and $\mathcal{A}_1(\nu_m) \geq \kappa^m D_m$. Approximation of each finite configuration by the vertical orbit transfers both constants to a finite atomic half-plane probability measure λ_m with $A_q(\lambda_m) \leq 2D_m$ and $A_1(\lambda_m)/A_q(\lambda_m) \geq \kappa^m/8$. Choose m_j so $\kappa^{m_j}/8 \geq 4^j$ and define

$$\mu = \sum_{j\geq 1} \frac{2^{-j}}{A_q(\lambda_{m_j})} \lambda_{m_j}.$$

Since $A_q(\lambda_m) \geq 1$, this measure has mass and A_q at most one. The m_jth summand alone shows $A_1(\mu) \geq 2^{-j} A_1(\lambda_{m_j})/A_q(\lambda_{m_j}) \geq 2^j$. Thus $A_1(\mu) = \infty$.

Book draft, Chapter 4, "Different Carleson classes below two". □

2. Hellinger–Bessel inequalities

2.1. **Weak-star density of evaluation functionals**

Which known dual elements are of this form? Is the class weak-$*$ dense in $(\mathcal{H}^1)^*$?

Søvik, *Where Nehari Survives*, Chapter 2, Question 7. The scoped question is whether the controlled class is weak-$*$ dense.

Answer. Yes.

Answer: Answered: yes. **Evidence:** Sketch.

Proof. Let Λ be the weak-star closed span of evaluation functionals on the real prime curve at real parts $\sigma > 1$. If $F \in H^1(\mathbb{T}^\infty)$ annihilates Λ, its Dirichlet series $f(s)$ vanishes for every real $s > 1$. The series converges absolutely and is analytic on $\operatorname{Re} s > 1$. The identity theorem and uniqueness of Dirichlet coefficients give $F = 0$. Therefore the pre-annihilator of Λ is zero, and the Hahn–Banach separation theorem implies $\Lambda = (H^1)^*$. Each evaluation is controlled, so the controlled class is weak-star dense. □

Full proof

2.2. **Norm-closed sums of evaluations**

Which known dual elements are of this form?

Søvik, *Where Nehari Survives*, Chapter 2, Question 7. We answer the scoped norm-closure question: is the class of controlled evaluation functionals norm closed?

Answer. No.

Answer: Answered: no. **Evidence:** Sketch.

Proof. Put $X = \mathbb{D}^\infty \cap \ell^2$, $K_s(z) = \prod_j (1 - \overline{s_j} z_j)^{-1}$, and $N_s = \|K_s\|_2$. Define

$$H(r,s) = \int_{\mathbb{T}^\infty} \frac{|K_r K_s|}{N_r N_s} \, \mathrm{d}m.$$

A controlled functional has a representation $L(F) = \int_X F(s) w(s) \, \mathrm{d}\mu(s)$, where μ is positive, $|w(s)| \le N_s^{-2}$, and $S(\mu) = \operatorname{ess\,sup}_{r \in \operatorname{supp}\mu} \int_X H(r,s) N_s^2 \, \mathrm{d}\mu(s) < \infty$.

If $\mu = 0$, then $L = 0$ and the growth estimate is immediate. Otherwise choose $r_0 \in \operatorname{supp}\mu$ with $\int_X H(r_0, s) N_s^2 \, \mathrm{d}\mu(s) \le S(\mu)$. Since $K_{r_0} K_s$ is analytic with constant coefficient one, $H(r_0, s) \ge (N_{r_0} N_s)^{-1}$. Therefore $\int_X N_s \, \mathrm{d}\mu(s) \le N_{r_0} S(\mu)$, and every controlled functional obeys

$$|L(F)| \le C \sup_{s \in X} |F(s)| N_s^{-3}$$

where $C = N_{r_0} S(\mu)$. Indeed, $|L(F)| \le \int_X |F(s)| N_s^{-2} \, \mathrm{d}\mu(s)$, and factoring $|F(s)| N_s^{-2} = (|F(s)| N_s^{-3}) N_s$ gives the estimate.

Now let $a_k = 1 - 2^{-k}$, $r_k = (a_k, 0, 0, \ldots)$, and $L = \sum_{k \ge 1} 2^{-k} k^{-2} \delta_{r_k}$. The exact point-evaluation norm is $\|\delta_{r_k}\| = N_{r_k}^2 = (1 - a_k^2)^{-1} \le 2^k$, so $\sum_k 2^{-k} k^{-2} \|\delta_{r_k}\| \le \sum_k k^{-2} < \infty$. Thus the truncations converge in the dual norm.

For $0 < \rho < 1$, let $F_\rho(z) = (1 - \rho z_1)^{-3/2}$, a bounded analytic function in $H^1(\mathbb{T}^\infty)$. For $s \in X$,

$$|F_\rho(s)| \le (1 - |s_1|)^{-3/2} \le 2^{3/2} (1 - |s_1|^2)^{-3/2} \le 2^{3/2} N_s^3,$$

since $N_s^2 = \prod_j (1 - |s_j|^2)^{-1}$. The weighted supremum is uniformly bounded. But $L(F_\rho) = \sum_k 2^{-k} k^{-2} (1 - \rho a_k)^{-3/2}$ tends to infinity by monotone convergence: its kth limiting summand is $2^{k/2}/k^2$. Thus L cannot be controlled, although it is a norm limit of finite sums of evaluations. □

Full proof

2.3. **The factor two in two-point stability**

Is the factor 2 in the two-point stability corollary necessary?

Søvik, *Where Nehari Survives*, Chapter 1, first open question, following the two-point stability corollary.

Answer. No. For every $p > 0$, the sharp leading coefficient is 4, not 8.

Answer: Answered: no. **Evidence:** Sketch.

Proof. Set $k = \lceil p/2 \rceil$ and $a = 2k/p \ge 1$. In Hardy and Bergman geometry put $E(r,s) = \sum_j \rho(r_j, s_j)^2$, where ρ is pseudohyperbolic distance; in Gaussian geometry put $E(r,s) = \sum_j |r_j - s_j|^2$. Set $\kappa = 1$ in Hardy and Gaussian geometry and $\kappa = \beta$ in Bergman geometry

of parameter $\beta > 1$. Write K_r for the reproducing kernel and $c(r,s) = |K_s(r)|/\sqrt{K_r(r)K_s(s)}$ for its normalized overlap. If the normalized point-evaluation efficiencies

$$e_r(F) = \frac{|F(r)|^p}{K_r(r)\|F\|_p^p}, \qquad e_s(F) = \frac{|F(s)|^p}{K_s(s)\|F\|_p^p}$$

of $F \in H^p$ are at least $1-\eta$, where $0 \le \eta < 1 - 2^{-1/a}$, the integer power $G = F^k$ belongs contractively to a Hilbert analytic space with normalized kernel overlap $c(r,s)^a$. The two evaluation vectors in that Hilbert space then imply

$$c(r,s)^a \ge 2(1-\eta)^a - 1, \qquad \kappa E(r,s) \le -\frac{2}{a}\log\bigl(2(1-\eta)^a - 1\bigr).$$

Its expansion at zero is $4\eta + O_p(\eta^2)$. Sharpness of the leading coefficient follows by taking $F = 1$ in one variable at $r = -t, s = t$ as $t \downarrow 0$: then $\kappa E/\eta \to 4$ in each geometry. The complete statement and proof are in the book. □

Full theorem and proof

3. Nevanlinna classes

3.1. **Extending boundary composition to the completed Nevanlinna class**

It remains an open question whether Lemma 4.4 can be extended to the full Nevanlinna class $\mathcal{N}$.

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, Remark 4.5, pp. 27–28.

Answer. No. For a symbol satisfying the Gordon–Hedenmalm hypotheses, a natural scalar composition of a completed-class input need not belong to the completion.
Answer: Answered: no. **Evidence:** Sketch.

Proof. Choose a smooth $g \ge 0$ on $\mathbb{T}$ with mean $3/2$ that vanishes on an open arc, and put

$$\psi(z) = \frac{1}{2} + \int_{\mathbb{T}} \frac{\xi + z}{\xi - z} g(\xi)\,\mathrm{d}m(\xi), \qquad \Phi(s) = \psi(2^{-s}).$$

Smoothness gives a bounded boundary extension; $\operatorname{Re}\psi \ge 1/2$ on the closed disc, and $\operatorname{Re}\psi > 1$ for $|z| < 1/2$. Hence Φ has characteristic zero, maps $\mathbb{C}_1$ into $\mathbb{C}_1$, and has relatively compact range in $\mathbb{C}_0$.

Let $A = 64$, $a_k = A^k$, $N_k = \lceil \exp(\exp(A^{8k})) \rceil$, $L_k = \log N_k$, and

$$F(s) = \sum_{k \ge 1} \frac{a_k}{\sqrt{N_k}} P_{N_k}(s), \qquad P_N(s) = \sum_{n \le N} n^{-s}.$$

Harper's low-moment estimate gives $\|P_{N_k}\|_{\mathscr{H}^1} \le C\sqrt{N_k}(\log\log N_k)^{-1/4}$, so the block norms are summable and $F \in \mathscr{H}^1$. Its ordinary series converges absolutely on $\operatorname{Re} s > 1/2$. Comparison of sums and integrals gives $P_N(s) = N^{1-s}/(1-s) + O_T(1)$ uniformly for $1/2 \le \operatorname{Re} s \le 3/4$ and $|\operatorname{Im} s| \le T$. The kth main term dominates on the band $\alpha/L_k \le \varepsilon \le 1/L_k$: earlier terms sum to at most $a_k/63$, later terms to less than $a_k/100$, and the error is bounded. Consequently, for each fixed $\alpha, T > 0$,

$$|F(1/2 + \varepsilon + it)| \ge b_T a_k \quad (\alpha/L_k \le \varepsilon \le 1/L_k,\ |t| \le T)$$

for all large k.

Suppose some completed-class element h had canonical scalar evaluations agreeing with $F(\Phi(s))$ on a right half-plane. Its holomorphic Bohr lift B_h is defined on $\ell^1 \cap \mathbb{D}^\infty$. The slice $b(z) = B_h(z, 0, \ldots)$ lies in the classical disc Nevanlinna class: the completed gauge bounds the weighted radial means of $\log(1 + |B_h|)$, and successive submean inequalities in coordinates after the first, followed by the uniformly small ℓ^1 tail, bound the radial means of $\log(1 + |b|)$.

For fixed sufficiently large σ, the prime vertical orbit is dense in the torus of radii $(p_j^{-\sigma})_j$, since the prime logarithms are rationally independent. Continuity of the Bohr lift therefore extends the assumed agreement from that orbit to all phases. Integrating over coordinates after the first and using the holomorphic mean-value identity gives $b(2^{-\sigma}\chi_1) = G(2^{-\sigma}\chi_1)$, where $G = F \circ \psi$. Thus $b = G$ on $\mathbb{D}$ by the identity theorem.

On a closed positive-measure arc I inside the zero arc of g, the Poisson formula gives $\operatorname{Re}\psi(r\zeta) - 1/2 \asymp 1 - r$, uniformly for $\zeta \in I$, while $|\operatorname{Im}\psi(r\zeta)|$ is bounded. Taking $r_k = 1 - 1/(CL_k)$ puts the real margin in the band above, so $|G(r_k\zeta)| \geq b_T A^k$ on I. Hence the radial means of $\log(1 + |G|)$ diverge, contradicting $G = b \in N(\mathbb{D})$. This excludes a natural completed-class composition output and therefore any continuous extension that agrees with composition. The proof uses no surjectivity assumption for the completed Bohr lift. □

4. Paley–Wiener spaces

4.1. **Uniform separation of extremal zero sets**

The zero set of any solution of (1.6) is uniformly discrete for all $0 < p < \infty$.

Conjecture 9.6, p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces.* Uniform discreteness concerns distinct zero locations for each fixed extremizer.
Result. Yes, for every $p > 0$ and every extremizer.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. For $p \geq 1/2$ this is the published theorem of Brevig, Chirre, Ortega-Cerdà and Seip. Here is the argument for $0 < p < 1/2$. Their extremal identities imply that all zeros are real, each has multiplicity at most $1/p$, distinct zero locations are a finite union of separated sets, and for distinct zeros a, b,

$$\int_{\mathbb{R}} \frac{x^2|\varphi(x)|^p}{(x-a)(x-b)}\,\mathrm{d}x = 0.$$

If gaps between distinct zeros tended to zero, the finite-union counting bound and multiplicity bound would give a fixed integer K bounding the number of zeros, with multiplicity, in every unit interval. Starting from a shrinking gap, extend a cluster whenever the next zero is within a factor R of the current cluster diameter, where R tends to infinity slowly. The unit-interval bound forces this process to stop with at most K zeros; the two outer gaps then dominate the cluster diameter.

Strip those k zeros from φ, recenter at the cluster midpoint, and scale the cluster to $[-1, 1]$. On a window of fixed scaled radius, the remaining factor has logarithmic modulus $su + o(1)$ uniformly, for some real slope s that need not be bounded. The orthogonality integral on that window is therefore, up to a relative $o(1)$ error, the signed integral of $e^{su}|P(u)|^p/(u^2-1)$, where P is a monic polynomial of degree $2 \leq k \leq K$ with all roots in $[-1, 1]$. A uniform polynomial estimate says that a sufficiently large fixed outer part of the window has strictly more mass than the interval between the two cluster endpoints, for every such P and every slope s. The remaining integrand is nonnegative outside that interval, contradicting the orthogonality identity. Thus no shrinking gaps exist. The full proof and its uniform polynomial estimate appear in the proof note. □

4.2. **The distributional reproducing identity**

Theorem 8.1 extends to the full range $0 < p < \infty$.

Conjecture 9.7, p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces.* For an extremizer φ put $\Phi = |\varphi|^{p-2}\varphi/\|\varphi\|_p^p$; the assertion is that Φ is tempered and $f(0) = \int f\Phi$ for every $f \in \mathcal{S}(\mathbb{R}) \cap PW^2$.
Result. The identity and temperedness hold for every $p > 0$ and every extremizer.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. For $p \geq 1$ this is the source's Theorem 3.6, and for $1/2 \leq p < 1$ it is Theorem 8.1. For $0 < p < 1/2$, uniform separation from Conjecture 9.6 and simplicity of the zeros give a

polynomial lower bound $|\varphi(x)| \geq c\,\mathrm{dist}(x, Z(\varphi))(1+|x|)^{-1-4/p}$. Consequently Φ is locally integrable and its mass on unit intervals grows at most polynomially, so it defines a tempered distribution.

For real Schwartz $f \in PW^2$, vary the extremizer through $\varphi + \epsilon(f - f(0)\varphi)$, preserving its value at zero. The derivative of the p-th power integral at $\epsilon = 0$ exists despite $0 < p < 1$: the zero-set bound makes $|\varphi|^{p-1}|f - f(0)\varphi|$ integrable, and the pointwise first-order remainder is dominated and tends to zero after division by ϵ. Minimality forces the derivative to vanish, giving $\int f\Phi = f(0)$. Splitting a complex Schwartz function into its real and imaginary parts proves the complex identity. The detailed argument is in the proof note. □

4.3. **Limits of the first positive extremal zero**

We have $\lim_{p\to\infty} t_1(p) = 1/2$ and $\lim_{p\to 0^+} p\,t_1(p) > 0$.

Conjecture 9.10, p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. For $p < 1$, the source assumes uniqueness when it uses the notation $t_1(p)$.
Result. Both limits exist, and the small-p limit is finite and strictly positive under the source's uniqueness assumption.
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: Uniqueness of the extremal for all sufficiently small $p > 0$.

Proof. For the large-p limit, the normalized extremizers converge locally uniformly to $\cos(\pi z)$, whose first positive zero is $1/2$; Hurwitz's theorem gives the zero limit. For small p, assume that the normalized extremizer φ_p is unique for $0 < p < p_0$. Set $w_p(\tau) = |\varphi_p(\tau/p)|^p$ and $b(p) = pt_1(p)$. Write $\mathcal{C}_p = \sup_{\|f\|_p=1} |f(0)|^p$, with $f \in PW^p$. The published extremal-constant asymptotic gives $c_0 = \lim_{p\to 0} 2\mathcal{C}_p/p \in (0,\infty)$, so $\int w_p = p/\mathcal{C}_p \to 2/c_0$. Hölder's inequality applied to products of extremizers at nearby exponents makes $\sqrt{w_p}$ Cauchy in L^2, hence w_p converges in L^1. Removing the first symmetric zero pair from an extremizer gives a uniform positive lower bound on $b(p)$; an integral comparison gives a finite upper bound. Continuity of $b(p)$ then shows its cluster set is a closed interval. If it had endpoints $\alpha < \beta$, a comparison of the limiting zero potentials at the two endpoints would force $W(\tau) = 0$ for $\tau < \alpha$ and an incompatible strict sign on part of (α, β). Hence the cluster interval is a singleton. This proves convergence to a number in $(0,\infty)$. These arguments are written out in the small-p proof; the large-p proof is in the point-evaluation note. □

4.4. **Extremal zeros as a uniqueness set**

$Z(\varphi_p) \cup \{0\}$ is a uniqueness set for PW^p if and only if $p \geq 1$.

Conjecture 9.11, p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*. For $p < 1$ the source uses its standing uniqueness assumption to name φ_p.
Result. The zero set is a uniqueness set exactly when $p \geq 1$.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. For $p \geq 1$, the result is the published uniqueness theorem in the point evaluation note. For $0 < p < 1$, let φ be any extremizer and put $g(z) = z\varphi(z)$. We show $g \in PW^p$. If $A = \|\varphi\|_p^p$, minimality applied to $\varphi_\lambda(z) = \varphi(\lambda z)\,\mathrm{sinc}(\pi(1-\lambda)z)$ gives, for every $R > 0$,

$$\int_{|x|\geq R} |\varphi(x)|^p \,\mathrm{d}x \leq \frac{2A}{(\pi R + 2)(1 - 2^{-p})}.$$

Indeed, set $b = \pi(1-\lambda)/\lambda = 2/R$ in the minimality inequality and use $|\,\mathrm{sinc}(by)| \leq 1/2$ when $|y| \geq R$. On dyadic annuli $2^n \leq |x| < 2^{n+1}$ this tail estimate yields

$$\int_{2^n\leq|x|<2^{n+1}} |x\varphi(x)|^p \,\mathrm{d}x \leq C_p 2^{-n(1-p)}.$$

The series converges for $p < 1$, so g is an entire function of type at most π in L^p. It is nonzero, vanishes at zero, and vanishes at every zero of φ, with at least the same multiplicity. Thus the set is not a uniqueness set. The proof of the $p \geq 1$ direction is in the proof note. □

4.5. **Finite perturbations can destroy Fourier interpolation**

> For example, does moving (but not removing) finitely many interpolation points preserve the interpolation property?

The unnumbered question after Conjecture 7.5, Section 7, preprint p. 96, of Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko and Maryna Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, Ann. Math. 196 (2022), 983–1082. The interpolation data at each radius are $f, f', \widehat{f}, \widehat{f'}$; the canonical radii are $r_n = \sqrt{2n}$ in dimensions 8 and 24.

Result. No. In each of those two systems a collision-free finite move makes the data fail to determine the radial Schwartz function.

Answer: Answered: no. **Evidence:** Sketch.

Proof. The source's interpolation basis has radial Schwartz functions a_j, b_j with cardinal value and derivative data at the canonical radii; their Fourier transforms have zero canonical data. They are real, and radial Fourier inversion gives $\widehat{\widehat{h}} = h$. For a fixed sign $\epsilon \in \{1, -1\}$, set $u_j = a_j + \epsilon\widehat{a}_j$ and $v_j = b_j + \epsilon\widehat{b}_j$. Then $\widehat{u}_j = \epsilon u_j$, $\widehat{v}_j = \epsilon v_j$, and the two value and derivative rows at each canonical radius form the identity matrix.

Fix n and define $W(r) = u_n(r)v_n'(r) - u_n'(r)v_n(r)$. Cardinality gives $W(r_n) = 1$ and $W(r_m) = 0$ for $m \neq n$. For the centered dilation operator $D = x \cdot \nabla + d/2$, Fourier transformation gives $\widehat{Dh} = -D\widehat{h}$. Parseval's identity and the opposite Fourier parities imply $\int_{\mathbb{R}^d} u_n Dv_n = 0$. In radial coordinates, integration by parts converts this to $\int_0^\infty r^d W(r)\, \mathrm{d}r = 0$. Since W is positive near r_n, it is negative at some positive noncanonical radius t.

Choose M with $t < r_{M+1}$. On the finite span of u_j, v_j for $n_0 \leq j \leq M$, the canonical evaluation matrix is the identity. Replace r_n by t; expansion along all unchanged rows makes the determinant $W(t) < 0$. Sorting the radii only permutes blocks of two rows and does not change the sign. The straight path from the canonical ordered nodes to these ordered nodes stays positive, distinct, and below r_{M+1}. Continuity therefore gives an intermediate moved system with singular evaluation matrix. A nonzero real kernel vector gives a nonzero radial Schwartz function whose values and derivatives vanish at every moved node and every unchanged tail node; Fourier parity makes the Fourier data vanish as well. All four data are moved together, without collisions. □

5. Bernstein–Nikolskii inequalities

5.1. **Uniqueness of the radial L_1^d extremal**

> Prove that the radial extremal function in the problem L_1^d is unique not only among radial functions.

Problem 7, printed p. 37, of Sergei Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. The class consists of integrable entire functions of spherical exponential type at most one on $\mathbb{R}^d$; the reconstructed question is whether its normalized radial minimizer is unique without the radial restriction.

Result. The normalized radial minimizer is the unique minimizer in the full class for every $d \geq 1$.

Answer: Answered: yes. **Evidence:** Sketch.

Proof. Write φ_d for the radial minimizer, normalized by $\varphi_d(0) = 1$, and $m_d = \|\varphi_d\|_{L^1(\mathbb{R}^d)}$. Given any admissible minimizer f, replace it by the real part and then its central even part $g(x) = (f(x) + f(-x))/2$. For almost every unit vector ω, the even entire function $t \mapsto g(t\omega)$ has finite radial integral with weight $|t|^{d-1}$. Its even power series defines a radial entire function on $\mathbb{C}^d$ of spherical type at most one. It is therefore an admissible radial competitor.

Radial uniqueness and polar integration force equality in the raywise lower bound for almost every ω; hence $g = \varphi_d$. The central averaging inequality then shows that the odd part $h = f - g$ satisfies $|h(x)| \le |\varphi_d(x)|$ on every real ray.

We use the following division fact. If Φ is even entire of exact type $\sigma > 0$, has only simple real zeros, and is of bounded type in both half-planes, while H is entire of type at most σ, of bounded type in both half-planes, and $|H(x)| \le |\Phi(x)|$ on the real axis, then $H = c\Phi$. Indeed, every zero cancels in $q = H/\Phi$, so q is entire, of bounded type in both half-planes, and bounded on the real axis. Evenness and exact type give mean type σ for Φ in each half-plane. Additivity of mean type gives nonpositive mean type for q in each half-plane. Krein's theorem then makes q zero type; the Phragmén–Lindelöf principle for a zero-type entire function bounded on the real axis makes it constant.

The radial profile $\Phi(z) = \varphi_d(ze_1)$ has exact type one: otherwise a slight dilation would reduce its L^1 norm while preserving admissibility. Its zeros are simple and real by the radial extremal theorem of Dai–Gorbachev–Tikhonov. The restrictions of f and φ_d to each real line have bounded type in both half-planes, by Fourier inversion for functions with compact spectral support. Apply the division fact to $H(z) = h(z\omega)$ and Φ on each ray. It gives $h(z\omega) = c_\omega \varphi_d(z\omega)$; since h is odd and φ_d even, $c_\omega = 0$. Thus $h = 0$ on all real rays and $f = \varphi_d$. □

5.2. **Four-digit accuracy of the monotone-delta constant**

The upper bound is sharp "at least up to the first four significant digits."

The paragraph after Theorem 1, Section 1.2, p. 2, of Andrés Chirre, Dimitar K. Dimitrov, Emily Quesada-Herrera and Mateus Sousa, *An extremal problem and inequalities for entire functions of exponential type*, Proc. Amer. Math. Soc. 152 (2024). The constant is $A_1 = \inf_f \|f\|_1 / f(0)$ over nonnegative radially decreasing real entire functions of exponential type at most 2π.

Result. $1.2771345 < A_1 < 1.277135042105$, which fixes the requested four significant digits. **Answer:** Answered: yes. **Evidence:** Sketch.

Proof. The source reduces the extremal problem to the reciprocal norm of a positive Gram operator Q on ℓ^2. Let P_d project onto the first d coordinates and write

$$Q = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix}, \qquad \lambda_d = \|A\|.$$

The published estimates on the matrix entries give $\|B\| < 3/(2d)$ and $\|C\| < 1/d$ when $d = 3010$, while the source's certified finite-matrix calculation gives $\lambda_d < \ell = 783002554181/10^{12}$. For a unit vector (x, y) in the two blocks,

$$\langle Q(x,y),(x,y)\rangle \le \ell\|x\|^2 + 2b\|x\|\|y\| + c\|y\|^2, \qquad b = \frac{3}{2d}, \quad c = \frac{1}{d}.$$

The largest eigenvalue of this 2×2 form is at most $\ell + b^2/(\ell - c)$, and exact rational comparison at $d = 3010$ yields $\|Q\| < 10^7/12771345$. Since $A_1 = 1/\|Q\|$, the lower bound follows. The published certified trial function gives the upper bound 1.277135042105. This proves the stated enclosure, not the stronger nine-digit conjecture elsewhere in the paper. □

Part 2. **Where Nehari's theorem survives: structured Helson matrices**

6. Embeddings and boundary values

6.1. **All-exponent compactness for curve embeddings**

Søvik, the earlier critical-line chapter question on compactness of curve embeddings; the cited source location is a stale line reference and its exact original wording is unavailable. In the book's setting, let R be a prime set with $0 < \theta \le 1$ and $\zeta_R(1+t) = \prod_{p\in R}(1-p^{-1-t})^{-1} \asymp t^{-\theta}$ for $0 < t \le 1$ (the density hypothesis of the book). Let Q be its complement with orders $k_q \ge 2$, and let ν be a finite positive measure on $(1/2, \infty)$. Write $\mu_{\nu,R}$ for the measure

obtained by averaging the Q-coordinates $\omega_q q^{-s}$ over roots of unity of order k_q and using the real prime curve $(p^{-s})_{p\in R}$. The target is whether $\nu((1/2,1/2+t]) = o(t^\theta)$ implies compactness of $H^p(\mathbb{T}^\infty) \to L^p(\mu_{\nu,R})$ for every $p > 0$.

Answer. Yes, under those density, averaging, and finiteness hypotheses, the boundary-mass decay implies compactness for every $p > 0$.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. Proof outline. Root-of-unity averaging gives the common Carleson estimate in the book's all-p corollary. The little-o hypothesis makes the embedding norm of the part with $1/2 < s \le 1/2 + \delta$ tend to zero as $\delta \downarrow 0$. For fixed $\delta > 0$, the image of the remaining curve and root phases is a compact subset of $\mathbb{D}^\infty \cap \ell^2$: its coordinate-square tails are uniformly summable. The book's Montel lemma gives a subsequence of every bounded H^p sequence converging uniformly on this compact set, also for $0 < p < 1$, so restriction to the remaining finite measure is compact. The small norm of the near-boundary part then proves compactness of the full embedding. □

Full proof

7. Nehari theorems and Hankel forms

7.1. **Positive Nehari theorem for Helson matrices**

> In general there are Carleson measures for $H^2(\mathbb{T}^\infty)$ which are not Carleson measures for $H^1(\mathbb{T}^\infty)$, but we do not know if these can be chosen to be concentrated on $(-1,1)^\infty$.

Perfekt and Pushnitski, *On Helson matrices: moment problems, non-negativity, boundedness, and finite rank*, question after Remark 4.11, p. 15.

Answer. Yes. A finite positive measure supported on the real product cube is H^2-Carleson but not H^1-Carleson.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. For $x = (4/5, 4/5, 4/5)$, let $\sigma = (\delta_x + \delta_{-x})/2$. The two-point Szegő Gram matrix gives the exact constant

$$A = C_2(\sigma) = \frac{1}{2}\big((1-(4/5)^2)^{-3} + (1+(4/5)^2)^{-3}\big).$$

The positive rational function $U(z) = \prod_{j=1}^3 (1-(4/5)z_j)^{-1} + \prod_{j=1}^3 (1+(4/5)z_j)^{-1}$ satisfies $U(\pm x) = 2A$ and $\|U\|_2^2 = 4A$. For $U^2 - \varepsilon$, the right derivative at $\varepsilon = 0$ of $\int (U^2-\varepsilon)\,\mathrm{d}\sigma - A\|U^2-\varepsilon\|_1$ equals $Ac_0 - 1$, where $c_0 = \int \mathrm{Re}(\overline{U}/U)\,\mathrm{d}m$. A finite interval certificate in the linked proof gives $c_0 > 1/A$. Hence a small positive ε yields a strict gain; uniform Taylor approximation gives a polynomial f, positive on the real cube, with $\|f\|_1 = 1$ and $\int f\, d(\sigma/A) = a > 1$.

For each m, place $(\sigma/A)^{\otimes m}$ in a disjoint block of $3m$ coordinates and call it τ_m. Tensorization gives $C_2(\tau_m) = 1$ and τ_m has mass A^{-m}. The finite measure $\mu = \sum_{m\ge1} m^{-2}\tau_m$ has finite mass and $C_2(\mu) \le \sum m^{-2}$. The positive test $f^{\otimes m}$ on the mth block has integral at least $m^{-2}a^m$, while its H^1 norm is one. Thus $C_1(\mu) = \infty$. The complete interval certificate and tensor argument are in the linked proof. □

Full proof and certificate

7.2. **Are the apicality criteria necessary for nonsaturation?**

> It is natural to ask which, if any, of these two conditions may also be necessary for φ not to be saturated.

Brevig and Seip, *Maximal norm Hankel operators*, §3, after Problem 1. Their two sufficient conditions are apicality on a high-modulus set and nonempty interior of a low-modulus set.

Answer. Neither condition is necessary; separate outer examples disprove the low-set-interior and apicality conditions, respectively.
Answer: Answered: no. **Evidence:** Sketch.

Proof. For failure of the low-set-interior condition, let $A \subset \mathbb{T}$ be closed, nowhere dense, and of positive measure. Choose an outer function φ with $|\varphi| = 1/2$ on A and $|\varphi| = 1$ on $\mathbb{T} \setminus A$. Its nonconstant logarithmic modulus is bounded, so the criterion proved in the cited note implies that φ is not saturated. For $1/2 < \delta < 1$, the low-modulus set is A; for $\delta \le 1/2$ it is empty. Thus no low-modulus set has interior.

For failure of apicality, use a separate example: set $a = \pi/8$, let $v(e^{it}) = -(-t)^{-1/2}$ on $-a < t < 0$ and zero elsewhere, and write $V = \widetilde{v}$. On $0 < t < a$, the conjugate integral gives $V(t) = -t^{-1/2} + O(1)$ and $V'(t) = t^{-3/2}/2 + O(1)$. For $E = \{e^{it} : 0 < t < a,\ \cos V(t) \ge 1/2\}$, each sufficiently small dyadic interval $[r, 2r]$ contains a subset of E of measure at least cr, and the integral of $|V|^2$ over that subset is at least a fixed positive constant. Hence $m(E) > 0$ but $V \notin L^2(E)$.

Let $s = (\log 2)\mathbf{1}_{\mathbb{T}\setminus E}$, $F = \exp \mathcal{H}(v)$, $G = \exp \mathcal{H}(s)$, and $\varphi = F/G$. Then G is invertible, $|\varphi| = 1$ on E, $|\varphi| \le 1/2$ off E, and $\operatorname{Re}(\varphi G) = \cos V \ge 1/2$ on E. The top-set criterion proved in the note implies nonsaturation. If $\log|\varphi| = u + w$ with $u \in L^\infty$ and $\widetilde{w}$ bounded on any top set, then on E we have $\widetilde{w} = V - \widetilde{s+u}$. Since $s + u \in L^2$, its conjugate is in L^2; this would force $V \in L^2(E)$, a contradiction. So this example is not apical at any level. The note contains the full arguments and the explicit strict distance estimate. □

Full proofs

7.3. **Power invariance of apical weights**

> Is it true that w is an apical Helson–Szegő weight if and only if w^κ is an apical Helson–Szegő weight for all positive numbers κ? In other words, we ask if (7) could be replaced simply by (27) $\operatorname{ess\,sup}_{L^+_w(\delta)} |\widetilde{v}| < \infty$.

Brevig and Seip, *Maximal norm Hankel operators*, Problem 2 and condition (27).

Answer. Yes; a finite localized conjugate bound can be made arbitrarily small.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. Write $\log w = u + v$ with $u \in L^\infty$ real and $v \in L^1$ real, and let $E = \{w \ge \delta \|w\|_\infty\}$. Assume $C = \operatorname{ess,sup}_E |\widetilde{v}| < \infty$. Since $v = \log w - u$ is bounded above, move a constant between u, v so $v \le 0$. Let F be the analytic function with real part the Poisson extension of v, normalized by $F(0) = \int v \in \mathbb{R}$. On E, both v and $\widetilde{v}$ are bounded, so $|F| \le K$ almost everywhere there for some $K < \infty$.

For $N > 0$ set $B_N = NF/(N - F)$. Because $\operatorname{Re} F \le 0$, $|N - F| \ge N$ and $|B_N| \le N$ in the disc. Define $u_N = u + \operatorname{Re} B_N$ and $v_N = v - \operatorname{Re} B_N$. Then $\log w = u_N + v_N$, with $u_N \in L^\infty$ and $v_N \in L^1$. The normalization makes the conjugate of $\operatorname{Re} B_N$ equal to $\operatorname{Im} B_N$, and on E

$$|\widetilde{v}_N| = |\operatorname{Im}(F - B_N)| \le |F - B_N| = \frac{|F|^2}{|N - F|} \le K^2/N.$$

Choose $N > K^2/\eta$ for any prescribed $\eta > 0$. This proves the strengthened finite-bound criterion. Since $\log(w^\kappa) = \kappa \log w$ and $E_{w^\kappa}(\delta^\kappa) = E_w(\delta)$, the criterion applies to every positive power; applying it to w^κ with exponent $1/\kappa$ proves the converse. □

Full proof

7.4. **Norm-attaining vectors in complete Pick spaces**

> It would be interesting to know what can be said of the norm attaining vectors and Hilbert points in this context.

Bampouras and Brevig, *Norm attaining vectors and Hilbert points*, paragraph after Theorem 4, p. 4. We answer the scoped question whether norm-attaining vectors for $(\mathcal{H}, \mathcal{H} \odot \mathcal{H})$ are exactly the $\mathcal{H}$-extremal vectors.

Answer. Yes, for every separable normalized complete Pick space.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. Let x_0 be the normalization point, $\mathcal{A}$ the multiplier algebra, and $\mathcal{I} = \{q \in \mathcal{A} : q(x_0) = 0\}$. Normalize the kernel by $k_{x_0} = 1$. A unit vector f is extremal when $\langle qf, f\rangle = q(x_0)$ for all $q \in \mathcal{A}$. For $u \in \mathcal{H}$, write $M_u = \overline{\mathcal{A}u}$ and $N_u = \overline{\mathcal{I}u}$.

We use the complete Pick factorization $u = a/(1-b)$, with $a, b \in \mathcal{A}$, $\|M_b\| \le 1$, and $b(x_0) = 0$. The needed cancellation is this: if L is a closed multiplier-invariant subspace and $(1-b)v \in L$, then $v \in L$. Indeed, for $r < 1$, $v_r = (1-rb)^{-1}(1-b)v$ lies in L. The identity

$$(I - rM_b)^{-1}(I - M_b) = r^{-1}\big(I - (1-r)(I - rM_b)^{-1}\big)$$

shows $\|v_r\| \le 4\|v\|$ for $r \ge 1/2$. Positivity of the two-point multiplier kernel gives $|b(x)|^2 \le 1 - k(x,x)^{-1} < 1$, so $v_r(x) \to v(x)$. A weakly convergent subsequence therefore has limit v, and L is weakly closed. Applying cancellation to the factorization gives $M_u = M_a$ and $N_u = N_a$.

A consequence is the product transfer: if $g, h, gh \in \mathcal{H}$, $u \in N_g$, and $uh \in \mathcal{H}$, then $uh \in N_{gh}$. To see this, write $g = a/(1-b)$, $h = c/(1-d)$, and $u = e/(1-t)$ in the same form. Cancellation gives $N_g = N_a$ and $N_{gh} = N_{ac}$. Since $e = (1-t)u \in N_a$, choose $q_j \in \mathcal{I}$ with $q_j a \to e$. Multiplying by c gives $ec \in N_{ac}$. The identity $ec = (1-t)(1-d)uh$ and two cancellations prove the claim.

Suppose now that f is a unit extremal vector and factor it exactly as $f = gh$ with $\|g\|\|h\| = \|f\|_{\mathcal{H}\odot\mathcal{H}}$. Neither factor belongs to its own N-subspace, since the product transfer would put f in N_f, contrary to extremality. Let $\gamma = \|P_{N_g^\perp} g\| > 0$ and $\varphi = \gamma^{-1} P_{N_g^\perp} g$. Then $\varphi \in M_g$, $\|\varphi\| = 1$, and $q\varphi \in N_g$ for every $q \in \mathcal{I}$. Thus φ is extremal, hence a contractive multiplier by the complete Pick theorem. Write $g = \gamma\varphi + u$ with $u \in N_g$. The product transfer gives $uh \in N_f$, so $\varphi h \in M_f$. Repeating the projection for h gives $h = \delta\psi + v$, where ψ is a unit extremal contractive multiplier and $v \in N_h$. Apply the product transfer to (h, φ) to get $\varphi v \in N_{\varphi h} \subset N_f$. Therefore $f - \gamma\delta\varphi\psi \in N_f$. Pairing with $f \perp N_f$ yields

$$1 \le \gamma\delta \le \|g\|\|h\| = \|f\|_{\mathcal{H}\odot\mathcal{H}}.$$

The reverse inequality follows from $f = f \cdot 1$, so f is norm attaining.

Conversely, normalize a norm-attaining f so that $\|f\| = 1$. Its coefficient functional $L_f(u) = \langle u, f\rangle$ extends to the weak product with norm one. With inner products linear in the first variable, the associated contractive conjugate-linear Hankel operator A is defined by $\langle g, Ah\rangle = L_f(gh) = \langle gh, f\rangle$; in particular, $A1 = f$ and $\langle f, Aq\rangle = \langle fq, f\rangle$. For $q \in \mathcal{I}$ we have $q \perp 1$. Expanding $\|A(1+zq)\|^2 = \|f + \overline{z}\, Aq\|^2 \le \|1 + zq\|^2$ for arbitrary complex z forces the coefficient $\langle f, Aq\rangle$ of z to vanish. Thus $\langle qf, f\rangle = 0$ for $q \in \mathcal{I}$. Subtracting the constant value $q(x_0)$ proves $\langle qf, f\rangle = q(x_0)\|f\|^2$ for every multiplier; undoing the normalization gives the claimed extremality. □

7.5. **A countable signed curve test for positive Nehari**

> Wanted: an example where the signed matrix is bounded by cancellation but has no bounded symbol, or a proof that none exists for curve measures.

Søvik, the cancellation question in the multiplicative-Hilbert discussion. The answer addresses the narrower countable sign-rotated curve target stated with the question.

Answer. Yes. There is a countably atomic positive measure on sign-rotated prime-curve points with bounded H^2 embedding and unbounded absolute H^1 embedding.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. For small positive vectors x with $\max_j x_j$ small and $\sum_j x_j^2$ close to 3, the two-atom measure $\sigma_x = (\delta_x + \delta_{-x})/2$ has exact H^2 constant A_x and a real polynomial, positive on the real cube, whose H^1 moment ratio exceeds ρA_x for a fixed $\rho > 1$. Tensoring m such blocks gives gain ρ^m.

For each fixed $s > 1/2$, the sum $\sum_p p^{-2s}$ is finite, but it grows without bound as $s \downarrow 1/2$. Choose $s_m > 1/2$ close enough to the boundary to pack m disjoint prime blocks, each with squared energy close to 3. Put independent signs on the blocks and fix the remaining signs positive. The exact kernel Gram calculation and the complementary-coordinate evaluation cost give a finite measure ν_m with $C_2(\nu_m)$ known exactly and positive tests whose absolute H^1 ratio is at least $\rho^m C_2(\nu_m)$. Choose m_j with $\rho^{m_j} \geq 2 \cdot 4^j$ and set $\mu = \sum_j 2^{-j} C_2(\nu_{m_j})^{-1} \nu_{m_j}$. Then its mass and C_2 are at most one, while the positive test from block m_j gives $C_1(\mu) \geq 2^j$. All atoms are on actual sign-rotated prime-curve points, and the full block and mixture estimates are in the linked proof. □

Full construction

7.6. **Compact and noncompact inverse Hankel choices**

It is unknown if it is possible to choose $q \notin \operatorname{Range} R$ so that $I - Q$ is compact.
It is also unclear whether it is possible to find q so that $I - Q$ is noncompact.

Megretskii, Peller, and Treil, *The Inverse Spectral Problem for Self-Adjoint Hankel Operators*, §III.7, p. 295; repeated by Peller, *Hankel Operators and Their Applications*, Chapter 12, §13, p. 537.

Answer. Both choices exist for every admissible operator with trivial kernel, while preserving Peller's extra integrability condition.
Answer: Answered: both choices exist. **Evidence:** Sketch.

Proof. Fix Peller's self-adjoint Γ with $\ker \Gamma = \{0\}$ satisfying conditions (C1)–(C3). In his construction, W is multiplication by t on $\mathcal{K} = \int^{\oplus} E(t)\, \mathrm{d}\rho(t)$, J is his unitary multiplication operator with $J^2 = I$, and $R = JW$. The first spectral level $\mathcal{K}_0 = \{\varphi(t)e_1\}$ is identified with $L^2(\rho)$. An admissible vector has the form $q = Jq_0$, where $q_0 \in \mathcal{K}_0$, $\|q_0\|_2 = 1$, $q_0 \neq 0$ ρ-almost everywhere, and $q_0 \notin \operatorname{Range} W$. Since J is unitary and commutes with W, this last condition is equivalent to $q \notin \operatorname{Range} R$. Peller's contraction Q is defined by $QW^{1/2} = |A|^{1/2}$, with $A^2 = W^2 - (\cdot, p)p$ and $p = Rq$. Peller's Lemma 10.6 reduces the calculation to the first, W-cyclic spectral level: $Q = Q_0 \oplus I$ and $Q_0 M^{1/2} = |A_0|^{1/2}$ on $L^2(\rho)$, where M is multiplication by s, $p = sq_0$, and $A_0^2 = M^2 - (\cdot, p)p$. It is therefore enough to study $I - Q_0$.

Since $\ker \Gamma = \{0\}$, W is injective; by (C2) it is noninvertible. Hence zero is an accumulation point of its spectrum. Replacing Γ by $\Gamma/\|\Gamma\|$ leaves Q unchanged, so assume $\|W\| \leq 1$. Start with any admissible $q_0^\sharp$, including Peller's additional condition: for almost all $t \in \sigma_2^{(\pm)}$,

$$\int_{\sigma(W)\setminus\sigma_2^{(\pm)}} \frac{|sq_0^\sharp(s)|^2}{|s^2 - t^2|}\, \mathrm{d}\rho(s) < \infty.$$

This is Peller's condition (14.8); the spectral sets $\sigma_2^{(\pm)}$ are those in his construction. They lie in the set where ρ and the scalar spectral measure of $|A|$ have the same null sets, so the almost-everywhere qualification is independent of the admissible choice. Let $I_k = (2^{-k-1}, 2^{-k}]$ for $k \geq 0$ and write $k_1 < k_2 < \cdots$ for the indices with $\rho(I_{k_l}) > 0$. The spectrum has positive ρ-mass arbitrarily close to zero, so this list is infinite and $k_l \geq l - 1$. Multiply $q_0^\sharp$ by a positive constant on each I_{k_l} to prescribe positive band masses $m_l = \int_{I_{k_l}} |q_0|^2\, \mathrm{d}\rho$ summing to one. This preserves its norm and null set; the range condition is verified for each choice below. The additional integral condition is preserved as explained below.

For the compact choice, take $m_l = 1/(l(l+1))$. For every n,

$$\int \frac{|q_0(s)|^2}{s^2}\,\mathrm{d}\rho(s) \geq \sum_{l>n} 4^{k_l} m_l \geq \frac{4^n}{n+1},$$

so $q_0 \notin \operatorname{Ran} W$. For $t > 0$ put

$$g_t(s) = \frac{s q_0(s)}{s^2+t}, \qquad d(t) = 1 - ((M^2+t)^{-1}p, p) = \int \frac{t}{s^2+t}\,\mathrm{d}\mu(s), \quad \mathrm{d}\mu = |q_0|^2\,\mathrm{d}\rho.$$

The rank-one resolvent identity and the integral formula for the fourth root then give

$$M^{1/2} - |A_0|^{1/2} = c\int_0^\infty \frac{t^{1/4}}{d(t)}(\cdot, g_t) g_t\,\mathrm{d}t, \qquad c = \frac{\sin(\pi/4)}{\pi}.$$

This is $(I - Q_0)M^{1/2}$. Dividing its kernel in the second variable by $y^{1/2}$ suggests the following kernel for $I - Q_0$:

$$K(x,y) = cxy^{1/2} q_0(x)\overline{q_0(y)}\,\mathcal{I}(x,y), \qquad \mathcal{I}(x,y) = \int_0^\infty \frac{t^{1/4}\,\mathrm{d}t}{(x^2+t)(y^2+t)d(t)}.$$

Let $\nu(t)$ count the indices l with $2^{-k_l} > \sqrt{t}$. Every later band lies below $\sqrt{t}$, so

$$d(t) \geq \frac{1}{2}\sum_{l>\nu(t)} m_l = \frac{1}{2(\nu(t)+1)}.$$

Splitting the integral defining $\mathcal{I}$ at a^2, where $a = \max(x,y)$, and using $t^{1/4}/(\min(x,y)^2 + t) \leq t^{-3/4}$ gives $\mathcal{I}(x,y) \leq Ca^{-3/2}(1+\nu(a^2))$. Indeed, for $t \geq a^2$ we have $\nu(t) \leq \nu(a^2)$, and for $t < a^2$ the number of intervening dyadic bands gives $\nu(t) \leq \nu(a^2) + 1 + \log_2(a^2/t)$; the integral of $u^{-3/4}(1 + \log_2(1/u))$ on $(0,1)$ is finite. If $x \in I_{k_l}$ and $y \in I_{k_{l'}}$ with $l \leq l'$, then $\nu(a^2) \leq l$ and $\min(x,y)/a \leq 2^{1-(l'-l)}$. Consequently

$$\int_{I_{k_l}\times I_{k_{l'}}} |K(x,y)|^2\,\mathrm{d}\rho(x)\,\mathrm{d}\rho(y) \leq C\,2^{1-(l'-l)}\frac{(1+l)^2}{l^2 l'^2}.$$

The same bound holds with l, l' interchanged, since $x^2 y/\max(x,y)^3 \leq \min(x,y)/\max(x,y)$. The sum over both triangles is finite: $(1+l)^2/l^2 \leq 4$, $l'^{-2} \leq l^{-2}$ for $l' \geq l$, and the sum over $l' - l$ is geometric. Thus K is square integrable.

Let T_K be its Hilbert–Schmidt integral operator. For $F = M^{1/2}f$ and $h \in L^2(\rho)$, Tonelli and Cauchy–Schwarz justify interchanging the resolvent and kernel integrals, because

$$\begin{aligned}\int_0^\infty \frac{t^{1/4}}{d(t)} \iint |g_t(x) g_t(y) f(y) h(x)|\,\mathrm{d}\rho(y)\,\mathrm{d}\rho(x)\,\mathrm{d}t \\ = c^{-1}\iint |K(x,y)F(y)h(x)|\,\mathrm{d}\rho(y)\,\mathrm{d}\rho(x) < \infty.\end{aligned}$$

Hence $(I - Q_0)F = T_K F$. Since $\operatorname{Ran} M^{1/2}$ is dense, the two bounded operators agree everywhere. Therefore $I - Q_0$ is Hilbert–Schmidt, and $I - Q = (I - Q_0) \oplus 0$ is compact.

For the noncompact choice, take $m_l = c_0 2^{-k_l}/l^2$, with c_0 chosen so that the masses sum to one. Then

$$\int \frac{|q_0(s)|^2}{s}\,\mathrm{d}\rho(s) < \infty, \qquad \int \frac{|q_0(s)|^2}{s^2}\,\mathrm{d}\rho(s) = \infty,$$

so $q_0 \in \operatorname{Ran} W^{1/2}$ but $q_0 \notin \operatorname{Ran} W$. Write $q_0 = W^{1/2}h$ and $C = Q^*Q$. The identity $CWC = W - (\cdot, W^{1/2}q_0)W^{1/2}q_0$ gives

$$\|W^{1/2}Ch\|^2 = \|q_0\|^2 - |\langle W^{1/2}h, q_0\rangle|^2 = 1 - 1 = 0,$$

so $Ch = 0$ and $Qh = 0$. To see that A is injective, equality in $\|Ax\|^2 = \|Wx\|^2 - |\langle Wx, q_0\rangle|^2$ would force Wx to be a multiple of q_0; since $q_0 \notin \operatorname{Ran} W$, this is possible only for $x = 0$. Taking adjoints in $QW^{1/2} = |A|^{1/2}$ gives $W^{1/2}Q^* = |A|^{1/2}$, so injectivity of A and $W^{1/2}$ gives $\ker Q^* = \{0\}$. If $I - Q$ were compact, Q would be Fredholm of index zero. Its dense range would then force $\ker Q = \{0\}$, contrary to $Qh = 0$ with $h \neq 0$.

Finally, the rescaling is constant on each dyadic band and takes only finitely many values on compact subsets of $(0, \infty)$. Near a fixed $t > 0$ it is bounded above and below; away from $(t/2, 2t)$, $s^2/|s^2 - t^2| \leq 4/3$. Thus the extra integral condition is finite for one vector exactly when it is finite for the other. □

7.7. **A converse Jensen-measure criterion**

An open problem is to obtain a converse to Lemma 5.1, characterizing $\sup_{\tau \in J_\phi} \|\tau_a\|$ in terms of A-subharmonic functions.

Cole and Gamelin, *Representing Measures and Hardy Spaces for the Infinite Polydisk Algebra*, §5, after Lemma 5.1, printed p. 119. Precisely, if J_ϕ is the Jensen measures for a character ϕ and τ_a is the part absolutely continuous with respect to a probability μ, determine $\lambda = \sup_{\tau \in J_\phi} \tau_a(X)$ by A-subharmonic functions.

Answer. The converse holds: λ is the smallest admissible limit in the criterion of Lemma 5.1, even for continuous A-subharmonic functions.

Answer: Answered: yes. **Evidence:** Sketch.

Proof. Let M_A be the character space. Let $\mathcal{S}$ be the cone of finite positive sums of constants and the continuous truncations $\ell_{f,N} = \max(\log|\widehat{f}|, -N)$, with $f \in A$. Jensen measures are exactly the probabilities τ on X such that $u(\phi) \leq \int u \, \mathrm{d}\tau$ for every $u \in \mathcal{S}$: the reverse implication follows by letting $N \to \infty$ in the truncated logarithmic inequalities. For real $h \in C(X)$ set

$$P(h) = \inf_{v \in -\mathcal{S},\ v|_X \geq h} v(\phi).$$

This is a finite, monotone sublinear functional with $P(h + c) = P(h) + c$. Hahn–Banach gives a linear functional $L \leq P$ attaining $P(h)$ at any selected h. Positivity follows from $L(-g) \leq P(-g) \leq 0$ for $g \geq 0$, and $L(1) = 1$. Riesz representation then gives a Jensen probability τ and proves $P(h) = \sup_{J_\phi} \int h \, \mathrm{d}\tau$.

For $M > 0$, minimax applied to $0 \leq h \leq 1$ yields

$$\sup_{\tau \in J_\phi} (\tau \wedge M\mu)(X) = \inf_{0 \leq h \leq 1} \left(P(h) + M \int (1 - h) \, \mathrm{d}\mu \right).$$

As $M \uparrow \infty$, the left side increases to λ. Choose $M_n = 2^n$ and continuous h_n within 2^{-n} of the infimum; by the dual formula choose $v_n \in -\mathcal{S}$ majorizing h_n on X and nearly attaining $P(h_n)$. Since $h_n \geq 0$, Jensen's inequality makes $v_n \geq 0$ throughout M_A. The minimizing bound implies $\sum_n \int (1 - v_n)_+ \, \mathrm{d}\mu < \infty$. Tonelli gives $(1 - v_n)_+ \to 0$ almost everywhere, so $u_n = -v_n$ satisfy $u_n \leq 0$, $\limsup u_n \leq -1$ almost everywhere, and $u_n(\phi) \to -\lambda$. Cole and Gamelin's criterion gives the reverse inequality for any admissible sequence, proving sharpness. □

8. Hankel forms and Schatten classes

8.1. **Sparse-prime eigenvalues above the critical band**

eigenvalues above Λ_α

Inferred question. Motivated by the Bergman-space result in Theorem 1.1(5) of Ma, Lu, and Zu, *On the spectra of multiplicative Hilbert operators on Bergman spaces of Dirichlet series*, we ask whether the positive critical matrix on a regular sparse-prime semigroup can have eigenvalues above its essential band or embedded in it. This is our extension, not a question posed by those authors.

The answer depends on the prime set. Finite deletions can produce arbitrarily many eigenvalues above the band; for primes congruent to 1 modulo 4 there are none. Under a stated C^2 regularity condition, there are no embedded eigenvalues.
Answer: Answered: depends on the prime set. **Evidence:** Sketch.

Book draft, critical essential-spectrum and sparse-prime eigenvalue results

Proof. Assume $0 < \theta \le 1$. Let $\mathcal{S}_R$ be the semigroup generated by R, assume $t^\theta \zeta_R(1+t) \to c_R > 0$, and set

$$A_R(m,n) = \frac{\Gamma(\theta)}{\sqrt{mn}(\log(mn))^\theta}, \qquad L_R = \frac{c_R \Gamma(\theta/2)^2}{\Gamma(\theta)}.$$

Gamma integration factors $A_R = N^*N$, with synthesis operator

$$(Na)(u) = \sum_{n \in \mathcal{S}_R \setminus \{1\}} a_n n^{-1/2-u}$$

on $L^2((0,\infty), u^{\theta-1}\,\mathrm{d}u)$. Its companion $T = NN^*$ has kernel $\zeta_R(1+u+v)-1$. The critical Mellin comparison gives essential spectrum $[0, L_R]$, and the nonzero eigenvalues of A_R and its companion coincide.

For any regular initial set R_0, a finite deletion F changes the boundary constant to $c_R = c_{R_0} \prod_{q \in F}(1-1/q)$ while preserving θ. Since $\sum_{q \in R_0} q^{-1} = \infty$, the new band edge can be made smaller than the least eigenvalue of a fixed positive principal block indexed by $p, p^2, \dots, p^k$, with $p \notin F$. The min-max principle then gives at least k eigenvalues above the band. Taking $R = \{p : p \equiv 1 \pmod 4\}$ instead, Euler-factor comparison gives $\zeta_R(1+t) - 1 \le c_R t^{-1/2}$ for all $t > 0$. The Mellin transform of the majorizing kernel shows $\|A_R\| = L_R$, so no eigenvalue lies above the band; the same regularity argument excludes eigenvalues in the band.

For embedded eigenvalues in general, assume $\Phi(t) = t^\theta \zeta_R(1+t)$ extends as a C^2 function to $[0,1]$ with $\Phi(0) = c_R$. An eigenfunction f of the companion at $E \in (0, L_R]$ has Mellin transform satisfying

$$(c_R B(x, \theta - x) - E)\mathcal{M}f(x) = -\mathcal{M}g(x),$$

where g is the difference between the exact and homogeneous kernels. The beta factor can equal E/c_R only on the critical line $\operatorname{Re} x = \theta/2$. Plancherel rules out poles on that line; a bootstrap across it gives the boundary estimates needed to differentiate the eigenvalue equation. Put $G(u) = f'(u) + (\theta-1)f(u)/u$. The commutator identity gives

$$E\langle f', G\rangle = -\langle TG, G\rangle \le 0, \qquad \langle f', G\rangle \ge \frac{\theta}{2-\theta}\|f'\|_2^2.$$

Thus $0 \ge E\langle f', G\rangle \ge E\theta(2-\theta)^{-1}\|f'\|_2^2$, forcing $f' = 0$. The infinite mass of $u^{\theta-1}\,\mathrm{d}u$ then gives $f = 0$. Thus the band has no eigenvalues under this regularity hypothesis. The C^2 assumption is essential to the stated scope of this last conclusion. □

8.2. **Unimodality of a multiplicative Hilbert norm**

> Does there exist a unique $\alpha_{\min} > 1$ such that $\|T_\alpha\|$ is strictly decreasing on $(1, \alpha_{\min})$ and strictly increasing on $(\alpha_{\min}, \infty)$?

Ma, Lu, and Zu, *On the Spectra of Multiplicative Hilbert Operator on Bergman Spaces of Dirichlet Series*, §6.2, Question 1, p. 73 (version 3).

Yes. The norm has a unique minimum at some $\alpha_{\min} \in (8/5, 3)$.
Answer: Answered: yes. **Evidence:** Sketch.

Conditional on: arXiv:2608.22882v3.

Book draft, "Unimodality in the Bergman parameter" subsection

Proof. The source's integral representation writes $N(\alpha) = \|T_\alpha\|$ as the norm of a positive kernel operator, with essential edge Λ_α. On $[8/5, 3]$, the logarithm of the kernel is convex

in α pointwise; positivity then makes $\log N(\alpha)$ strictly convex. Interval bounds on the differentiated Euler sums show that N is strictly decreasing on $(1, 8/5]$ and strictly increasing on $[3, \infty)$, while the one-sided slopes at $8/5$ and 3 point into the interval. Strict convexity gives a unique interior minimizer, and the endpoint monotonicity extends the decrease and increase to the full intervals $(1, \alpha_{\min})$ and $(\alpha_{\min}, \infty)$. The interval certificates and kernel estimates are in the cited chapter. □

8.3. **A universal Mellin constant for positive Helson matrices**

> Let the companion kernel of a positive Helson matrix grow like $c\,x^{-\kappa}$ at the boundary, with the fixed critical weight $u^{\kappa-1}\,\mathrm{d}u$. Is its critical essential spectrum always $[0, c\,\Gamma(\kappa/2)^2/\Gamma(\kappa)]$?

Søvik, book draft, Chapter 5, problem on the universal positive-Helson constant.

For $\kappa > 0$ and $c > 0$, under the literal limit $x^\kappa h(x) \to c$, the interval is the asserted one. Comparability alone does not force that interval.

Answer: Answered: yes under a limit; no under mere comparability. **Evidence:** Sketch.

Book draft, the companion-kernel theorem and translated-kernel limits

Proof. Assume $\kappa > 0$ and $c > 0$. Write $\mathrm{d}\nu(u) = u^{\kappa-1}\gamma(u)\,\mathrm{d}u$, with $\gamma(u) \to b > 0$, and assume the companion kernel has $x^\kappa h(x) \to c$. Under the unitary change $u = e^{-s}$, the integral kernel becomes

$$K_0(s-t)\,\varphi(\Lambda(s,t))w(s)w(t), \qquad K_0(r) = (2\cosh(r/2))^{-\kappa},$$
$$\varphi(L) = e^{-\kappa L}h(e^{-L}), \qquad w(s) = \gamma(e^{-s})^{1/2}.$$

Here $\Lambda(s,t) = -\log(e^{-s} + e^{-t})$. As $s, t \to \infty$, $\Lambda(s,t) \to \infty$, so $\varphi(\Lambda(s,t)) \to c$ and $w(s)w(t) \to b$. Truncating to a tail therefore differs, in essential spectrum, by a norm-small operator from the half-line convolution model with kernel $bcK_0(s-t)$; the part meeting a fixed initial interval is Hilbert–Schmidt. The Fourier multiplier of convolution by K_0 is

$$m(\xi) = \int_{\mathbb{R}} K_0(r)e^{-ir\xi}\,\mathrm{d}r = \frac{|\Gamma(\kappa/2 + i\xi)|^2}{\Gamma(\kappa)}.$$

It is positive, even, and decreases from $m(0) = \Gamma(\kappa/2)^2/\Gamma(\kappa)$ to zero. The half-line model has essential spectrum $[0, bcm(0)]$, which gives the formula; in the question $b = 1$.

If $x^\kappa h(x)$ is merely comparable to a positive constant, its logarithmic profile φ can oscillate instead of converging. A periodic profile gives a translation-periodic limiting kernel; Floquet decomposition then produces bands, and the certified oscillating example in the book has a gap. Thus comparability does not imply the interval formula. □

8.4. **A Besov criterion for curve-measure Schatten classes**

> For noninteger p, converting the corresponding spectral condition into a one-dimensional Besov criterion on F remains open.

Søvik, curve-measure Schatten endpoint note, paragraph after the integer trace formulas. Let ν be a finite positive Borel measure on $(0, \infty)$; then $F(t) = \nu((0,t])$ is finite and increasing, with $F(0+) = 0$. We interpret the passage as asking whether, for

$$M_\nu = \left(\int_0^\infty (mn)^{-1/2-u}\,\mathrm{d}\nu(u)\right)_{m,n\ge 1}.$$

$M_\nu \in S_p$ can be characterized by a one-dimensional Besov condition on F for noninteger $p > 0$.

Yes, with a one-sided endpoint Besov condition of smoothness one; no fixed global unweighted Besov space gives the same criterion.

Answer: Answered: yes, at an endpoint. **Evidence:** Sketch.

Book draft, "Distribution functions and endpoint smoothness" section

Proof. For $I_j = (e^{-j-1}, e^{-j}]$, put $a_j = e^j \nu(I_j)$ and $b_j = e^j F(e^{-j})$. Decomposing the tail into these intervals gives

$$b_j = \sum_{k \geq j} e^{-(k-j)} a_k, \qquad a_j \leq b_j.$$

The geometric convolution is bounded on ℓ^p for every $p > 0$ (Minkowski when $p \geq 1$, and p-subadditivity when $0 < p < 1$). Hence $a \in \ell^p$ iff $b \in \ell^p$. For $t \in I_j$, monotonicity of F gives $e^{-1} b_{j+1} \leq F(t)/t \leq e b_j$. Integrating on each logarithmic interval therefore shows

$$M_\nu \in S_p \quad \Longleftrightarrow \quad \int_0^1 \left(\frac{F(t)}{t} \right)^p \frac{\mathrm{d}t}{t} < \infty,$$

using the all-exponent dyadic criterion for the curve matrix.

For $0 < r < \infty$, let $A_r(F; h)$ be the normalized L^r average of F on $(0, h)$, and set $A_\infty(F; h) = \sup_{0<t<h} |F(t)|$. Since F is increasing, $2^{-1/r} F(h/2) \leq A_r(F; h) \leq F(h)$, with the evident endpoint version. Thus the displayed integral is equivalent to the one-sided first-difference Besov seminorm of smoothness one and summability p, localized at 0; the polynomial of degree less than one is forced to be $F(0+) = 0$.

A global unweighted Besov membership condition cannot replace it. An interior atom makes F a step function, which belongs only below the usual jump smoothness threshold (or at its weak endpoint). In the remaining smoothness range, take $\mathrm{d}\nu(u) = \beta u^{\beta-1} \mathbf{1}_{(0,1)} \, \mathrm{d}u$ with $0 < \beta < 1$, so $F(t) = t^\beta$ has the required smoothness but the associated matrix is unbounded: testing on $x_n = n^{-1/2}/\sqrt{H_N}$ for $2 \leq n \leq N$ yields a quadratic form bounded below by $c H_N (\log N)^{-\beta} \to \infty$. The endpoint-localized criterion answers the question while the global obstruction specifies its limitation. □

8.5. **A sparse-prime logarithmic Weyl law**

Our conditional question is: under which boundary regularity assumptions does the critical sparse-prime matrix with measure $u^{\theta-1}(1 + \log(1/u))^{-\alpha} \, \mathrm{d}u$ have an exact eigenvalue asymptotic? The all-prime model of Miheisi and Pushnitski provides the comparison case in Theorem 1.1.

Suppose $0 < \theta \leq 1$, $\alpha > 0$, and for an integer $k > \alpha$, $r(t) = t^\theta \zeta_R(1 + t) - c_R$ satisfies $|r^{(\ell)}(t)| \leq C_\ell t^{\eta - \ell}$ on $0 < t \leq 2$ for $0 \leq \ell \leq k$ and some $\eta > 0$. Then the positive eigenvalues satisfy $\lambda_n \sim \kappa_\theta(\alpha) n^{-\alpha}$, with the Mellin constant below.
Answer: Answered: $\lambda_n \sim \kappa_\theta(\alpha) n^{-\alpha}$. **Evidence:** Sketch.
Conditional on: the finite-derivative bounds stated above.

Proof. Let $\mathrm{d}\nu(u) = u^{\theta-1}(1 + \log(1/u))^{-\alpha} \, \mathrm{d}u$ on $(0, 1]$. The companion operator on $L^2(\nu)$ has kernel $\zeta_R(1 + u + v)$. Under $u = e^{-x}$ the unitary image on $L^2(0, \infty)$ has kernel

$$w(x) w(y) K_\theta(x - y) \{ c_R + r(e^{-x} + e^{-y}) \}, \quad w(x) = (1 + x)^{-\alpha/2}, \quad K_\theta(t) = (2 \cosh(t/2))^{-\theta}.$$

The model term has Fourier multiplier

$$m_\theta(\xi) = \frac{c_R |\Gamma(\theta/2 + i\xi)|^2}{\Gamma(\theta)}.$$

Factoring by $m_\theta(D)^{1/2}$ converts its nonzero eigenvalues to those of $m_\theta(D)^{1/2} \mathbf{1}_{(0,\infty)}(1 + X)^{-\alpha} m_\theta(D)^{1/2}$. Smooth the step at the origin; the compact-block error has singular values $O(n^{-M})$ for each fixed M, by Taylor approximation of K_θ on $[0, 1]^2$. The one-sided Weyl law for a polynomially decaying position weight and exponentially decaying multiplier then gives

$$\kappa_\theta(\alpha) = \left(\frac{1}{2\pi} \int_{\mathbb{R}} m_\theta(\xi)^{1/\alpha} \, \mathrm{d}\xi \right)^\alpha.$$

For the exact kernel, the chain rule and the assumed derivative bounds give $|\partial_x^\ell E(x, y)| \leq C_\ell e^{-c(x+y)}$ for $\ell \leq k$, where $c = \min(\eta, \theta)/2$. Truncate at $L = (k/c) \log n$ and divide $[0, L]$ into n intervals; Taylor polynomials of degree $k - 1$ in x give a rank at most kn approximation and error $O((L/n)^k + e^{-cL})$. Hence $s_n(E) = O(n^{-k} (\log n)^k) = o(n^{-\alpha})$, so the leading eigenvalue

constant is unchanged. Finally the synthesis factorization N^*N and companion NN^* transfers the asymptotic to the Helson matrix. The hypotheses on the finite derivatives are part of the theorem; a boundary limit alone is not claimed to suffice. □

8.6. **The spectrum of an unbounded Helson operator**

Inferred question. For the measure model in Ma, Lu, and Zu's Example 3.6, p. 21, with parameter $0 < c < 1$, do the associated unbounded Helson operators have spectrum $[0, \infty)$ and purely absolutely continuous spectrum of multiplicity one on $(0, \infty)$? The source establishes the operator model and domain facts, not this spectral conclusion.

Conditional on the closed-operator identification and adjoint-core result in Theorem 1.2, p. 3, and Lemma 2.7, p. 12, of the same arXiv preprint, the spectrum is $[0, \infty)$, purely absolutely continuous on $(0, \infty)$, with multiplicity one.

Answer: Answered: yes. **Evidence:** Sketch.

Conditional on: arXiv:2610.12252v1.

Book draft, "An unbounded family with simple continuous spectrum" section

Proof. Assume $0 < c < 1$. Write $g(u) = u^{c-1}e^{-u}/\Gamma(c)$ and use the synthesis operator $(Na)(u) = \sum_n a_n n^{-u-1/2}$. Conditional on the cited domain results, $H_c = N^*N$ and its companion $A = NN^*$, after a unitary change of density, has kernel $\sqrt{g(u)g(v)}\zeta(1 + u + v)$. Compare it with the positive homogeneous operator K whose kernel is

$$\frac{(uv)^{(c-1)/2}}{\Gamma(c)(u + v)}.$$

Their difference is the sum of kernels containing $e^{-(u+v)/2} - 1$ and $\zeta(1 + u + v) - (u + v)^{-1}$. Both are Hilbert–Schmidt: for the first, the squared norm reduces to

$$\frac{B(c, c)}{\Gamma(c)^2} \int_0^\infty t^{2c-3}(1 - e^{-t/2})^2 \, \mathrm{d}t < \infty,$$

and for the second the remainder $0 \le \zeta(1 + t) - 1/t \le 1$ and $\int g = 1$ suffice. The first integral converges at zero because $c > 0$, and at infinity because $c < 1$. The common closed-form domain then gives $A = K + C$ with compact C.

The unitary dilation $U_\lambda f(u) = \lambda^{1/2} f(\lambda u)$ satisfies $U_\lambda^{-1} K U_\lambda = \lambda^{1-c} K$. Since K is nonzero and positive, scaling makes its spectrum $[0, \infty)$. Compact perturbation preserves this essential spectrum; positivity of A then yields $\sigma(A) = \sigma_{\mathrm{ess}}(A) = [0, \infty)$. Polar decomposition transfers its positive spectrum to $N^*N = H_c$.

For spectral type, the book's argument sharpens the Hilbert–Schmidt difference to trace class by writing both kernels as sums or integrals of rank-one Laplace kernels and comparing the sum with its integral interval by interval. The homogeneous operator K has trivial kernel by injectivity of the Laplace transform. For $X = \delta^{-1} \log K$, its unitary dilations and e^{isX} satisfy the Weyl relations. Von Neumann uniqueness identifies this pair with a direct sum of Schrödinger representations; the dilation generator has simple spectrum under the logarithmic change of variables, so the sum has multiplicity one. Thus K is unitarily equivalent to multiplication by λ in $L^2((0, \infty), \mathrm{d}\lambda)$. Kato–Rosenblum then gives the same absolutely continuous multiplicity for A. The commutator with the self-adjoint dilation generator is δK plus a compact error ($\delta = 1 - c$); its strict Mourre estimate away from zero excludes singular continuous spectrum. Differentiating the eigenvalue equation and integrating against the derivative gives a nonnegative kernel form equal to minus a sum of strictly positive weighted derivative terms, so there are no positive eigenvalues. These steps prove pure absolute continuity and multiplicity one on $(0, \infty)$, subject throughout to the stated domain identification. □

Part 3. Where Nehari's theorem fails: Schatten classes

9. Hankel forms and Schatten classes

9.1. Rank-fourteen obstruction to same-rank evaluation approximation

> We conjecture that it is possible to take N_m as the rank of H, i.e. that the set of continuous Hankel forms of rank $\le r$ coincides with the closure (in any reasonable topology) of the set $\{(f,g) \mapsto \sum_{j=1}^{r} c_j f(z_j) g(z_j) : c_1, \dots, c_r \in \mathbb{C}, z_1, \dots, z_r \in \Omega\}$. (For $n = 1$, this follows easily from (14.3), but we have been unable to find a proof in higher dimension.)

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, Remark 14.2, p. 126.

No in general: a continuous rank-fourteen Hankel form in six variables is not a pointwise limit of sums of at most fourteen point-evaluation forms.

Answer: Answered: no. **Evidence:** Sketch.

Proof. Jelisiejew's Theorem 1.1, parts 3 and 4, DOI 10.1016/j.laa.2018.08.002, gives a non-cone cubic $C = \sum_{|\alpha|=3} c_\alpha u^\alpha$ in six variables whose contraction-apolar algebra has Hilbert function $(1, 6, 6, 1)$ and is nonsmoothable. Convert contraction to ordinary differentiation by putting $F(u) = \sum_{|\alpha|=3} c_\alpha u^\alpha / \alpha!$, where $\alpha! = \prod_{j=1}^{6} \alpha_j!$. For $p(z) = \sum_\beta p_\beta z^\beta$, the coefficient of u^γ in $p(\partial)F$ is

$$\frac{1}{\gamma!} \sum_\beta p_\beta c_{\gamma+\beta}.$$

Thus the ordinary differential annihilator of F is exactly the contraction annihilator of C. Let $A = \mathbb{C}[z_1, \dots, z_6]/\operatorname{Ann}(F)$ and $\lambda(p) = (p(\partial)F)(0)$. Its Gorenstein pairing $\lambda(pq)$ has rank $\dim A = 14$. On the standard Fock space $\mathcal{F}^2(\mathbb{C}^6)$, with $\|f\|_{\mathcal{F}^2}^2 = \sum_\alpha \alpha! |f_\alpha|^2$ for $f(z) = \sum_\alpha f_\alpha z^\alpha$, define $\Gamma(f,g) = \lambda(fg)$. This pairing uses only degrees through three, so it is a bounded Hankel form of rank fourteen.

Assume that sums of at most fourteen point-evaluation forms approach Γ pointwise. Choose a quotient basis $p_1 = 1, p_2, \dots, p_{14}$ of A with each $\deg p_i \le 3$. Its Gram matrix and the six mixed-moment matrices are

$$G_{ij} = \lambda(p_i p_j), \qquad (V_k)_{ij} = \lambda(p_j z_k p_i), \qquad M_k = G^{-1} V_k.$$

The Gorenstein pairing makes G invertible. Every entry of G and V_k is a moment of degree at most seven, so pointwise convergence forces the approximating matrices $G_m, V_{k,m}$ to converge to G, V_k. Eventually G_m is invertible. After combining repeated support points and deleting zero weights, write an approximant as $\lambda_m = \sum_{s=1}^{r_m} a_s \operatorname{ev}_{z_s}$. If $E_{is} = p_i(z_s)$ and $D = \operatorname{diag}(a_s)$, then $G_m = EDE^{\mathsf{T}}$. Its rank is fourteen, while $r_m \le 14$, so $r_m = 14$, all weights are nonzero, and E is invertible. Moreover

$$M_{k,m} = G_m^{-1} V_{k,m} = E^{-\mathsf{T}} \operatorname{diag}(z_{1,k}, \dots, z_{14,k}) E^{\mathsf{T}}.$$

These matrices commute, and $p_i(M_{1,m}, \dots, M_{6,m}) e_1 = e_i$ for every i. Passing to the limit preserves these polynomial equations.

Any commuting tuple $M_1, \dots, M_6$ satisfying $p_i(M_1, \dots, M_6) e_1 = e_i$ defines a length-fourteen quotient: the map $q \mapsto q(M_1, \dots, M_6) e_1$ is onto, and its kernel is an ideal because the matrices commute. The equations specify a closed quotient-basis chart. The approximating evaluation tuples are reduced points of this chart, so their limit would make A a flat limit of reduced length-fourteen schemes, that is, smoothable. This contradicts the choice of C. Only moments of degree at most seven were used, so a neighborhood defined by finitely many pointwise tests already excludes all sums of at most fourteen evaluations. This refutes the several-variable conjecture even in the pointwise topology and leaves the one-variable assertion unaffected. □

9.2. **Operator ideals with the Berger–Coburn phenomenon**

Are there nontrivial operator ideals in $B(H^2(\mathbb{T}), L^2(\mathbb{T}))$ or in $B(A^2(\mathbb{B}_n), L^2(\mathbb{B}_n))$ for which the Berger-Coburn phenomenon holds?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, Open Problem 1, p. 4.

No: in either setting the only ideal with the phenomenon is the whole operator space.

Answer: Answered: no. **Evidence:** Sketch.

Three open problems of Hu and Virtanen

Proof. First, any ideal in $B(X, Y)$ that contains one noncompact operator T equals $B(X, Y)$. Indeed, some spectral projection E of T^*T away from zero has infinite-dimensional range. Choose an isometry $V : X \to EX$. Then TV is bounded below, so it has a bounded left inverse $L : Y \to X$. For every $S \in B(X, Y)$,

$$S = (SL)TV,$$

which belongs to the ideal property whenever T does.

It remains to exhibit a bounded symbol with $H_f = 0$ and $H_{\bar{f}}$ noncompact, in each of the Hardy and Bergman spaces. Let $g(z) = (1 - z)^i$ on the disc, take its boundary values on $\mathbb{T}$, and on the ball set $f(z) = g(z_1)$. The function is bounded and holomorphic, so $H_f = 0$. For real $0 < t < 1$, $(1 - t^2)|g'(t)| = 1 + t \geq 1$. Testing $H_{\bar{f}}$ on normalized reproducing kernels at t (or $(t, 0, \ldots, 0)$) and changing variables by the automorphism interchanging that point with zero shows its norm is bounded below by the first Taylor coefficient of g after this change of variables. The displayed derivative estimate makes this lower bound uniform as $t \to 1$. The kernels converge weakly to zero, so $H_{\bar{f}}$ is not compact.

If an ideal has the Berger–Coburn property, it contains $H_{\bar{f}}$ because it contains $H_f = 0$. Since $H_{\bar{f}}$ is noncompact, the first paragraph forces the ideal to be all of $B(X, Y)$. The whole space plainly has the property. □

9.3. **Schatten exponents for the Berger–Coburn phenomenon on Fock spaces**

Let F^2_ϕ be a doubling Fock space. For which values of p does the Berger-Coburn phenomenon hold for $S_p(F^2_\phi, L^2_\phi)$?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, Open Problem 2, p. 9.

Every doubling Fock space has the phenomenon at $p = 2$ and fails it for $0 < p \leq 1$. For each $p > 1$ with $p \neq 2$ there are doubling weights for which it fails; for constant-Laplacian weights it holds for all $1 < p < \infty$.

Answer: Answered: holds on every space iff $p = 2$; some spaces iff $p > 1$. **Evidence:** Sketch.

Three open problems of Hu and Virtanen

Proof. For bounded symbols, the cited local-oscillation characterization says $H_f \in S_p$ exactly when

$$\int_{\mathbb{C}} \rho(z)^{-2} G(f)(z)^p \, \mathrm{d}A(z) < \infty,$$

where ρ is the doubling scale and $G(f)$ is the local L^2 distance to holomorphic functions. At $p = 2$, the known characterization of simultaneous membership of H_f and $H_{\bar{f}}$ gives the Berger–Coburn property on every doubling space. On constant-Laplacian weights, the Hu–Virtanen theorem gives it for all $1 < p < \infty$.

The counterexamples use $f(z) = 1/z$ for $|z| \geq 1$ and $f(z) = 0$ inside the unit disc. The characterization yields $H_f \in S_p$ but $H_{\bar{f}} \notin S_p$ for $0 < p \leq 1$ on every doubling Fock space. For $1 < p < 2$, take $\varphi(z) = |z|^m$ with an even integer $m \geq 2p/(2 - p)$; for $p > 2$, take $\varphi(x + iy) = y^{2n+2}$ with an integer $n > 1/(p - 2)$. The same symbol has finite local-distance integral for f and divergent integral for $\bar{f}$ under these respective weights, so the property fails. These ranges, together with the universal $p = 2$ case and the constant-Laplacian positive result, give the claimed every-space and some-space classifications. □

9.4. **Compactness and conjugate Fock Hankel operators**

Let F^2_ϕ be a doubling Fock space and $f \in L^\infty$. Is it true that if $H_f : F^2_\phi \to L^2_\phi$ is compact, then $H_{\bar f}$ is also compact?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, Open Problem 3, p. 9.

Yes, for every doubling Fock weight and every bounded symbol.

Answer: Answered: yes. **Evidence:** Sketch.

Three open problems of Hu and Virtanen

Proof. Let $\rho(z)$ be the radius for which the doubling measure of $D(z, \rho(z))$ is one. For fixed $r > 0$, write $G_r(f)(z)$ for the normalized L^2 distance on $D(z, r\rho(z))$ from f to holomorphic functions, and $\mathrm{MO}_r(f)(z)$ for the corresponding distance to constants. Compactness of H_f and weak convergence to zero of normalized kernels imply $\|H_f k_z\|_2 \to 0$. On a fixed small disk around z, the weighted normalized kernel has size comparable to $\rho(z)^{-1}$ and has no zeros. Dividing its projection by the kernel therefore gives a local holomorphic approximant to f, with error bounded by $C\|H_f k_z\|_2$. Hence $G_{r_0}(f)(z) \to 0$ for a fixed $r_0 > 0$.

The doubling geometry gives local comparability of ρ and $\rho(z) = o(|z|)$. For any sequence $z_j \to \infty$, rescale $f_j(\zeta) = f(z_j + \rho(z_j)\zeta)$. The local approximants just obtained, on a countable cover of each fixed disk, have vanishing L^2 error and locally bounded holomorphic representatives because f is bounded. Montel's theorem yields a subsequential entire limit; it is bounded by $\|f\|_\infty$, so Liouville's theorem makes it constant. Thus the local mean oscillations tend to zero. The $p = q = 2$ compactness criterion for doubling Fock spaces says that this vanishing mean oscillation makes both H_f and $H_{\bar f}$ compact. □

9.5. **Hankel operators on weighted Bergman A^1**

For $1 < p < \infty$, the problem whether $H^\omega_f : A^1_\omega \to L^1_\omega$ is bounded

Duan, Guo, Hong, and Yan, *Toeplitz and Hankel operators on Bergman spaces with doubling weights*, final paragraph before the references.

Yes. Every symbol satisfying the source's logarithmic local-oscillation condition gives a bounded map $A^1_\omega \to L^1_\omega$ for radial two-sided doubling weights ω.

Answer: Answered: yes. **Evidence:** Sketch.

Hankel operators with logarithmic-BMO symbols

Proof. Let $\ell(z) = 1 + \log(1/(1 - |z|))$ and let M be the logarithmic-BMO seminorm. A Whitney partition and smoothing of the local averages produce a differentiable F with $|\nabla F| \lesssim M/(\delta\ell)$, where $\delta = 1 - |z|$. This decomposition does not assert a pointwise bound on $h = f - F$. Its needed residual estimate, for every analytic g, is

$$\int_{\mathbb{D}} \ell|h|\,|g|\,\omega \,\mathrm{d}A \lesssim M\|g\|_{A^1_\omega}.$$

For the smooth part, solve $\bar\partial u = g\,\bar\partial F$ by the canonical Bergman kernel

$$S_N(z, w) = \frac{(1 - |w|^2)^N}{(z - w)(1 - z\bar w)^N},$$

with N larger than the tail exponent of ω. The weighted column estimate gives

$$\int \ell|T_N(g\bar\partial F)|\omega \lesssim M\|g\|_{A^1_\omega}.$$

The far region follows by summing the geometric shells $2^{j(b-N)}$; near the diagonal, integrate first in the unweighted input variable and use analytic local averages, which is needed for narrow radial peaks. Applying the actual weighted projection makes the solution canonical, and its A^1_ω bound yields $\|(I - P_\omega)(fg)\|_{L^1_\omega} \lesssim M\|g\|_{A^1_\omega}$. Density extends the estimate to all $g \in A^1_\omega$. This proves the full statement for the stated two-sided doubling class, including unbounded symbols. □

9.6. **Why the threshold $p = 4$ appears in two settings**

> It would be interesting to understand whether the appearance of the same threshold in these two settings reflects some connection between them. In particular, the endpoint case $p = 4$ seems especially interesting, since its Schatten norm can be explicitly computed.

Bampouras, *Nehari's theorem for Schatten classes of Hankel forms on the polydisc and Paley–Wiener spaces*, end of §1.

The two thresholds arise from different mechanisms. This does not rule out a further relation between the settings.

Answer: Answered: the mechanisms differ. **Evidence:** Sketch.

Proof. Let Γ be either the positive lattice $\mathbb{N}_0^d$ or a bounded open convex set $\Omega \subset \mathbb{R}^n$. In both settings, expanding $\|H_\varphi^* H_\varphi\|_{S_2}^2$ gives the same quartic integral

$$\|H_\varphi\|_{S_4}^4 = \int_{a+d=b+c} \overline{\widehat{\varphi}(a)}\widehat{\varphi}(b)\widehat{\varphi}(c)\overline{\widehat{\varphi}(d)} W_\Gamma(a,b,c,d),$$

where W_Γ is the measure of the allowed common shift, explicitly

$$W_\Gamma(a,b,c,d) = \big|\{\zeta \in \Gamma : a-\zeta,\ b-\zeta,\ \zeta+c-a \in \Gamma\}\big|, \qquad d = b+c-a.$$

Replacing W_Γ by 1 gives $\|\varphi\|_4^4$.

For $\Gamma = \mathbb{N}_0^d$, direct counting gives $W_\Gamma = \prod_{j=1}^d (1 + \min(a_j, b_j, c_j, d_j)) \geq 1$. More precisely, the constant monomial contributes the unweighted quartic form, and each other monomial contributes a nonnegative fourth power. Hence $\|\varphi\|_4 \leq \|H_\varphi\|_{S_4}$. This positive constant-term contribution is the torus mechanism.

For $\Gamma = \Omega$, put $\omega = \chi_\Omega * \chi_\Omega$ and $L_x = \{\xi \in \Omega : x - \xi \in \Omega\}$. If Z is the admissible-shift set, then $Z \subset L_a \cap L_b$. Translation by $c-a$ maps Z into $L_c \cap L_d$, because for $\zeta \in Z$ the point $\zeta' = \zeta + c - a$ satisfies $c - \zeta' = a - \zeta$ and $d - \zeta' = b - \zeta$. Thus its measure is bounded by each of $\omega(a), \omega(b), \omega(c), \omega(d)$. Choose a fixed nonzero $\beta \in C_c^\infty(B(0,1))$ and $\widehat{\varphi}(\xi) = \beta((\xi - z)/\rho)$, with $B(z,\rho) \subset 2\Omega$ and z at distance ρ from the boundary. The Schur test and the Hilbert–Schmidt identity give

$$\|H_\varphi\|_{S_4}^4 \leq a\rho^{3n}\|\beta\|_1^2\|\beta\|_2^2, \qquad \|\varphi\|_4^4 = \rho^{3n}\|\check{\beta}\|_4^4,$$

where $a = \sup_{B(z,\rho)} \omega$. Since ω vanishes on the boundary, $a \to 0$ as $\rho \downarrow 0$. Therefore no uniform $L^4 \leq CS_4$ inequality holds for this Paley–Wiener model.

The Paley–Wiener statement is method-level. Let $\Omega \subset \mathbb{R}^2$ be a bounded convex set with C^2 boundary and positive curvature, with uniform interior and exterior rolling-ball radii. Fix $\beta \in C_c^\infty(B(0,1))$, choose caps $\widehat{\varphi}_i(\xi) = \beta((\xi - z_i)/\rho)$ contained in 2Ω, and require the input lenses

$$D_i = \{\xi \in \Omega : \xi + \eta \in \operatorname{supp} \widehat{\varphi}_i \text{ for some } \eta \in \Omega\}$$

to be pairwise disjoint. For $\psi = \sum_{i=1}^N \varphi_i$ the narrow counting test is $h = \varphi = \psi$, and

$$Q_p(\psi,\psi) = \frac{\|\psi\|_2^2}{\|\psi\|_1 \|H_\psi\|_{S_p}}.$$

Under these hypotheses, $Q_p \leq C_p$ for $1 \leq p \leq 4$, while suitable such configurations make Q_p unbounded for every $p > 4$.

Here are the estimates giving the planar upper bound. Choose w_0 with $\beta(w_0) \neq 0$, put $x_i = z_i + \rho w_0$, $m_i = x_i/2$, $h_i = \operatorname{dist}(m_i, \partial\Omega)$, and $L_i = \Omega \cap (x_i - \Omega)$. The rolling-ball condition and containment of each cap give $h_i \geq c\rho$; the lenses L_i are disjoint and their areas $\omega_i = |L_i|$ sum to S. Subdivide a fixed central contraction of each lens into disjoint squares of side $s\rho$, with $s > 0$ sufficiently small. There are at least $c\omega_i\rho^{-2}$ such squares. The normalized indicators of a square and its reflection through $x_i/2$ form orthonormal families. On each pair the matrix coefficient of H_ψ has size at least $c\rho^2$: only cap i contributes, and β varies by

at most half its nonzero value. The Schatten diagonal inequality therefore yields, for every $p \geq 1$,

$$\|H_\psi\|_{S_p}^p \geq c\rho^{2(p-1)}S. \tag{1}$$

Group the caps by depth: I_j consists of those with $h_i \in [\delta_j, 2\delta_j), \delta_j = 2^j c\rho$, and h_i below a fixed small rolling-ball radius; the remaining deep class has bounded cardinality. For a shallow lens, rolling balls give $|L_i| \geq c\delta_j^{3/2}$ and $L_i \subset \{u : \operatorname{dist}(u, \partial\Omega) \leq 4\delta_j\}$. The area of this boundary layer is $O(\delta_j)$, so disjointness gives $N_j\delta_j^{1/2} \leq C$, where $N_j = |I_j|$. Also $S \geq c\sum_j N_j\delta_j^{3/2}$ and each lens has area at least $c\rho^{3/2}$. Set $A = 1 + \sum_j \sqrt{N_j}$. These bounds imply

$$A^2 \geq cN, \qquad A^8\rho^2 \leq CN^2S. \tag{2}$$

For the second inequality, write

$$\sqrt{N_j} = N_j^{1/4}(N_j\delta_j^{3/2})^{1/8}(N_j\delta_j^{1/2})^{1/8}\delta_j^{-1/4}$$

and apply Hölder with exponents $4, 8, 8/5$; the geometric sum $\sum_j \delta_j^{-2/5} \leq C\rho^{-2/5}$ gives $\sum_j \sqrt{N_j} \leq CN^{1/4}S^{1/8}\rho^{-1/4}$. The deep class and the additive 1 in A are absorbed using its bounded size and $S \geq c\rho^{3/2}$.

The needed lower Khintchine estimate is

$$\|\psi\|_1 \geq cN^{3/2}A^{-2}. \tag{3}$$

To see why curvature matters, first fix one depth class. The identity $\|\psi_j\|_4^4 = \|\widehat{\psi}_j * \widehat{\psi}_j\|_2^2$ bounds this fourth moment by $C\rho^6 E_j$, where E_j counts quadruples satisfying $|m_i + m_{i'} - m_k - m_{k'}| < 2\rho$. On each of finitely many boundary arcs write the boundary as $(a, g(a))$, with $g'' \geq \kappa > 0$. Disjoint lenses and the boundary-layer area estimate show that an interval of length $\lambda \geq \sqrt{\delta_j}$ contains at most $C\lambda/\sqrt{\delta_j}$ boundary coordinates. For fixed i, k, the quadruple relation puts $a_{k'}$ within $O(\rho)$ of $a_{i'} + a_i - a_k$; its vertical component forces

$$|g(a_i) + g(a_{i'}) - g(a_k) - g(a_{i'} + a_i - a_k)| \leq C\delta_j.$$

Split each depth class into finitely many such arcs and apply the triangle inequality in L^4, so it suffices to count quadruples lying in one graph chart. The derivative of the expression inside absolute values has constant sign and magnitude at least $\kappa|a_i - a_k|$. If $|a_i - a_k| \geq \sqrt{\delta_j}$, the local packing bound leaves only $O(1)$ choices for i', k'. If it is smaller, there are only $O(1)$ choices of k for each i; for each such pair there are at most N_j choices of i', and cap separation leaves only $O(1)$ choices of k'. Thus $E_j \leq CN_j^2$; classes at fixed positive depth contain only $O(1)$ caps and satisfy the same estimate trivially. Consequently $\|\psi_j\|_4 \leq C\sqrt{N_j}\rho^{3/2}$ and, summing classes, $\|\psi\|_4 \leq CA\rho^{3/2}$. Disjoint cap supports give $\|\psi\|_2^2 = N\rho^2\|\beta\|_2^2$. Interpolation $\|\psi\|_2^2 \leq \|\psi\|_1^{2/3}\|\psi\|_4^{4/3}$ proves (3).

Since $\|\psi\|_2^2 \asymp N\rho^2$, (1) and (3) give

$$Q_p(\psi, \psi)^p \leq C_p \frac{A^{2p}\rho^2}{N^{p/2}S}.$$

For $p \leq 4$, (2) bounds the right side by $C_p N^{p/2-2} \leq C_p$. This proves boundedness through the endpoint for this counting family; it does not prove the Paley–Wiener Nehari inequality.

For the reverse threshold, take $\rho = \delta$ caps at depth comparable to δ along a curved boundary arc, with centers separated tangentially by a sufficiently large multiple of $\sqrt{\delta}$. Then $N \asymp \delta^{-1/2}$, the corresponding lenses are disjoint, and $\omega = |\Omega \cap (x - \Omega)| \leq C\delta^{3/2}$ on every cap. Apply weighted Cauchy–Schwarz with an integer $k > 1$:

$$\|\psi\|_1 \leq \left(\int_{\mathbb{R}^2} (1 + \rho^2|x|^2)^{-k}\,\mathrm{d}x\right)^{1/2} \left(\int_{\mathbb{R}^2} (1 + \rho^2|x|^2)^k |\psi(x)|^2\,\mathrm{d}x\right)^{1/2} \leq C\sqrt{N}.$$

The first factor is $O(\rho^{-1})$. By Plancherel, the square of the second factor is bounded by a constant times $\sum_{|\alpha|\le k}\rho^{2|\alpha|}\|\partial^\alpha\widehat{\psi}\|_2^2$. The differentiated cap supports remain disjoint, and each cap contributes $O(\rho^2)$ to this sum; hence it is $O(N\rho^2)$. For one cap the Schur and Hilbert–Schmidt bounds are $\|H_{\varphi_i}\| \le C\rho^2$ and $\|H_{\varphi_i}\|_{S_2}^2 \le C\omega\rho^2$. The operators act on pairwise orthogonal D_i blocks, so for $p \ge 2$,

$$\|H_\psi\|_{S_p}^p = \sum_i \|H_{\varphi_i}\|_{S_p}^p \le CN\delta^{3/2}\rho^{2(p-1)}.$$

Together with $\|\psi\|_2^2 \asymp N\rho^2$, this gives $Q_p \ge cN^{1/2-1/p}(\rho^2/\delta^{3/2})^{1/p} \asymp N^{1/2-2/p}$, which diverges exactly for $p > 4$. The lattice threshold has a different explanation: its positive constant-term contribution to the quartic form gives $\|\varphi\|_4 \le \|H_\varphi\|_{S_4}$, whereas the Paley–Wiener cap construction above rules out the analogous L^4 bound. □

9.7. **A finite-rank form missed by point jets on a punctured surface**

For the one complex dimensional Riemann surface of Example 3.4 it seems reasonable to conjecture that every finite rank symbol is of the form $b(\cdot) = \sum_{k=1}^N a_k Z^{n_k} K_{z_k}(\cdot)$ for some constants a_k, z_k, n_k.

Deck and Gross, *Hankel forms on complex manifolds*, Remark 6.10, p. 94.

No. On the literal punctured surface, the constant symbol gives a bounded rank-one Hankel form that is not a finite sum of the proposed point jets.

Answer: Answered: no. **Evidence:** Sketch.

Proof. On the Riemann surface of $z^{1/n}$, the coordinate $w = z^{1/n}$ identifies the surface with $\mathbb{C}\setminus\{0\}$, while its spectral algebra is $\mathbb{C}[w]$. For the lifted Gaussian measure with parameter $a > 0$, angular integration gives

$$\|w^k\|^2 = (2a)^{k/n}\Gamma(1 + k/n), \qquad k \ge 0.$$

The monomials are orthogonal, so evaluation at the omitted point is bounded: $|P(0)| = |\langle P, 1\rangle| \le \|P\|$. Therefore $\Gamma_1(P, Q) = \langle PQ, 1\rangle = P(0)Q(0)$ is a bounded rank-one Hankel form.

A finite sum of point jets at nonzero points $w_1, \dots, w_s$ annihilates

$$P(w) = \prod_{j=1}^s (1 - w/w_j)^{m_j+1},$$

where m_j bounds the jet order at w_j. Yet $P(0) = 1$, so the functional $P \mapsto P(0)$ does not lie in the span of these jets. This refutes the literal point-support conjecture; it does not classify finite-rank symbols on other analytic spaces. □

9.8. **Failure of clipped-moment monotonicity for Gaussian products**

Does the clipped-moment monotonicity proved for affine Gaussian symbols extend to general products?

Søvik, *An analytic clipped-symbol proof of the affine Gaussian fourth-Schatten bound*, sentence after the affine corollary. For independent standard complex Gaussians, let $R_n = |Z_1 \cdots Z_n|$, define $A_n(k) = \mathbb{E}(1 - k^2R_n^2)_+$, $K_n(k) = \mathbb{E}[R_n \min(kR_n, 1)]$, and $Q_n = A_n^4 + 4A_n^2K_n^2 + 2^nK_n^4$.

No: for every $n \ge 194$, $Q_n'(k) < 0$ for some $k > 0$.

Answer: Answered: no. **Evidence:** Sketch.

Book draft, "Clipped moments of products" subsection

Proof. Put $B = \mathbb{E}[R_n^2\mathbf{1}_{kR_n<1}]$ and

$$\Phi = A_n^2K_n + 2^{n-1}K_n^3 - kA_n^3 - 2kA_nK_n^2.$$

Differentiating under the expectation gives $A_n' = -2kB$ and $K_n' = B$, hence $Q_n' = 8B\Phi$ with $B > 0$. If $\varepsilon_n = \mathbb{E}R_n = \Gamma(3/2)^n$, then $\varepsilon_n - 1/(4k) \le K_n \le \varepsilon_n$ and $\Phi \le 2^n\varepsilon_n^3 - kA_n^3$.

For $n \geq 3999$, let $s = 3/20$, $x = 1 - s$, $m = n\psi(x)$, $w = \sqrt{2n\psi'(x)}$, and choose $\log k = -(m + w + 1)/2$. Under the exponential change of measure with density $e^{-sY}/\Gamma(x)^n$, where $Y = \log R_n^2$, the summands of Y are independent log-Gamma variables with mean m and variance $w^2/2$. Chebyshev gives probability at least $1/2$ to $|Y - m| \leq w$, on which $1 - k^2 e^Y \geq 1 - e^{-1}$. Therefore

$$A_n \geq \frac{1 - e^{-1}}{2} \Gamma(x)^n e^{s(m-w)}.$$

Substitution in the bound for Φ reduces negativity to $n\rho > c_1\sqrt{n} + c_0$, where

$$\rho \in [0.0317484, 0.0317485], \qquad c_1 \in [1.944943, 1.944944], \qquad c_0 \in [3.955466, 3.955468].$$

At $n = 3999$ the difference is greater than 0.0127, and it increases thereafter.

For $194 \leq n \leq 3998$, take $k = e^{0.4276n}$. Mellin inversion gives, for $0 < c < 1$,

$$A_n = \frac{1}{2\pi} \int_{\mathbb{R}} \operatorname{Re} \frac{k^{-2(c+it)} \Gamma(1 - c - it)^n}{(c + it)(c + 1 + it)} \, \mathrm{d}t.$$

With $c = 3/20$, the tail $|t| > 2$ is bounded using $|(c + it)(c + 1 + it)| \geq t^2$ and

$$\left| \frac{\Gamma(x + iy)}{\Gamma(x)} \right|^2 = \prod_{j \geq 0} \left(1 + \frac{y^2}{(x + j)^2} \right)^{-1}.$$

Outward-rounded Gauss–Legendre integration on $[-2, 2]$, with the tail added, certifies $A_n > 0$ and makes $A_+^2 \varepsilon_n + 2^{n-1} \varepsilon_n^3 - k A_-^3 - 2k A_- K_-^2 < 0$ for every integer in that finite range, where $K_- = \varepsilon_n - 1/(4k)$. The interval certificate is included with the book's computation. Thus $\Phi < 0$ throughout both ranges, proving the claim. The cases $3 \leq n \leq 193$ are not settled by this argument. □

9.9. **A decaying first moment for high-dimensional Gaussian characters**

The open question is whether some such character still has a first moment that decays with n.

Søvik, *Non-Gaussian finite-torus block routes*, immanant discussion.

For every integer $n \geq 2$, a sign-twisted standard character of dimension $n-1$ has normalized Gaussian immanants Φ_n with $\mathbb{E}|\Phi_n| \to 0$.

Answer: Answered: yes. **Evidence:** Sketch.

Book draft, "Gaussian immanants" section

Proof. Let Z be an $n \times n$ matrix of independent standard complex Gaussian entries, and set

$$\chi_n(\sigma) = \operatorname{sgn}(\sigma)(\operatorname{fix}(\sigma) - 1), \qquad \Phi_n(Z) = \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} \chi_n(\sigma) \prod_i Z_{i,\sigma(i)}.$$

The standard representation on the coordinate-sum-zero subspace is irreducible, so its sign twist has dimension $n-1$ and character χ_n. Orthogonality of distinct permutation monomials gives $\mathbb{E}|\Phi_n|^2 = 1$.

Write P for the projection onto diagonal traceless matrices in $\mathfrak{sl}_n(\mathbb{C})$ and $\operatorname{Ad}_Z(X) = ZXZ^{-1}$. Expanding the determinant along the diagonal shows

$$\sqrt{n!}\, \Phi_n(Z) = \det Z \, \operatorname{Tr}(P \operatorname{Ad}_Z).$$

Let $W = Z^*Z$, $d = n^2 - 1$, and tilt the Gaussian law by $|\det Z|$; denote expectation under this probability law by $\mathbb{E}_*$. Biunitary invariance writes $Z = UDV$ with independent Haar unitary factors on each singular-value orbit. Schur averaging for the irreducible conjugation representation on $\mathfrak{sl}_n$ yields

$$\mathbb{E}_{U,V} |\operatorname{Tr}(P \operatorname{Ad}_{UDV})|^2 = \frac{(n-1)(\operatorname{Tr} W \operatorname{Tr} W^{-1} - 1)}{d^2}.$$

Conditional Jensen and Cauchy–Schwarz therefore imply

$$\mathbb{E}|\Phi_n| \le D_n\sqrt{\frac{n-1}{d^2}\mathbb{E}_*(\operatorname{Tr} W \operatorname{Tr} W^{-1} - 1)}, \qquad D_n = \frac{\mathbb{E}|\det Z|}{\sqrt{n!}}.$$

Conditioning successively on Gaussian columns gives $\mathbb{E}_* \operatorname{Tr} W \operatorname{Tr} W^{-1} = 2n^3 + n^2 - 2n$. Consequently the square-root factor is $\sqrt{(2n+1)/(n+1)}$, which is bounded. Gaussian orthogonalization also gives

$$D_n = \prod_{j=1}^{n} \frac{\Gamma(j+1/2)}{\Gamma(j)\sqrt{j}} \asymp n^{-1/8},$$

since the logarithm of the jth factor is $-1/(8j) + O(j^{-2})$. Thus $\mathbb{E}|\Phi_n| = O(n^{-1/8})$. □

9.10. **Dimension growth under the canonical dual normalization**

The question tracked here is whether the best Schatten-to-symbol constant exhibits exponential dimension growth under the canonical dual norm $D_d(\varphi) = \sup_{\|f\|_{L^1}\le 1} |\langle f, \varphi\rangle|$.

For every $p > 4$, the corresponding best constant is exactly the book's $C_p(d)$ and satisfies $C_p(d) \ge A_p e^{c_p d}$. This statement uses the specified dual normalization; it does not assert a dimension-uniform comparison with an unspecified product-BMO norm.

Answer: Answered: exponential growth for $p > 4$. **Evidence:** Sketch.

Book draft, dimension-growth corollary

Proof. Hahn–Banach duality for analytic H^1 gives

$$D_d(\varphi) = \sup_{\|f\|_{L^1}\le 1} |\langle f, \varphi\rangle| = \inf_{P_+\psi = \varphi} \|\psi\|_{L^\infty}.$$

Thus the best constant in $D_d(\varphi) \le K\|H_\varphi\|_{S_p}$ is the same $C_p(d)$ used for the bounded-symbol Nehari inequality in the book. The finite-block construction supplies a block of dimension d_p with ratio $R_p > 1$. Tensoring k disjoint copies multiplies the ratio to R_p^k; for $k = \lfloor d/d_p \rfloor$, this gives $C_p(d) \ge A_p e^{c_p d}$ after absorbing the bounded remainder in the constant. The argument establishes this normalized statement for $p > 4$; it does not identify it with other conventions for product BMO. □

Part 4. The borderline: the multiplicative Hilbert matrix

10. Nehari theorems and Hankel forms

10.1. **An explicit bounded symbol for the multiplicative Hilbert matrix**

> Can the Bessel operator V be dualized into an explicit bounded completion, i.e. an explicit symbol?

Søvik, *Where Nehari Survives*, Chapter 9, the explicit-symbol question.

Answer. Yes. A specified $B \in L^\infty(\mathbb{T}^\infty)$ has norm at most $10^5 + 7$ and represents the coefficients $c_1 = 1$ and $c_n = 1/(\sqrt{n}\log n)$ for $n \ge 2$.

Answer: Answered: yes. **Evidence:** Sketch.

Proof. For $\sigma_j = e^{-j-\theta}$, $0 < \theta < 1$, put $r^j = (p^{-1/2-\sigma_j})_p$. The construction localizes the evaluation density ψ_{r^j, D_j^θ} to $T_{r^j}(D_j^\theta)$, where D_j^θ is invariant under common rotations and has measure at least $1/2$. Its modulus on that image is $\|K_{r^j}\|_2^2/m(D_j^\theta) = \zeta(1+2\sigma_j)/m(D_j^\theta)$. Since $\zeta(1+2\sigma_j) \le 1 + 1/(2\sigma_j)$, we have $\sigma_j|\psi_{r^j, D_j^\theta}| \le 1 + 2\sigma_j$.

Hoeffding bounds make the images disjoint when scales differ by at least $L = 10^5$, so at most L contribute at any point. Since $\|K_{r^j}\|_2^2 = \prod_p (1 - (r_p^j)^2)^{-1} = \zeta(1+2\sigma_j) \le 1 + 1/(2\sigma_j)$ and $m(D_j^\theta) \ge 1/2$, localization gives $\sigma_j|\psi_{r^j, D_j^\theta}| \le 1 + 2\sigma_j$. Thus the weighted sum is bounded by $L+4$. The Fourier tail $\sum_{n\ge 2} \overline{z(n)}/(n^{3/2}\log n)$ has norm less than 3, giving $\|B\|_\infty \le 10^5 + 7$. The evaluation identity turns the integral of the scale terms into $\int_0^1 f(1/2+\sigma)\,\mathrm{d}\sigma$; the Fourier

tail supplies the missing $n^{-3/2}/\log n$ contribution, yielding the coefficient $1/(\sqrt{n}\log n)$. The linked chapter gives the full construction. □

Full construction

10.2. **A certified three-prime constant gap**

Close $1.0617 \le C_2/\pi_2^* \le 1.0878$, and certify a lower bound for C_3.

Søvik, *Where Nehari Survives*, Chapter 9, earlier wording of open item 3; the current chapter records that the three-prime benchmark is exceeded. We answer the scoped question: is $C_3/\pi_3^* > 1$? The separate two-prime interval remains open. Here $\mathcal{S}_3$ is the semigroup generated by $2, 3, 5$, C_3 is the norm of $\ell(f) = \sum_{n\in\mathcal{S}_3\setminus\{1\}} a_n/(\sqrt{n}\log n)$ on the corresponding $\mathscr{H}_0^1$ subspace, and π_3^* is the norm of $(1/(\sqrt{mn}\log(mn)))$ with rows $m \in \mathcal{S}_3$ and columns $n \in \mathcal{S}_3 \setminus \{1\}$.

Answer. The three-prime component is yes: a rational interval certificate proves $C_3 > 1.599$ and $\pi_3^* < 1.5624$, hence $C_3/\pi_3^* > 1.0234$. The separate two-prime interval remains open. **Answer:** Answered: yes. **Evidence:** Sketch.

Proof. Proof outline. The polynomial witness in the first three prime variables has a coefficient pairing with the multiplicative Hilbert functional that gives $C_3 > 1.5990217$, with the logarithmic terms bounded by rational Taylor remainders. For the upper bound, truncate the rectangular three-prime Hankel matrix to $mn \le 10^{12}$ and call the truncation T; it has 3429 rows and 3428 columns. Positive dyadic Schur weights seeded by the truncated matrix's SVD vectors are verified by 80-bit interval arithmetic, giving $\|T\| \le 1.5617957$. The omitted matrix has Hilbert–Schmidt norm less than 0.000529, so $\pi_3^* < 1.5624$. The certificate combines the unrounded bounds to give $\pi_3^* < 1.56232458$. Therefore $C_3/\pi_3^* > 1.02348$. This certificate settles the three-prime strict gap; the separate two-prime interval remains open. □

Full certificate

11. Coefficient inequalities

11.1. **H^1 partial-sum norm growth**

Determine the precise asymptotic growth of the norm of the partial sum operator $S_N : \sum_{n=1}^{\infty} a_n n^{-s} \mapsto \sum_{n=1}^{N} a_n n^{-s}$ when $N \to \infty$ for $p = 1$ (or more generally, for $p \le 1$ or $p = \infty$).

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," Problem 4.5, p. 6 (2016). The answer below treats the $p = 1$ case.

Result. $\|S_N\|_{\mathcal{H}^1\to\mathcal{H}^1} \asymp \log N/\log\log N$.

Answer: Answered: $\asymp \log N/\log\log N$. **Evidence:** Sketch.

Proof. The upper bound is Bondarenko, Brevig, Saksman, and Seip, Theorem 5.2. The proof of the matching lower bound is in the public note on the H^1 partial-sum norm; here is its main construction.

For a nonnegative trigonometric polynomial u of mean one, choose a vector λ having nonzero pairing with every nonzero frequency, and let $P_\lambda u$ retain the frequencies with $\lambda\cdot\gamma \le 0$. Write $G = 2P_\lambda u - 1 = u + iv$. The nonconstant frequencies of G lie strictly in one half-space, so their sums never contain zero. Consequently, for every polynomial A, $\int A(G+\varepsilon)\,\mathrm{d}m = A(1+\varepsilon)$. Expanding the principal logarithm of $G+\varepsilon$ in a uniformly convergent power series and applying this identity gives

$$\int (u+\varepsilon)\log(u+\varepsilon)\,\mathrm{d}m \le (1+\varepsilon)\log(1+\varepsilon) + \frac{\pi}{2}\|v\|_1.$$

Letting $\varepsilon \downarrow 0$ and using $\|P_\lambda u\|_1 \ge \|v\|_1/2$ yields $\|P_\lambda u\|_1 \ge \pi^{-1}\int u\log u\,\mathrm{d}m$.

Take $Q_d = \prod_{j=1}^{d} p_j$ for the first d primes and $f_d(z) = 2^{-d}\prod_{j=1}^{d}(1+z_j)^2$. Its norm is one, and the cutoff S_{Q_d} is the half-space projection of $u_d = \prod_j(1+\cos\theta_j)$. Since $\int u_d \log u_d = d(1-\log 2)$, we obtain $\|S_{Q_d} f_d\|_1 \ge d(1-\log 2)/\pi$. The elementary bound $p_d \ll d\log(d+1)$

gives $\log Q_d \ll d\log(d+1)$. For arbitrary large N, choose $d \asymp \log N/\log\log N$ small enough that $Q_d^2 \le N/4$, and put $b = \lfloor N/Q_d \rfloor$. Then $b > Q_d$ and $\lfloor N/b \rfloor = Q_d$. Multiplication by b^{-s} preserves the Hardy norm and transfers the lower bound from Q_d to N. The leading constant is not determined. □

11.2. **Structured absolute-coefficient inequality from polytopes**

Is there a dimension-free absolute inequality $\sum_n |a_n| c_n \lesssim \|f\|_1$ with explicit c_n comparable to $1/(\sqrt{n}\log n)$ on structured sets?

Source: Søvik, "Polytopes versus tori, and Hardy's inequality," Question 31 in the monograph draft. We answer the scoped coefficient question, the final sentence of that source question. **Result.** For every $A > 2$ the fixed support set

$$S_A = \{1,\dots,15\} \cup \{n \ge 16 : |\Delta(n)| \ge A\sqrt{\log\log n\, \log\log\log n}\},$$
$$\Delta(n) = \sum_j (-1)^{j+1} v_{p_j}(n),$$

satisfies, with a constant independent of the prime dimension,

$$|a_1| + \sum_{n\in S_A, n\ge 2} \frac{|a_n|}{\sqrt{n}\log n} \le C_A \|f\|_{\mathcal{H}^1}.$$

Answer: Answered: yes. **Evidence:** Sketch.

Lemma 1. *There is an absolute C such that, for $X \ge e^e$ and $|u| \le u_0 = \log(3/2)$,*

$$\sum_{n\le X} \frac{d(n)}{n} e^{u\Delta(n)} \le C\exp(2\cosh u\, \log\log X).$$

Proof. Writing $\varepsilon_j = (-1)^{j+1}$, nonnegativity of the terms gives

$$\sum_{n\le X} \frac{d(n)}{n} e^{u\Delta(n)} \le \prod_{p_j \le X} (1 - e^{\varepsilon_j u}/p_j)^{-2}.$$

All ratios in this finite Euler product are at most $3/4$. Its logarithm is therefore

$$2\sum_{p_j\le X} \frac{e^{\varepsilon_j u}}{p_j} + O(1) = 2\cosh u \sum_{p\le X} \frac{1}{p} + 2\sinh u \sum_{p_j\le X} \frac{\varepsilon_j}{p_j} + O(1),$$

uniformly for the stated u range: the remaining terms are bounded by a constant times $\sum_p p^{-2} < \infty$. The alternating sum has absolute value at most $1/2$, since $1/p_j$ decreases. The prime harmonic bound $\sum_{p\le X} 1/p \le \log\log X + C_0$ follows from Theorem 5, equation (3.18), of Rosser and Schoenfeld, enlarging the constant on the finite initial range. This proves the lemma. □

Proof. We prove the finite arithmetic bound

$$B_A = \sum_{\substack{n\in S_A \\ n\ge 2}} \frac{d(n)}{n\log^2 n} < \infty. \tag{1}$$

For $k \ge 2$, restrict to $e^k \le \log n < e^{k+1}$ and put $h_k = A\sqrt{k\log k}$ and $X_k = \exp(e^{k+1})$. Membership in S_A implies $|\Delta(n)| \ge h_k$, because $L\log L$ increases for $L \ge 1$. For $0 < u \le u_0$, the elementary exponential bound for each of the two tails, followed by Lemma 1, gives

$$\sum_{\substack{n\in S_A \\ e^k \le \log n < e^{k+1}}} \frac{d(n)}{n\log^2 n} \le C\exp\bigl(-2k - uh_k + 2(k+1)\cosh u\bigr).$$

For sufficiently large k, choose $u = h_k/(2(k+1)) \le u_0$. Using $\cosh u = 1 + u^2/2 + O(u^4)$, the exponent is

$$2 - \frac{h_k^2}{4(k+1)} + O\left(\frac{h_k^4}{(k+1)^3}\right) = 2 - \frac{A^2}{4}\log k + o(1).$$

Thus, for every r with $1 < r < A^2/4$, the block sum is at most $C_{A,r}k^{-r}$. The finitely many initial blocks are finite sets of integers; summing proves Equation (1).

Helson's coefficient inequality, recorded in equation (22) of Bondarenko, Brevig, Saksman, and Seip, for $p = 1$, is

$$\sum_{n\ge1} \frac{|a_n|^2}{d(n)} \le \|f\|_{\mathcal{H}^1}^2.$$

Cauchy–Schwarz, including the constant term with weight one, now gives the desired bound with $C_A = (1 + B_A)^{1/2}$. All weights and the set S_A were fixed independently of f and its finite prime dimension. □

Part 5. Related function-theoretic questions

12. Extremal coefficient problems

12.1. **No disconnected equality points below terminal threshold**

Fix $k \ge 2$ and $0 < p < 1$. Is there some t_0 such that $\Phi_k(p,t) = \Phi_1(p,t)$ holds for every $t_0 \le t \le 1$?

Source: Brevig, Grepstad, and Instanes, "F. Wiener's trick and an extremal problem for H^p," §6, Question 2. The answer below treats the sharper threshold question for $k \ge 3$.

Question answered here. For each $0 < p < 1$ and $k \ge 3$, is the equality set $\{t \in [0,1] : \Phi_k(p,t) = \Phi_1(p,t)\}$ exactly one terminal interval $[\tau_k, 1]$?

Result. Yes, conditionally; the threshold is unique and satisfies $t_\star < \tau_k < 1$, where $t_\star = (1 + q^{-2})^{-1/p}$ and $q = 2/p - 1$.

Answer: Answered: yes. **Evidence:** Sketch.

Conditional on: Brevig, Grepstad, and Instanes, "F. Wiener's trick and an extremal problem for H^p," Lemma 4, and Theorems 1 and 2(a) of Ø. Søvik, "A coefficient problem in H^p with prescribed value at the origin: zeros and the terminal interval" public manuscript.

Proof. For $t \in (0,1)$, write a canonical extremal as $f = gh^q$, where $q = 2/p - 1$, $g = IO^{p/2}$, and $h = O^{p/2}$ for its inner and outer factors. Then g, h have H^2 norm one, h is outer, g/h is inner, and after choosing phases $g(0) = s > 0$, $h(0) = a > 0$, and $t = sa^q$. Lemma 4 of Brevig, Grepstad, and Instanes gives an outer polynomial h of degree at most k. For a nonconstant extremal, optimizing the coefficients of g with its constant coefficient fixed and its H^2 norm one forces g to have degree k and gives the reflected-coefficient identity

$$\mu g^*(z) = h(z)^q + O(z^k), \qquad \mu > 0,$$

where $g^*(z) = z^k\overline{g(1/\overline{z})}$. In particular $\mu\overline{g_k} = a^q$ and $g_k \ne 0$. The function $\Phi_k(p,\cdot)$ is continuous on $(0,1)$: normal families and Fatou's lemma give upper semicontinuity, while replacing an extremal $f(z)$ by $f(\rho z) + (u - t)$ for $\rho < 1$ and u near t gives lower semicontinuity as $\rho \uparrow 1$. The normalized polynomial pairs lie in a compact coefficient set. A coefficientwise limit at parameters tending to t is again canonical: the outer factors have $a^q \ge t > 0$, remain zero-free in $\mathbb{D}$ by Hurwitz's theorem, and satisfy $|g| = |h|$ on $\mathbb{T}$. The limit is extremal by continuity, and $g_k \ne 0$ persists by the cited canonical-factor lemma. Thus the multipliers form a nonempty compact subset of $(0,\infty)$ at each $t \in (0,1)$. Write its endpoints as $\mu_{\min}(t)$ and $\mu_{\max}(t)$.

We first verify the envelope derivative needed below. For a fixed canonical pair, set $c_k = [z^k]h^q$. If u is near t, define

$$s_u = \frac{u}{a^q}, \qquad g_u = s_u + \sqrt{\frac{1-s_u^2}{1-s^2}}(g-s).$$

Then $\|g_u\|_{H^2} = 1$ and $g_u h^q$ is admissible at u by Hölder's inequality. Equality in the coefficient Cauchy–Schwarz inequality for the original extremal gives

$$A_{g,h}(u) \coloneqq \operatorname{Re}[z^k](g_u h^q) = s_u \operatorname{Re} c_k + \mu\sqrt{(1-s_u^2)(1-s^2)}.$$

At $u = t$, this equals $\Phi_k(p,t)$, and direct differentiation yields

$$A'_{g,h}(t) = \frac{\Phi_k(p,t) - \mu}{t}.$$

The Taylor remainder is uniform over canonical extremals when t stays in a compact subinterval of $(0,1)$: $s \le a$ implies $s \le t^{1/(q+1)} < 1$, while $a \ge t^{1/q} > 0$, and the coefficients and multipliers range over a compact set. Transporting extremals forward from t and backward from $t+\delta$ therefore gives

$$\partial_t^+ \Phi_k(p,t) = \frac{\Phi_k(p,t) - \mu_{\min}(t)}{t}.$$

The same comparison in the opposite direction gives $\partial_t^- \Phi_k(p,t) = (\Phi_k(p,t) - \mu_{\max}(t))/t$.

For $t > 2^{-1/p}$ put $T = t^p > 1/2$ and $\beta = \sqrt{t^{-p}-1}$. The binomial $F_k(z) = t(1+\beta z^k)^m$ is admissible and has coefficient $B(t) = tm\beta$. Comparing the kth Fourier coefficients of $|g|^2 = |h|^2$ on $\mathbb{T}$ gives $h_k = (s/a)g_k$. Since $s/a \le 1$ and $t = (s/a)a^{q+1}$,

$$\mu = \frac{t}{a|h_k|} \ge \frac{t}{\sqrt{a^2(1-a^2)}} \ge \frac{t}{\sqrt{T(1-T)}} =: \mu_{\mathrm{bin}}(t).$$

The last inequality uses $a^2 \ge T > 1/2$. Equality holds only if $s = a$ and h has no nonconstant coefficient except h_k; positivity of the reflected coefficient then gives $f = F_k$. Conversely F_k attains equality. Direct differentiation also gives $B'(t) = (B(t) - \mu_{\mathrm{bin}}(t))/t$.

Consequently the nonnegative excess

$$E_k(t) = \frac{\Phi_k(p,t) - B(t)}{t}$$

has right derivative

$$\partial_t^+ E_k(t) = \frac{\mu_{\mathrm{bin}}(t) - \mu_{\min}(t)}{t^2} \le 0,$$

strictly so wherever $E_k(t) > 0$. The function is continuous; a continuous function with nonpositive right derivative everywhere is nonincreasing (apply the right-derivative test to $E_k(t) - ct$ for $c > 0$). Hence the zero set of E_k is a terminal interval, with no later return to positivity.

Theorem 2(a) of Søvik's public manuscript gives equality with Φ_1 near 1. We verify strict inequality for $0 < t < t_\star + \epsilon$ directly. Put $\nu = q+1 = 2/p$ and

$$U(z) = \frac{z + z^2 + \cdots + z^{k-1}}{\sqrt{k-1}}.$$

Then $\|U\|_{H^2} = 1$, $[z^k]U^2 = 1$, and $[z^k]U^3 = (k-2)/(2\sqrt{k-1}) > 0$. For $\beta = \sqrt{t^{-p}-1} < 1$ and small $r > 0$, the polynomial

$$h_r(z) = t^{p/2}\bigl(1 + rU(z) + \sqrt{\beta^2 - r^2}\, z^k\bigr)$$

is zero-free on the closed disc by continuity from $1 + \beta z^k$, and $\|h_r\|_{H^2} = 1$. Hence $h_r^{2/p}$ is admissible. Terms involving both U and z^k have degree larger than k, and expansion gives

$$\frac{[z^k]h_r^{2/p}}{t} = \nu\beta + \frac{\nu}{2}(q - 1/\beta)r^2 + \binom{\nu}{3}\frac{k-2}{2\sqrt{k-1}}r^3 + O(r^4).$$

At $t = t_\star$, $\beta = 1/q$, so the quadratic term vanishes and the cubic term is positive. This strict improvement persists on a neighborhood of $t_\star$. For $t_p < t < t_\star$, where t_p is the degree-one transition, $1/q < \beta < 1$ and the quadratic term is positive. For $0 \le t \le t_p$, Theorem 12 of Brevig, Grepstad, and Instanes supplies a degree-one extremal F with a zero in the disc, including at $t = t_p$. The function $F(z^k)$ has at least k zeros and therefore is not extremal by Theorem 1 of Søvik's public manuscript. Thus $\Phi_k(p,t) > \Phi_1(p,t)$ throughout this interval. The endpoint $t = 0$ is also strict directly: put $\nu = q + 1$, $a = q^{-1/2}$, and

$$G(z) = \frac{(1+az)^\nu}{(1+a^2)^{1/p}}, \qquad F(z) = zG(z^{k-1}).$$

Then $\|F\|_p = 1$, $F(0) = 0$, and $[z^k]F = \nu a/(1+a^2)^{1/p} > 1 = \Phi_1(p,0)$. Indeed the square of this coefficient is $q^q/(q+1)^{q-1}$; its logarithm vanishes at $q = 1$ and has derivative $2/(q+1) - \log(1+1/q) > (q-1)/(q(q+1)) > 0$. The identity $\Phi_1(p,0) = 1$ follows by writing $f = zh$ and using $|h(0)| \le \|h\|_p \le 1$, with equality for $f = z$. To locate the degree-one branch, Brevig, Grepstad, and Instanes's Lemma 11 and equation (29) give $\alpha_p < \alpha_2 = 1/q$ and $t_p = \alpha_p(1+\alpha_p^2)^{-1/p}$. The function $\alpha \mapsto \alpha(1+\alpha^2)^{-1/p}$ is increasing on $[0, 1/q]$, since its derivative has the sign of $1 - q\alpha^2 > 0$ there. Hence $t_p < t_\star/q < t_\star$, and Brevig, Grepstad, and Instanes's Theorem 12 gives $B(t) = \Phi_1(p,t)$ on $[t_\star, 1]$. These facts give a left endpoint $\tau_k \in (t_\star, 1)$ and the stated equivalence. If a nonbinomial extremal existed at $t > \tau_k$, its multiplier would exceed $\mu_{\mathrm{bin}}(t)$; the left-derivative formula would then give $\partial_t^- \Phi_k(p,t) < B'(t)$, contradicting $\Phi_k = B$ near t. At $t = 1$ the constant function is the sole extremal. □

12.2. **Sharp terminal-threshold leading constant**

For $0 < p < 1$,

$$k^\alpha\big(1 - T_k(p)\big) \longrightarrow \frac{1}{p}\Big(\frac{q^q}{A_p(q^2-1)^{(q-1)/2}}\Big)^{2/q} \qquad (k \to \infty).$$

Source: Søvik, "A coefficient problem in H^p with prescribed value at the origin: zeros and the terminal interval," §6, Conjecture 1. Here $q = 2/p - 1$, $\alpha = 2(1-p)/(2-p)$, and $A_p = \lim_{n\to\infty} n^{1-1/p} C(n,p)$ is assumed to exist, as in the source. For $n \ge 1$ and $0 \le t \le 1$, define

$$\begin{aligned} C(n,p) &= \sup\{|[z^n]f| : \|f\|_{H^p} \le 1\},\\ \Phi_n(p,t) &= \sup\{|[z^n]f| : \|f\|_{H^p} \le 1,\ f(0) = t\},\\ T_n(p) &= \inf\{a \in [0,1] : \Phi_n(p,t) = \Phi_1(p,t) \text{ for every } t \in [a,1]\}. \end{aligned}$$

Result. The displayed limit holds under the source's assumption that A_p exists.
Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: the existence of $A_p = \lim_{n\to\infty} n^{1-1/p} C(n,p)$.

Put $m = 2/p = q + 1$, $\gamma = (q-1)/2 = 1/p - 1$, and $\delta = 1 - t^p$. We use normalized arc length on the circle. The elementary Poisson-kernel test for $C(n,p)$ gives $A_p > 0$ whenever the assumed limit exists.

Lemma 2. *Let*

$$\mathcal{L}(F) = e^{2\pi y}\int_{\mathbb{R}} F(x+iy)e^{-2\pi i x}\,\mathrm{d}x, \qquad \|F\|^p_{H^p(\mathbb{C}_+)} = \sup_{y>0}\int_{\mathbb{R}} |F(x+iy)|^p\,\mathrm{d}x.$$

The integral is independent of $y > 0$, and $|\mathcal{L}(F)| \le A_p\|F\|_{H^p(\mathbb{C}_+)}$.

Proof. The subharmonic point bound $|F(x+iy)|^p \le C_p y^{-1}\|F\|^p_{H^p(\mathbb{C}_+)}$ implies absolute integrability on every horizontal line. Cauchy's theorem on rectangles whose vertical sides tend to infinity proves independence of y. Normalize $\|F\|_{H^p(\mathbb{C}_+)} \le 1$. For $a, \epsilon > 0$ and an integer $M > 1/p$ put $G(z) = F(z+ia)(1 - i\epsilon z)^{-M}$. It has norm at most one and enough horizontal decay to periodize:

$$P_n(z) = \sum_{j\in\mathbb{Z}} G(z+nj), \qquad f_n(e^{2\pi i z/n}) = n^{1/p}P_n(z).$$

The sum is locally uniform and vanishes as $\operatorname{Im} z \to \infty$, so f_n is analytic in the disc with a removable singularity at zero. Subadditivity of the pth power gives $\|f_n\|_{H^p} \le 1$. Unfolding its nth Fourier coefficient gives $n^{-\gamma}[w^n]f_n = \mathcal{L}(G)$. The assumed coefficient limit therefore yields $|\mathcal{L}(G)| \le A_p$. Let $\epsilon \downarrow 0$ and then $a \downarrow 0$. □

For an admissible f with $t > 0$, take its inner/outer factorization in the form $f = gh^q$, where $g = Ih$, h is outer, $g(0) = s > 0$, $h(0) = a > 0$, and $\|g\|_2 = \|h\|_2 \le 1$. The branch of h^q is positive at zero. Write

$$G = g - s, \quad H = h - a, \quad v = [z^n]G, \quad w = [z^n]H,$$
$$G_0 = G - vz^n, \quad H_0 = H - wz^n,$$

and $E = \|G_0\|_2^2 + q\|H_0\|_2^2$. Here $sa^q = t$. As $\delta \to 0$, the identity $(1-(1-s^2))(1-(1-a^2))^q = (1-\delta)^m$ gives

$$(1 - s^2) + q(1 - a^2) = m\delta + O_p(\delta^2), \qquad E + |v|^2 + q|w|^2 \le m\delta + O_p(\delta^2). \tag{2}$$

Lemma 3. *For δ sufficiently small, put $R = f - t - a^qG - qsa^{q-1}H$. On every circle $|z| = r < 1$, and every measurable arc set J,*

$$\int_J |R| \,\mathrm{d}\mu \le C_p\delta + C_p \int_J (|G|^2 + |H|^2)(1 + |H|^{m-2}) \,\mathrm{d}\mu. \tag{3}$$

In addition, with $g_b = s + vz^n$, $h_b = a + wz^n$ and $R_0 = f - g_bh_b^q - G_0h_b^q - qg_bh_b^{q-1}H_0$,

$$|[z^n]R_0| \le C_pE\{1 + (nE)^\gamma\}. \tag{4}$$

Proof. For the first assertion set $u = H/a$ and $K = \log(h/a) - u$, with logarithm zero at the origin. The inequality $\log|1 + u| \le \operatorname{Re} u + |u|^2/2$ and $K(0) = 0$ imply $\int |\operatorname{Re} K| \,\mathrm{d}\mu \le C_p\|H\|_2^2 \le C_p\delta$. Davis's weak $(1,1)$ inequality for conjugate functions shows that the arc set $|\operatorname{Im} K| > 1$ has measure $O_p(\delta)$. When $|u| \le 1/2$ and this exceptional event does not occur, the analytic logarithm agrees with the principal logarithm at that point: if they differed by $2\pi i\ell$ with $\ell \ne 0$, then $|\operatorname{Im} K| > 1$. Taylor's formula consequently gives $|R| \le C_p(|G|^2 + |H|^2)$. On the exceptional part with $|u| \le 1/2$, all factors are bounded, since $|g| \le |h|$. On $|u| > 1/2$, $|f| \le |h|^m$ and $|G| \le |h| + s$ give $|R| \le C_p|H|^2(1 + |H|^{m-2})$. This proves (3).

For the second assertion repeat the argument with $u = H_0/h_b$ and $K = \log(h/h_b) - u$. For small δ, h_b is bounded above and away from zero, so the exceptional set now has measure $O_p(E)$. The resulting integral bound is

$$\int |R_0| \,\mathrm{d}\mu \le C_pE + C_p \int (|G_0|^2 + |H_0|^2)(1 + |H_0|^{m-2}) \,\mathrm{d}\mu.$$

At $r = e^{-2\pi/n}$ the H^2 point estimate gives $\sup_{|z|=r} |H_0(z)| \le C\sqrt{nE}$. Cauchy's coefficient formula proves (4). □

Lemma 4. *Suppose $n_j \to \infty$, $\delta_j \to 0$ and $n_j\delta_j \to \infty$. If $E_j/\delta_j \to e$ along a subsequence, then*

$$\limsup_j \frac{|[z^{n_j}]R_j|}{n_j^\gamma \delta_j^{m/2}} \le A_p(e/m)^{m/2}. \tag{5}$$

Proof. Set $\lambda_j = \sqrt{n_j\delta_j}$ and rescale on the upper half-plane:

$$U_j(z) = \lambda_j^{-1}G_{0,j}(e^{2\pi iz/n_j}), \qquad V_j(z) = \lambda_j^{-1}H_{0,j}(e^{2\pi iz/n_j}).$$

These are n_j-periodic, and on every horizontal line their weighted squared integral over a period is at most E_j/δ_j. The H^2 point estimate makes all real translates normal. Fix $\eta > 0$. Successively choose centers, separated by distances tending to infinity modulo n_j, at which $|V_j(x + i)| > \eta$. Passing to subsequences gives nonzero local limits (U^r, V^r). Each such

profile consumes at least $c\eta^2$ of the squared-integral budget by the half-plane point estimate. Disjoint windows and Fatou's lemma yield

$$\sum_r e_r \le e, \qquad e_r = \|U^r\|^2_{H^2(\mathbb{C}_+)} + q\|V^r\|^2_{H^2(\mathbb{C}_+)}.$$

Thus the extraction stops after finitely many profiles. Outside expanding windows about their centers, the limiting supremum of $|V_j(x+i)|$ is at most η.

The constants s_j, a_j and the removed degree-n_j terms divided by λ_j vanish locally. The functions $f_j(e^{2\pi iz/n_j})/\lambda_j^m$ therefore have analytic local limits F^r, by the bound $|f_j| \le |h_j|^m$, and $|F^r| = |U^r||V^r|^q$. Hölder's inequality and weighted arithmetic and geometric mean give $\|F^r\|_{H^p(\mathbb{C}_+)} \le (e_r/m)^{m/2}$. The same local limit holds for the rescaled R_j.

On the line of height one, Cauchy's formula becomes

$$\frac{[z^{n_j}]R_j}{\lambda_j^m/n_j} = e^{2\pi}\int_0^{n_j} \frac{R_j(e^{2\pi i(x+i)/n_j})}{\lambda_j^m} e^{-2\pi ix}\,\mathrm{d}x.$$

Fixed windows converge to integrals of F^r after taking subsequences of the phases. Increasing the windows gives, by Lemma 2, at most

$$\sum_r |\mathcal{L}(F^r)| \le A_p \sum_r (e_r/m)^{m/2} \le A_p (e/m)^{m/2}.$$

On the complement, $|H_j| \le (\eta + o(1))\lambda_j + O(\sqrt{\delta_j})$. The local-set form of (3) bounds its contribution by $C_p\lambda_j^{-(m-2)} + C_p\eta^{m-2}$ after normalization. Since $\lambda_j \to \infty$, let $\eta \downarrow 0$ to obtain (5). □

Proof. Put

$$B_p^* = \frac{q^q}{A_p(q^2-1)^{(q-1)/2}}, \qquad B = n^\gamma \delta^{q/2}.$$

First we identify the value transition. If $B \to B_0 \in (0,\infty)$, then $n\delta \to \infty$. Along any sequence of competitors we may pass to a subsequence on which $E/(m\delta) \to \rho \in [0,1]$. By Cauchy–Schwarz and (2), the linear part $a^q v + qsa^{q-1}w$ is at most $m\sqrt{(1-\rho)\delta} + o(\sqrt{\delta})$. Lemma 4 bounds the remainder, giving

$$\limsup \frac{\Phi_n(p,t)}{\sqrt{\delta}} \le \max_{0\le\rho\le 1}\{m\sqrt{1-\rho} + A_pB_0\rho^{m/2}\}. \tag{6}$$

If $B \to 0$, the H^2 point estimate and (3) give $|[z^n]R| \le C_p\delta\{1 + (n\delta)^\gamma\} = o(\sqrt{\delta})$, so the right side is m.

For the reverse bound, fix $\rho \in [0,1]$, take $\beta^2 = (1-\rho)\delta/t^p$ and $b(z) = t(1+\beta z^n)^m$. Then $\|b\|_{H^p}^p = 1 - \rho\delta$. Choose F_{n-1} of H^p norm one with its $(n-1)$st coefficient arbitrarily close to $C(n-1,p)$. By p-subadditivity, $b + (\rho\delta)^{1/p}zF_{n-1}$ is admissible. Its normalized coefficient tends to $m\sqrt{1-\rho} + A_pB_0\rho^{m/2}$. Thus equality holds in (6) for $B_0 > 0$. Writing $y = \sqrt{1-\rho}$, elementary minimization of $m(1-y)/(1-y^2)^{m/2}$ gives its minimum $q^q/(q^2-1)^{(q-1)/2}$ at $y = 1/q$. Consequently the limiting maximum equals m for $B_0 \le B_p^*$ and exceeds m for $B_0 > B_p^*$. The formula of Brevig, Grepstad, and Instanes (Theorem 12) gives $\Phi_1(p,t) = mt^{1-p/2}\sqrt{\delta}$ near $t = 1$. It follows that $\Phi_n(p,t) > \Phi_1(p,t)$ eventually when $B \to B_0 > B_p^*$.

We still need exact equality below the transition. The coefficient of z^n in f has the exact decomposition

$$[z^n]f = a^q v + qsa^{q-1}w + [z^n]R_0,$$

because G_0, H_0 have no terms of degree 0 or n, whereas g_b, h_b have only degrees divisible by n. Set $e_g = \|G_0\|_2^2$, $e_h = \|H_0\|_2^2$ and $T = (1-e_g)^{1/2}(1-e_h)^{q/2}$. Hölder's inequality gives

$\|g_b h_b^q\|_{H^p} \le T$, and $t \le T \le 1$. Applying the exact degree-one formula of Brevig, Grepstad, and Instanes (Theorem 12) to $g_b h_b^q/T$ gives

$$|a^q v + qsa^{q-1}w| \le T\Phi_1(p, t/T) = mt\sqrt{(T/t)^p - 1}.$$

Weighted arithmetic and geometric means yield $T^p \le 1 - E/m$. Hence $E \le m\delta$ and, with $\rho = E/(m\delta)$,

$$|a^q v + qsa^{q-1}w| \le \Phi_1(p, t)\sqrt{1-\rho}. \tag{7}$$

Fix $\epsilon \in (0, B_p^*)$. Suppose exact equality failed along $n_j \to \infty$ with $B_j \le B_p^* - \epsilon$ and competitors strictly improving on Φ_1. Necessarily $\delta_j \to 0$. After a subsequence, $B_j \to B_0$ and $\rho_j \to \rho$. If $\rho > 0$ and $B_0 > 0$, the profile proof of Equation (6) gives the strictly smaller upper value $m\sqrt{1-\rho} + A_p B_0 \rho^{m/2} < m$. If $\rho > 0$ and $B_0 = 0$, use the $B = 0$ remainder bound and (7) for the same contradiction. If $\rho_j \to 0$ and $E_j > 0$, then (4) gives

$$\frac{|[z^{n_j}]R_{0,j}|}{\rho_j\sqrt{\delta_j}} \le C_p\{\sqrt{\delta_j} + B_j\rho_j^{\gamma}\} \longrightarrow 0.$$

But (7) loses at least $\Phi_1(p, t_j)\rho_j/2$; if $E_j = 0$, the remainder vanishes. Both cases contradict strict improvement. Composition with z^n gives the reverse inequality $\Phi_n \ge \Phi_1$. Thus, for all large n, equality holds whenever $B \le B_p^* - \epsilon$.

For fixed $B > 0$ put $t_n(B) = (1 - B^{2/q}n^{-(q-1)/q})^{1/p}$. Exact subcritical equality for every $t \ge t_n(B_p^* - \epsilon)$ gives $T_n(p) \le t_n(B_p^* - \epsilon)$. Strict supercritical improvement at $t_n(B_p^* + \epsilon)$ gives $T_n(p) \ge t_n(B_p^* + \epsilon)$ directly from the definition of T_n. Therefore

$$(B_p^* - \epsilon)^{2/q} \le n^{(q-1)/q}(1 - T_n(p)^p) \le (B_p^* + \epsilon)^{2/q}.$$

Let $\epsilon \downarrow 0$. Since $T_n \to 1$, $1 - T_n^p \sim p(1 - T_n)$, and $(q-1)/q = \alpha$. This is the asserted limit. □

13. Hardy and Bergman extremals

13.1. **Separation of simplest-fraction sets in A_1^2**

In particular, are pairwise distances between these sets bounded away from 0 in the space A_1^2?

Question 2, p. 9, of Evgeny Abakumov, Alexander Borichev and Konstantin Fedorovskiy, *Chui's conjecture in Bergman spaces*, Math. Ann. 379 (2021), 1507–1532. Let SF_n consist of sums of n kernels $(z-a)^{-1}$ with $a \in \mathbb{T}$, counted with multiplicity, and equip A_1^2 with $\|f\|_1^2 = 2\int_{\mathbb{D}} |f(z)|^2(1-|z|^2)\,\mathrm{d}m_2(z)$, where $\mathrm{d}m_2 = \pi^{-1}\,\mathrm{d}x\,\mathrm{d}y$.
Result. For all $n, k \ge 1$, $\operatorname{dist}_{A_1^2}(SF_n, SF_{n+k}) \ge \pi/\sqrt{6}$.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. Take $f = \sum_{j<n}(z-a_j)^{-1} \in SF_n$ and $g = \sum_{j<n+k}(z-b_j)^{-1} \in SF_{n+k}$, and write $a_j = e^{i\theta_j}$, $b_j = e^{i\psi_j}$. Expanding each kernel in the disc gives $f - g = -\sum_{m\ge1} c_m z^{m-1}$, where $c_m = \sum_{j<n} a_j^{-m} - \sum_{j<n+k} b_j^{-m}$. Since the monomials are orthogonal in A_1^2 and $\|z^{m-1}\|_1^2 = 2/(m(m+1)) \ge 1/m^2$,

$$\|f - g\|_1^2 \ge \sum_{m\ge1} |c_m|^2/m^2.$$

Define $F(t) = \#\{j : \theta_j \le t\} - \#\{j : \psi_j \le t\} + kt/(2\pi)$ on $[0, 2\pi]$. Its mth Fourier coefficient is $c_m/(2\pi i m)$. Parseval therefore identifies the last sum with $2\pi^2 \min_{c\in\mathbb{R}} \int_0^{2\pi} |F(t) - c|^2\,\mathrm{d}t/(2\pi)$. Because $F(t) - kt/(2\pi)$ is integer-valued, $|F(t) - c| \ge \operatorname{dist}(kt/(2\pi) - c, \mathbb{Z})$. Averaging over the integer number k of periods gives a squared mean of at least $1/12$. Hence $\|f - g\|_1^2 \ge \pi^2/6$, uniformly in the two functions and their indices. □

13.2. **The inner-translate candidate for the backward-shift norm**

the extremal functions are determined by inner functions with a prescribed value at the origin

This implicit candidate is suggested by Theorem 1.1 and the general norm discussion in Section 1, p. 2, of Konstantinos Bampouras and Adrián Llinares, *The norm of the backward shift on H^4 is $\sqrt[4]{\varphi}$*, arXiv:2605.10469v1. For $2 < p < \infty$, the candidate is the supremum of the quotients obtained from $I - c$, where I is inner, $I(0) = a \in (0, 1)$, and $c \in \mathbb{C}$.
Result. The candidate is false at $p = 6$: $\Lambda(6) < \|B\|_{H^6 \to H^6}$.
Answer: Answered: no. **Evidence:** Sketch.

Proof. Write normalized arc measure as $\mathrm{d}m$ and let P_a be harmonic measure based at $a \in (0, 1)$. The boundary values of every inner function with $I(0) = a$ have distribution P_a: for powers this follows from $\int I^n \, \mathrm{d}m = a^n$, and trigonometric polynomial approximation gives the claim. Thus the candidate at $p = 6$ is

$$\Lambda(6)^6 = \sup_{0<a<1, c\in\mathbb{C}} \frac{N(a)}{D_a(c)}, \quad N(a) = \int_{\mathbb{T}} |\zeta - a|^6 \, \mathrm{d}P_a(\zeta), \quad D_a(c) = \int_{\mathbb{T}} |\zeta - c|^6 \, \mathrm{d}P_a(\zeta).$$

For fixed a, strict convexity and conjugation symmetry give a unique real minimizer c_a of D_a. Differentiation shows $D_a'(0) < 0 < D_a'(a)$, hence $0 < c_a < a$.

The maximum over a is attained in $(0, 1)$. To see this, the standard circle integral gives

$$N(a) = (1 - a^2)(1 + 4a^2 + a^4). \tag{8}$$

As $a \to 0$, uniqueness of the minimizer gives $c_a \to 0$, so the quotient tends to one. As $a \to 1$, the triangle inequality and $D_a(c_a) \le N(a)$ give $|c_a - a| \le 2N(a)^{1/6} \to 0$. For any sequence $a \to 1$, Fatou's lemma applied to

$$\frac{D_a(c_a)}{1 - a} = \int_{\mathbb{T}} \frac{(1 + a)|\zeta - c_a|^6}{|\zeta - a|^2} \, \mathrm{d}m(\zeta)$$

gives a lower limit at least $2 \int_{\mathbb{T}} |\zeta - 1|^4 \, \mathrm{d}m = 12$. Equation (8) gives $N(a)/(1 - a) \to 12$, hence $N(a)/D_a(c_a) \to 1$. Finally $N(1/2) = 99/64 > 1 = D_{1/2}(0)$, so the maximal quotient exceeds one and is attained at some $a \in (0, 1)$, with $c = c_a \in (0, a)$.

Suppose $\Lambda(6) = \|B\|_{H^6 \to H^6}$. Choose the one-zero Blaschke factor $I(z) = (a + z)/(1 + az)$ and put

$$A = 1 - a^2, \quad K = 1 - ac, \quad r = (c - a)/(1 - ac) \in (-a, 0),$$
$$h = I - a = \frac{Az}{1 + az}, \qquad f = I - c = \frac{K(z - r)}{1 + az}.$$

Then $Bf = h/z$ and $\|Bf\|_6 = \|h\|_6$; thus f would be extremal. Set $\lambda = \|h\|_6^6/\|f\|_6^6$. Varying f by arbitrary multiples of z^n, $n \ge 1$, in the extremal quotient shows that every strictly positive Fourier coefficient of $|h|^4 h - \lambda |f|^4 f$ vanishes. On $\mathbb{T}$ this function is the rational function

$$Q(z) = \frac{A^5 z^3 - \lambda K^5 (z - r)^3 (1 - rz)^2}{(1 + az)^3 (z + a)^2}.$$

Its only possible pole outside the closed disc is at $z_0 = -1/a$. The positive Fourier coefficients are precisely those of its exterior-pole principal part, so their vanishing forces the numerator to have a zero of order at least three at z_0. Both numerator terms are nonzero there, since $-a < r < 0$. The first and second logarithmic derivatives of $z^3/((z - r)^3(1 - rz)^2)$ must therefore vanish at z_0. The first gives

$$-3a + \frac{3a}{1 + ar} + \frac{2ar}{a + r} = 0, \qquad 3a^2 + ar = 2$$

because $r \neq 0$. Substitution into the second gives

$$0 = -3a^2 + \frac{3a^2}{(1 + ar)^2} + \frac{2a^2 r^2}{(a + r)^2} = \frac{a^2(9a^4 - 4)}{6(1 - a^2)^2}.$$

Hence $a^2 = 2/3$, and the first equation then gives $r = 0$, a contradiction. Thus the attained inner-translate quotient is strictly below the operator norm. □

14. Contractivity and hypercontractivity

14.1. **Asymptotic norm of the Bloch-to-Bergman inclusion**

"Conjecture 3.9. For any $\alpha > -1$,

$$\lim_{p\to\infty} \frac{C_\alpha(p)}{p/(2e)} = \frac{1}{\sqrt{(\alpha+1)(\alpha+2)}}.$$

"

Here $C_\alpha(p)$ is the norm of the inclusion from the Bloch space, with $\|f\|_{\mathcal{B}} = |f(0)| + \sup_{z\in\mathbb{D}}(1-|z|^2)|f'(z)|$, to A^p_α with probability measure $(\alpha+1)(1-|z|^2)^\alpha \,\mathrm{d}A(z)$, where $\mathrm{d}A = \mathrm{d}x\,\mathrm{d}y/\pi$. Source: Adrián Llinares, "Norms of Inclusions between Some Spaces of Analytic Functions," arXiv:2107.11315v2, Conjecture 3.9, §3, p. 9.
Result. No. For some $\eta < 1$ and $\varepsilon_0 > 0$, the limsup is at most $\eta/[2e\sqrt{(\alpha+1)(\alpha+2)}]$ whenever $-1 < \alpha < -1+\varepsilon_0$. The true limit and its existence remain undetermined.
Answer: Answered: no. **Evidence:** Sketch.

Theorem 1. *There are constants $\eta < 1$ and $\varepsilon_0 > 0$ such that, for every $-1 < \alpha < -1+\varepsilon_0$,*

$$\limsup_{p\to\infty} \frac{C_\alpha(p)}{p} \le \frac{\eta}{2e\sqrt{(\alpha+1)(\alpha+2)}}. \tag{9}$$

Consequently the conjecture is false. This does not determine the actual limit or assert its existence.

The key is a uniform strict local average deficit of the Bloch quotient, as in Ivrii and Kayumov, *Makarov's principle for the Bloch unit ball*, equation (1.7) and §1.2. We prove the needed qualitative deficit and its weighted version below.

Put

$$\mathrm{d}\nu(z) = \frac{\mathrm{d}A(z)}{(1-|z|^2)^2}, \qquad q_f(z) = (1-|z|^2)|f'(z)|.$$

Let $\mathcal{U}$ consist of the analytic functions f on $\mathbb{D}$ with $f(0) = 0$ and $q_f \le 1$. Fix $\rho = 1/2$ and define

$$B_w = \left\{ z : \left| \frac{z-w}{1-\overline{w}z} \right| < \rho \right\}, \qquad V = \nu(B_w) = \frac{\rho^2}{1-\rho^2}, \qquad (\mathcal{A}v)(w) = \frac{1}{V}\int_{B_w} v(z)\,\mathrm{d}\nu(z).$$

The averaging operator $\mathcal{A}$ is positive and self-adjoint with respect to ν: its kernel is symmetric because $z \in B_w$ if and only if $w \in B_z$, and V does not depend on the center.

Lemma 5. *There is a constant $A < 1$ such that $\mathcal{A}(q_f^2) \le A$ for every $f \in \mathcal{U}$.*

Proof. The normalized family $\mathcal{U}$ is compact for locally uniform convergence: the derivative bound gives a uniform bound on every compact subset, and it is preserved under limits. In particular

$$A := \max_{f\in\mathcal{U}} \frac{1}{V}\int_{|z|<\rho} q_f(z)^2\,\mathrm{d}\nu(z)$$

exists. If $A = 1$, an attaining function would have $q_f = 1$ throughout that disk, since $q_f \le 1$ and is continuous. Its derivative would be nonvanishing and $\log|f'(z)| = -\log(1-|z|^2)$ there. The left side is harmonic and the right side has strictly positive Laplacian, a contradiction. Thus $A < 1$; also $A > 0$, as is seen from $f(z) = z$. For an arbitrary center, compose f with a disk automorphism carrying 0 to that center and subtract its value at 0. The Bloch quotient and ν are invariant under that change of variables, proving the assertion. □

Lemma 6. *There are $B < 1$, $\varepsilon_1 > 0$, $0 < \tau_1 \le 1$, and $C < \infty$ such that, whenever $f \in \mathcal{U}$ is analytic on a neighborhood of $\overline{\mathbb{D}}$, $0 < \varepsilon \le \varepsilon_1$, and $0 \le \tau \le \tau_1$,*

$$\begin{aligned}\int_{\mathbb{D}} q_f(z)^2 e^{\tau \operatorname{Re} f(z)} (1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A(z)\\ \le B\int_{\mathbb{D}} e^{\tau \operatorname{Re} f(z)}(1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A(z) + C.\end{aligned} \tag{10}$$

All four constants are independent of f, ε, and τ in these ranges.

Proof. Set $L = \operatorname{atanh}\rho$. The Bloch derivative bound integrated along a hyperbolic geodesic gives $|f(z) - f(w)| \le L$ for $z \in B_w$. For

$$W(w) = e^{\tau \operatorname{Re} f(w)}(1-|w|^2)^{1+\varepsilon},$$

use the automorphism $z = (w+v)/(1+\overline{w}v)$ to obtain

$$\frac{\mathcal{A}W(w)}{W(w)} \ge e^{-\tau L} J_\varepsilon(w), \qquad J_\varepsilon(w) = \frac{1}{V}\int_{|v|<\rho} \left(\frac{1-|v|^2}{|1+\overline{w}v|^2}\right)^{1+\varepsilon} \mathrm{d}\nu(v). \tag{11}$$

The denominator in this integral is bounded away from zero, even for $|w| \le 1$. Hence $J_\varepsilon(w)$ is jointly continuous in $\varepsilon \ge 0$ and $w \in \overline{\mathbb{D}}$ near $\varepsilon = 0$. For $|w| = 1$, the angular average of $(1-|v|^2)/|1+\overline{w}v|^2$ is 1 on every circle $|v| = r$. It follows that $J_0(w) = 1$ on $|w| = 1$.

Choose $\kappa \in (A, 1)$, with A from Lemma 5. Joint continuity gives $r_0 < 1$ and $\varepsilon_1 > 0$ such that $J_\varepsilon(w) \ge \kappa$ when $r_0 < |w| < 1$ and $0 \le \varepsilon \le \varepsilon_1$. Choose $0 < \tau_1 \le 1$ small enough that

$$B \coloneqq \frac{A e^{\tau_1 L}}{\kappa} < 1.$$

By positivity, self-adjointness, and Lemma 5,

$$\int_{\mathbb{D}} q_f^2 \mathcal{A}W\,\mathrm{d}\nu = \int_{\mathbb{D}} W\mathcal{A}(q_f^2)\,\mathrm{d}\nu \le A\int_{\mathbb{D}} W\,\mathrm{d}\nu.$$

Combining this with (11) yields

$$\int_{|w|>r_0} q_f(w)^2 W(w)\,\mathrm{d}\nu(w) \le B\int_{\mathbb{D}} W\,\mathrm{d}\nu.$$

On the remaining disk, $q_f \le 1$ and $|f(w)| \le \operatorname{atanh} r_0$, so its contribution is bounded by

$$C = e^{\tau_1 \operatorname{atanh} r_0}\int_{|w|\le r_0} \frac{\mathrm{d}A(w)}{1-|w|^2} < \infty.$$

This is precisely (10). All integrals are finite for the stated functions, and the interchange of the nonnegative integrals is justified by Tonelli's theorem. □

Proof of Theorem 1. Choose D with $B < D < 1$, where B is as in Lemma 6. Take $0 < \varepsilon_0 \le \varepsilon_1$ small enough that

$$\tau = \tau(\varepsilon) \coloneqq \frac{2\sqrt{\varepsilon(1+\varepsilon)}}{\sqrt{D}} \le \tau_1 \qquad (0 < \varepsilon < \varepsilon_0). \tag{12}$$

Fix one such ε. We first prove

$$\sup_{f\in\mathcal{U}} \int_{\mathbb{D}} e^{\tau \operatorname{Re} f(z)}(1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A(z) < \infty. \tag{13}$$

Initially assume f is analytic past $\overline{\mathbb{D}}$, and write

$$F = e^{\tau \operatorname{Re} f}, \qquad M = \int_{\mathbb{D}} F(1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A, \qquad N = \int_{\mathbb{D}} F(1-|z|^2)^{\varepsilon}\,\mathrm{d}A,$$

and let E denote the left side of (10). Using the Euclidean Laplacian,

$$\Delta F = \tau^2 |f'|^2 F, \qquad \Delta(1-|z|^2)^{\varepsilon+1} = 4\varepsilon(\varepsilon+1)(1-|z|^2)^{\varepsilon-1} - 4(\varepsilon+1)^2(1-|z|^2)^{\varepsilon}.$$

Green's identity therefore gives

$$\tau^2 E = 4\varepsilon(\varepsilon+1)M - 4(\varepsilon+1)^2 N. \tag{14}$$

One may apply the identity on $|z| < r$ and let $r \uparrow 1$: F and its first derivatives are bounded, while the weight and its normal derivative are $O((1-r^2)^{\varepsilon+1})$ and $O((1-r^2)^{\varepsilon})$, respectively. Thus both boundary terms vanish; the interior integrals converge because $\varepsilon > 0$.

The radial Bloch bound gives

$$|f(z)| \le \operatorname{atanh}|z| \le \log 2 - \frac{1}{2}\log(1-|z|^2).$$

Consequently, since $\tau \le 1$,

$$N \le \frac{2^\tau}{1+\varepsilon-\tau/2} \le 4.$$

Combining (10) and (14) now gives

$$\bigl(4\varepsilon(\varepsilon+1) - B\tau^2\bigr)M \le \tau^2 C + 16(\varepsilon+1)^2.$$

The coefficient on the left is $4\varepsilon(\varepsilon+1)(1-B/D) > 0$ by (12). This bounds M uniformly over all the stated f. For arbitrary $f \in \mathcal{U}$, its dilates $f(rz)$ are analytic past $\overline{\mathbb{D}}$ and remain in $\mathcal{U}$, since

$$(1-|z|^2)r|f'(rz)| \le \frac{r(1-|z|^2)}{1-r^2|z|^2} \le 1.$$

Fatou's lemma proves (13). It also holds, with the same constant, for every rotation $e^{i\theta}f$.

For any $0 < s < \tau$, choose finitely many angles θ_j such that $\max_j \operatorname{Re}(e^{i\theta_j} w) \ge (s/\tau)|w|$ for every $w \in \mathbb{C}$. Then

$$e^{s|f(z)|} \le \sum_j e^{\tau \operatorname{Re}(e^{i\theta_j} f(z))}.$$

Thus (13) implies uniform exponential integrability of $e^{s|f|}$ over $\mathcal{U}$ against the probability measure $\varepsilon(1-|z|^2)^{\varepsilon-1}\,\mathrm{d}A$. If $\|f\|_{\mathcal{B}} \le 1$, then $f - f(0) \in \mathcal{U}$ and $|f(0)| \le 1$, so the same assertion holds over the full Bloch unit ball. Since $x^p \le (p/(es))^p e^{sx}$ for $x \ge 0$, there is a finite constant $K_{\varepsilon,s}$, independent of p and f, such that

$$C_{\varepsilon-1}(p) \le \frac{p}{es} K_{\varepsilon,s}^{1/p}.$$

First let $p \to \infty$, and then $s \uparrow \tau$. We obtain

$$\limsup_{p\to\infty} \frac{C_{\varepsilon-1}(p)}{p} \le \frac{1}{e\tau} = \frac{\sqrt{D}}{2e\sqrt{\varepsilon(1+\varepsilon)}}.$$

Taking $\eta = \sqrt{D} < 1$ and $\alpha = \varepsilon - 1$ proves (9). □

14.2. **Tight Hardy-to-Bergman inclusions on the unit ball**

Conjecture 22(b). Then $H^p \subset A^q_\alpha$ is a tight fitting if and only if

$$p < q \quad \text{and} \quad nq = (n+1+\alpha)p.$$

Furthermore, $\|f\|_{A^q_\alpha} = \|f\|_{H^p}$ if and only if $f(z) = c/(1-\langle z,a\rangle)^{2n/p}$, where $c \in \mathbb{C}$ and $a \in \mathbb{B}^n$.

Here $0 < p, q < \infty$, $\alpha > -1$, and $n \ge 1$. Set $M = n+1+\alpha > n$. A tight fitting is a proper, contractive, noncompact inclusion. Source: Bao, Ma, Yan, and Zhu, "Embedding and compact embedding between Bergman and Hardy spaces," Conjecture 22(b), p. 19.

Result. Conditional on the complex-hyperbolic isoperimetric hypothesis, tight fitting holds exactly when $p < q$ and $nq = (n+1+\alpha)p$, and equality occurs exactly for the stated kernel family $c(1-\langle z,a\rangle)^{-2n/p}$.

Answer: Answered: yes. **Evidence:** Sketch.

Conditional on: Li–Su, arXiv:2411.01911, Assumption 1.4 and Theorem 4.1; the geometric hypothesis is claimed to follow from Kalaj, arXiv:2607.20055.

Proof. First take $nq = Mp$ and put $r = p/n = q/M > 0$. Then $M > n$ gives $q > p$. For $f \neq 0$, normalize $\|f\|_{\mathcal{H}^p} = 1$, and define

$$u(z) = |f(z)|^r(1-|z|^2), \qquad \mathrm{d}v_g = \frac{\mathrm{d}v}{(1-|z|^2)^{n+1}}, \qquad \mu_f(t) = v_g(\{u > t\}).$$

Li and Su's Theorem 4.1, under Assumption 1.4, give

$$\mu_f(t) \leq \mu_1(t), \qquad \mu_1(t) = \begin{cases} (t^{-1}-1)^n, & 0 < t < 1, \\ 0, & t \geq 1. \end{cases} \tag{15}$$

Here the model is $f = 1$. The normalization is literal: Li–Su's invariant Bergman parameter is M, so their density $c_M(1-|z|^2)^M \,\mathrm{d}v_g$ equals the density defining A^q_α above. By layer cake and (15),

$$\|f\|^q_{A^q_\alpha} = c_M M \int_0^\infty t^{M-1}\mu_f(t)\,\mathrm{d}t \leq c_M M \int_0^1 t^{M-1}(t^{-1}-1)^n\,\mathrm{d}t = 1. \tag{16}$$

The last integral is the beta integral $B(M-n, n+1)$, and is finite since $M > n$. This proves contraction with norm one, since constants attain it.

Suppose equality holds in (16). The nonnegative difference in (15) then has zero integral against the strictly positive weight $t^{M-1}\,\mathrm{d}t$ on $(0,1)$; hence $\mu_f = \mu_1$ almost everywhere there. The sharp point-evaluation estimate (1.2) of Li and Su gives $u \leq 1$. It also gives uniform boundary decay for a fixed f, as follows. Approximate f in H^p by a polynomial P; density is stated immediately after their estimate (1.2). Point evaluation applied to $f - P$ yields

$$(1-|z|^2)|f(z)-P(z)|^{p/n} \leq \|f-P\|^{p/n}_{H^p}.$$

Together with $|x+y|^{p/n} \leq C_{p/n}(|x|^{p/n} + |y|^{p/n})$ and the boundedness of P, this implies $u(z) \to 0$ uniformly as $|z| \to 1$. Consequently u attains its maximum in $\mathbb{B}_n$. If its maximum were $m < 1$, then $\mu_f(t) = 0 < \mu_1(t)$ throughout $(m, 1)$, contradicting the almost-everywhere equality. There is therefore $a \in \mathbb{B}_n$ such that

$$|f(a)|^p(1-|a|^2)^n = 1. \tag{17}$$

Let φ_a be the standard involutive ball automorphism interchanging 0 and a, and set

$$F(w) = \frac{(1-|a|^2)^{n/p}}{(1-\langle w, a\rangle)^{2n/p}} f(\varphi_a(w)).$$

All powers use the holomorphic logarithm on the right half-plane. This transform is an H^p isometry. Indeed its boundary multiplier to the power p is the sphere Jacobian

$$J_{\varphi_a}(\zeta) = \frac{(1-|a|^2)^n}{|1-\langle \zeta, a\rangle|^{2n}},$$

so boundary change of variables preserves the L^p integral. This also holds for arbitrary H^p functions by polynomial density: apply the identity first to polynomials, then use point evaluation and completeness of H^p to pass to the limit. Thus $\|F\|_{H^p} = |F(0)| = 1$ by (17).

For every $0 < \rho < 1$, subharmonicity and the Hardy norm now imply

$$1 = |F(0)|^p \leq \int_{\mathbb{S}_n} |F(\rho\zeta)|^p\,\mathrm{d}\sigma(\zeta) \leq \|F\|^p_{H^p} = 1.$$

The equality of every centered mean implies that the continuous subharmonic function $|F|^p$ is harmonic: its positive Riesz measure has zero integral against every ball Green kernel. Near 0 the function F is nonzero, and there

$$\Delta|F|^p = p^2|F|^{p-2}\sum_{j=1}^n \Big|\frac{\partial F}{\partial w_j}\Big|^2.$$

Hence all derivatives vanish near 0, so F is constant by holomorphic continuation. Using $\varphi_a^2 = \mathrm{id}$ and

$$1 - \langle \varphi_a(z), a\rangle = \frac{1-|a|^2}{1-\langle z, a\rangle}$$

recovers exactly the proposed kernel family for f. Conversely these kernels are transforms of constants. The identity

$$1 - |\varphi_a(z)|^2 = \frac{(1-|a|^2)(1-|z|^2)}{|1-\langle z, a\rangle|^2}$$

and the invariance of $\mathrm{d}v_g$ show that the same transform is an A^q_α isometry when $Mq^{-1} = np^{-1}$. Thus every proposed kernel attains equality in the two norms; $f = 0$ is included by $c = 0$.

Finally, Theorem A(b), (c) of Bao, Ma, Yan, and Zhu show that for $p < q$ the inclusion exists precisely when $n/p \le M/q$, and it is noncompact precisely at equality. Their Theorem B(b), (c) makes all the inclusions with $p \ge q$ and $\alpha > -1$ compact. On the critical line the reverse inclusion $A^q_\alpha \subset H^p$ is impossible by their Theorem B(a), since it would require $\alpha \le -1$ (or $\alpha < -1$ when $p > 2$). Hence the critical inclusion is proper. These statements prove both directions of the asserted tight-fitting criterion. □

14.3. **Contractive projections on H^p for $0 < p < 1$**

> "It seems that the assumption $Pz^n \in H^\infty(\mathbb{T})$ in Theorem 3.8 arises from technical reasons. We conjecture that this assumption can be dropped. Specifically, we conjecture that if P is a contractive projection on $H^p(\mathbb{T})$ with $0 < p < 1$ satisfying $P1 = 1$, then it automatically maps bounded functions to bounded functions, and therefore $Pf = f(0)$ or P is the identity."

Source: Fu, Guo, and Li, "Contractive projections, conditional expectations, and idempotent coefficient multipliers on H^p spaces $(0 < p < 1)$," conjecture after Theorem 3.8 in version 2, pp. 15–16.

Result. Yes, conditional on their Theorem 3.8: every such projection is evaluation at zero or the identity.

Answer: Answered: yes. **Evidence:** Sketch.

Conditional on: Fu, Guo, and Li, arXiv:2503.06615v2, Theorem 3.8.

Proof. Set

$$A_p(r) = \frac{1}{2\pi}\int_0^{2\pi} |1 + re^{i\theta}|^p \, \mathrm{d}\theta, \qquad r \ge 0.$$

Subharmonicity gives $A_p(r) \ge 1$. The binomial expansion for $0 \le r < 1$ gives

$$A_p(r) = \sum_{k=0}^{\infty} \binom{p/2}{k}^2 r^{2k} = 1 + \frac{p^2}{4} r^2 + O_p(r^4).$$

Moreover, $0 \le A_p(r) - 1 \le C_p r^2$ for all $r \ge 0$: for $r \le 1$ this follows from the nonnegative series (including its finite limit at $r = 1$), and for $r \ge 1$ from $A_p(r) \le (1+r)^p \le 2^p r^p \le 2^p r^2$.

Fix $f \in H^2(\mathbb{T})$ and write $g = Pf \in H^p(\mathbb{T})$. For every $\varepsilon > 0$ and $0 \le \theta < 2\pi$, the hypotheses imply

$$\|1 + \varepsilon e^{i\theta} g\|_p^p \le \|1 + \varepsilon e^{i\theta} f\|_p^p.$$

Averaging over θ, and evaluating g on any fixed circle of radius $0 < \rho < 1$, gives

$$\int_{\mathbb{T}} (A_p(\varepsilon |g(\rho\zeta)|) - 1) \, \mathrm{d}m(\zeta) \le \int_{\mathbb{T}} (A_p(\varepsilon |f^*(\zeta)|) - 1) \, \mathrm{d}m(\zeta).$$

Indeed, the L^p integral on a fixed circle is at most the H^p norm, and the latter is the boundary integral for f. After division by ε^2, the left side converges by uniformity on the fixed circle, while dominated convergence applies on the right because $f^* \in L^2$ and $A_p(r) - 1 \le C_p r^2$. The quadratic limit therefore yields

$$\int_{\mathbb{T}} |g(\rho\zeta)|^2 \, \mathrm{d}m(\zeta) \le \|f\|_2^2.$$

Taking the supremum over ρ shows $g \in H^2$ and $P|_{H^2}$ is a contraction. It is still an idempotent fixing 1. A contractive idempotent on a Hilbert space is an orthogonal projection: if x belongs to its range and y to its kernel, then $\|x\|_2 \le \|x + ty\|_2$ for every complex t, which forces $\langle x, y\rangle_{H^2} = 0$. Thus the restriction is an orthogonal projection on H^2.

For each fixed n, the coefficient functional $\lambda_n(f) = \widehat{Pf}(n)$ is continuous on H^p. Indeed, Taylor coefficients are continuous on H^p: the subharmonic point-evaluation bound on $|z| = 1/2$, followed by Cauchy's formula, gives $|\widehat{h}(n)| \le C_{n,p}\|h\|_p$ for $h \in H^p$. The Duren–Romberg–Shields duality theorem Theorem 1 of Duren, Romberg, and Shields (in the disc-algebra form stated in Lemma 3.2 of Fu, Guo, and Li) assigns to λ_n a unique $a_n \in A(\mathbb{D})$ with $\lambda_n(z^k) = \overline{\widehat{a_n}(k)}$ for every $k \ge 0$. By orthogonality, with the convention that the H^2 inner product is linear in the first argument,

$$\lambda_n(z^k) = \langle Pz^k, z^n\rangle_{H^2} = \langle z^k, Pz^n\rangle_{H^2} = \overline{\widehat{Pz^n}(k)}.$$

Thus a_n and Pz^n have identical Taylor coefficients and are the same analytic function; hence $Pz^n \in A(\mathbb{D})$.

The inclusion $Pz^n \in A(\mathbb{D}) \subset H^\infty$ verifies exactly the extra hypothesis of Theorem 3.8 of Fu, Guo, and Li. Their conclusion proves that P is evaluation at zero or the identity. □

14.4. **Gegenbauer probability-measure hypercontractivity**

Conjecture. The same is true for the measures $c_\alpha(1 - x^2)^\alpha \, \mathrm{d}x$ on $(-1, 1)$, $\alpha > -1$ (Gegenbauer polynomials).

The preceding sphere conjecture asserts $\|T_\omega\|_{p,q} = 1$ for real $-\sqrt{(p-1)/(q-1)} \le \omega \le \sqrt{(p-1)/(q-1)}$, with $1 < p \le q < \infty$. Here T_ω multiplies the degree-k orthogonal-polynomial component by ω^k. Write λ for the source's α, and normalize $\mathrm{d}\mu_\lambda(x) = c_\lambda(1 - x^2)^\lambda \, \mathrm{d}x$ to have mass one. Source: Janson, "On hypercontractivity for multipliers on orthogonal polynomials," §6, conjecture, printed p. 108 (PDF p. 12).
Result. No. At $\lambda = -199/200$, $p = 2$, $q = 4$, $\omega = 1/\sqrt{3}$, and $f(x) = x^2 - 100/101$, the fourth power of the norm ratio is $3983735/3160809 > 1$.
Answer: Answered: no. **Evidence:** Sketch.

Proof. Put $d = 2\lambda + 3 > 1$. Substituting $y = x^2$ in the beta integral gives

$$\mathbb{E}x^{2k} = \frac{B(k + 1/2, \lambda + 1)}{B(1/2, \lambda + 1)} = \prod_{j=0}^{k-1} \frac{2j + 1}{d + 2j}.$$

Thus $Q = x^2 - 1/d$ is orthogonal to constants and to x, so it lies entirely in the degree-two component; hence $T_{1/\sqrt{3}}Q = Q/3$. The moments through degree eight give

$$\mathbb{E}Q^2 = \frac{2(d-1)}{d^2(d+2)}, \qquad \mathbb{E}Q^4 = \frac{12(d-1)(5d^2 - 13d + 12)}{d^4(d+2)(d+4)(d+6)}.$$

Consequently the fourth power of the norm ratio is

$$R(d) = \frac{(d+2)(5d^2 - 13d + 12)}{27(d-1)(d+4)(d+6)}, \quad R(d) - 1 = \frac{-2(11d^3 + 123d^2 + 196d - 336)}{27(d-1)(d+4)(d+6)}.$$

For $d = 101/100$ the cubic is negative, and direct substitution gives the stated rational ratio. All integrals converge because $\lambda > -1$. □

15. Bohr radii and monomial convergence

15.1. **Exact asymptotics of the Bohr radius of ℓ_p^n balls**

"Whether or not the equality from (19.2) holds for $1 < p < 2$ is unknown."

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," Chapter 19, p. 487. The inferred question is whether the normalized limit in (19.2) equals 1 for fixed $1 < p < 2$.

Result. No. If $s = 2(1 - 1/p)$, then

$$\limsup_{n\to\infty} \frac{K(B_p^n)}{((\log n)/n)^{1-1/p}} \le \exp\left(-\frac{1 - s + s\log s}{2}\right) < 1.$$

Answer: Answered: no. **Evidence:** Sketch.

Proof. The complete proof is in the public note on the Bohr radius of ℓ_p^n; we give the key estimate. For m-homogeneous tetrahedral polynomials, the pointwise variance on B_p^n is at most $m^{-m(2/p-1)}/m!$. The Kahane–Salem–Zygmund net argument therefore supplies signs with supremum norm at most

$$C\sqrt{n\log(1+m)}\, m^{-m(1/p-1/2)}/\sqrt{m!}.$$

At the flat point $z = n^{-1/p}(1,\dots,1)$ the sum of the absolute monomials is $\binom{n}{m} n^{-m/p}$. The definition of the Bohr radius and Stirling's formula imply, for $m = \lceil t\log n\rceil$,

$$\limsup_{n} \log \frac{K(B_p^n)}{((\log n)/n)^{1-1/p}} \le (1-1/p)\log t + \frac{1}{2t} - \frac{1}{2}.$$

Choose $t = p/(2(p-1)) = 1/s$. The right side is $-(1 - s + s\log s)/2 < 0$, proving the strict improvement over one. □

15.2. **Are the perfect squares a stationary set?**

Is the set S of perfect squares stationary?

Source: Hervé Queffélec, "Dirichlet Series and Function Theory in Polydiscs," Oberwolfach Report 06/2014, Problem Session, §3.1, Question 1, pp. 385–386 (reported by O. F. Brevig).

For a trigonometric polynomial $P(t) = \sum_k a_k e^{ikt}$, write $[P] = \mathbb{E}\bigl\|\sum_k \varepsilon_k a_k e^{ikt}\bigr\|_\infty$, where (ε_k) are independent Rademacher variables. A set $E \subset \mathbb{Z}$ is stationary if $[P] \le C\|P\|_\infty$ for all trigonometric polynomials supported in E, with C independent of P.

Result. No; even the odd positive squares are not stationary.

Answer: Answered: no. **Evidence:** Sketch.

Proof. Choose $w \in C_c^\infty((1/2,1))$, with $0 \le w \le 1$ and $w = 1$ on $[3/5, 4/5]$, and put

$$P_N(t) = \sum_{n\ge 1} \sin(\pi n/2) w(n/N) e^{in^2 t}.$$

The theta identity $\eta(\tau)^3 = \sum_{j\ge 0}(-1)^j(2j+1)e^{\pi i\tau(2j+1)^2/4}$ and the modular transformations of η imply

$$\left|\sum_{n\ge 1} n\sin(\pi n/2) e^{-yn^2} e^{in^2 t}\right| \le C y^{-3/4}$$

uniformly in t. Put

$$F_y(t) = \sum_{n\ge 1} \sin(\pi n/2) e^{-yn^2} e^{in^2 t}.$$

Integrating the weighted estimate against $v^{-1/2}$ gives $\|F_y\|_\infty \le C y^{-1/4}$. Set $m(u) = e^u w(\sqrt{u})$ on the support of $w(\sqrt{u})$ and zero elsewhere. The smooth cutoff $K_R(t) = \sum_{k\in\mathbb{Z}} m(k/R) e^{ikt}$ satisfies

$$|K_R(t)| \le C_w \min\{R, 1/(Rt^2)\}, \qquad -\pi < t \le \pi,$$

by taking two differences of its coefficients. Thus $\|K_R\|_1$ is uniformly bounded. The exact multiplier identity $P_N = F_{N^{-2}} * K_{N^2}$ gives $\|P_N\|_\infty \le C_w\sqrt{N}$.

Lefèvre's Lemma 2.1, *Colloquium Mathematicum* 76 (1998), p. 5, gives $[P] \ge c\delta\sqrt{M\log M}$ when at least M Fourier coefficients of P have modulus at least δ. There are at least $N/20$ odd $n \in [3N/5, 4N/5]$ whose coefficients in P_N have modulus one. Thus $[P_N] \ge c_w\sqrt{N\log N}$ and $[P_N]/\|P_N\|_\infty \to \infty$. Every frequency is an odd square, proving nonstationarity. □

16. Random multiplicative questions

16.1. **Divisor-restricted Gál form over square-free sets**

> « Le résultat suivant, relatif aux éléments de $\mathcal{A}$, laisse supposer que la taille maximale de $Q(M)$ lorsque M décrit les sous-ensemble de $\mathbb{N}_1$ de taille N vaut $\Gamma^*(N)^{1/2+o(1)}$. Cette question demeure ouverte en l'état. Rappelons la définition des normes $Q(M)$ en $(1 \cdot 6)$. »

In §5 the source uses $Q(M)$ for the divisor-restricted quadratic-form norm, which we denote by $\mathcal{Q}(M)$ below. The introduction uses the same notation $Q(M)$ for the full Gál-form norm; we keep that name for the latter in the proof. The quoted closing sentence points back to equation $(1 \cdot 6)$, which defines the full norm, so this is a source-notation discrepancy; the question's scope follows the restricted form defined immediately before the quoted passage. The target ranges over square-free N-element sets, not only $M \in \mathcal{A}$.

Source: de la Bretèche and Tenenbaum, "Sommes de Gál et applications," §5.1, p. 18 (version 8, 2022; published in Proceedings of the London Mathematical Society 119 (2019)). The inferred question concerns the maximum of the divisor-restricted form over square-free N-element sets.

For a finite set $M \subset \mathbb{N}$, define

$$
\begin{aligned}
Q(M) &= \sup_{\|c\|_2=1} \Big| \sum_{m,n\in M} c_m c_n \frac{(m,n)}{\sqrt{mn}} \Big|, \\
\mathcal{Q}(M) &= \sup_{\|c\|_2=1} \Big| \sum_{\substack{m,n\in M \\ n\mid m}} c_m c_n \sqrt{n/m} \Big|, \\
\Gamma^*(N) &= \sup_{\substack{|M|=N \\ M \text{ square-free}}} \frac{1}{N} \sum_{m,n\in M} \frac{(m,n)}{\sqrt{mn}}, \\
\mathscr{L}(x) &= \exp \sqrt{\frac{\log x \log_3 x}{\log_2 x}}.
\end{aligned}
$$

Here (m,n) is the greatest common divisor and $\log_j$ is the j-fold iterated logarithm; x is sufficiently large for the displayed expression.

Result. Yes. Over square-free sets the maximum is $\Gamma^*(N)^{1/2+o(1)}$.

Answer: Answered: yes. **Evidence:** Sketch.

Theorem 2. *For every finite $M \subset \mathbb{N}$ with $|M| = N \geq 2$,*

$$
\mathcal{Q}(M) \leq 1 + \sqrt{E(N^2) Q(M)}, \qquad E(x) = \prod_{p \leq x} (1 - 1/p)^{-1}.
$$

Corollary 1. *As $N \to \infty$, the maximum of $\mathcal{Q}(M)$ over square-free N-element sets is $\Gamma^*(N)^{1/2+o(1)}$.*

Proof of Theorem 2. Put $y = N^2$. Split the pairs (m,n) with $n \mid m$ according to whether m/n is y-smooth (every prime factor at most y) or not, and write the sum as $\Sigma_1 + \Sigma_2$ accordingly.

The rough part. If m/n has a prime factor larger than y, then $m/n > N^2$ and $\sqrt{n/m} < 1/N$. The diagonal pairs are smooth, so each row and each column of the matrix of Σ_2 has at most $N - 1$ nonzero entries, each less than $1/N$. The Schur test gives $|\Sigma_2| \leq \|a\|_2 \|b\|_2$.

The smooth part. Let P be the set of primes dividing some element of M, let $P_y = \{p \in P : p \le y\}$, and write $\beta_p(n)$ for the exponent of p in n. On $\mathbb{T}^P$ with Haar measure σ set

$$F(z) = \sum_{n \in M} b_n \prod_{p \in P} z_p^{\beta_p(n)}, \qquad G(z) = \sum_{m \in M} \overline{a_m} \prod_{p \in P} z_p^{\beta_p(m)},$$

$$K(z) = \prod_{p \in P_y} \frac{1}{1 - p^{-1/2} z_p}.$$

Expanding K in geometric series, the coefficient of $\prod_p z_p^{\beta_p(m)-\beta_p(n)}$ in K equals $\sqrt{n/m}$ when $n \mid m$ and m/n is y-smooth, and 0 otherwise. Indeed, it vanishes unless $\beta_p(m) \ge \beta_p(n)$ for $p \in P_y$ and $\beta_p(m) = \beta_p(n)$ for $p \in P \setminus P_y$. Hence

$$\Sigma_1 = \int_{\mathbb{T}^P} FK\,\overline{G}\,\mathrm{d}\sigma, \qquad |\Sigma_1| \le \|G\|_{L^2} \|FK\|_{L^2} = \|a\|_2 \, \|FK\|_{L^2}.$$

For $r = p^{-1/2}$ the Poisson kernel $P_p(z) = (1 - r^2)|1 - rz|^{-2}$ has Fourier coefficients $r^{|k|}$, by expanding two geometric series. Since $|1 - p^{-1/2} z_p|^{-2} = (1 - 1/p)^{-1} P_p(z_p)$,

$$\|FK\|_{L^2}^2 = \prod_{p \in P_y} \Big(1 - \frac{1}{p}\Big)^{-1} \sum_{n, n' \in M} b_n \overline{b_{n'}}\, \kappa(n, n'),$$

where $\kappa(n, n') = 0$ unless $\beta_p(n) = \beta_p(n')$ for all $p \in P \setminus P_y$, in which case

$$\kappa(n, n') = \prod_{p \in P_y} p^{-|\beta_p(n) - \beta_p(n')|/2}.$$

The Gál kernel $(n, n')/\sqrt{nn'}$ is the same product taken over all $p \in P$. Hence $0 \le \kappa(n, n') \le (n, n')/\sqrt{nn'}$ in both cases, and

$$\Big|\sum_{n, n'} b_n \overline{b_{n'}} \kappa(n, n')\Big| \le \sum_{n, n' \in M} |b_n| |b_{n'}| \frac{(n, n')}{\sqrt{nn'}} \le Q(M) \|b\|_2^2,$$

the last step by taking $c = |b|/\|b\|_2$ in the definition of $Q(M)$. As P_y consists of primes at most y, the product is at most $E(N^2)$, and $|\Sigma_1| \le \sqrt{E(N^2) Q(M)}\, \|a\|_2 \|b\|_2$. Adding the two parts proves the inequality. Taking $a = b = c$ with $\|c\|_2 = 1$ gives the bound for $\mathcal{Q}(M)$. □

Proof of Corollary 1. By Mertens' theorem $E(N^2) \sim 2e^\gamma \log N = \mathscr{L}(N)^{o(1)}$. Theorem 1.2 of de la Bretèche and Tenenbaum gives $\sup_{|M| = N} Q(M) = \mathscr{L}(N)^{2\sqrt{2} + o(1)}$, and Theorem 2 then gives the first bound. For square-free sets, equation (1.7) of de la Bretèche and Tenenbaum and the sentence following it give $\sup Q(M) \le (e^2 + 1)(\log N + 2) \max_{n \le N} \Gamma^*(n)$ over N-element $M \subset \mathbb{N}_1$. By equation (1.5) of de la Bretèche and Tenenbaum the right side is $\mathscr{L}(N)^{2 + o(1)}$, so Theorem 2 gives $\mathcal{Q}(M) \le \mathscr{L}(N)^{1 + o(1)}$ for these M. The matching lower bound is the lower bound in equation (5.5) of de la Bretèche and Tenenbaum, attained by square-free sets (proved in Bondarenko and Seip's resonance construction). Finally $\mathscr{L}(N)^{1 + o(1)} = \Gamma^*(N)^{1/2 + o(1)}$ by equation (1.5) of de la Bretèche and Tenenbaum. □

17. Composition operators

17.1. **Nonoptimality of a target exponent**

Bayart and Brevig write: "For $\dim(\varphi) \ge 3$, we conjecture that $(2^\alpha - 1)/d$ is not optimal." See the remark following Theorem 21 in their paper on composition operators on Dirichlet series. The statement concerns Dirichlet polynomials of degree one or two with unrestricted range.

Answer. The exponent is not optimal: every such symbol maps $\mathcal{D}_\alpha$ boundedly into $\mathcal{D}_\beta$ whenever $d\beta + 9/8 \ge 2^\alpha$, and compactly when the inequality is strict.

Answer: Answered: yes, the exponent is not optimal. **Evidence:** Sketch.

Proof. The proof reduces the embedding estimate to a local sublevel bound for the Bohr lift near a boundary-contact point. In radial and angular coordinates, the radial variables are $O(\varepsilon)$ and the angular variables satisfy $G(\theta) \le \varepsilon$ and $|H(\theta) - t| \le \varepsilon$. A one-variable sublevel lemma gives measure $O(\varepsilon^{1+1/m})$ once some transverse derivative of G of order $m \le 8$ is nonzero. The quadratic form of G yields orders two or four unless it is a square; in that degenerate case restriction to two variables and an explicit Taylor calculation show that an order at most eight derivative must be nonzero. The local mass is therefore $O(\varepsilon^{9/8})$. The composition embedding criterion gives boundedness when $d\beta + 9/8 \ge 2^\alpha$ and compactness above the threshold.

The full proof and the Taylor calculation are in the sharpness note. □

17.2. **A strictly singular noncompact composition operator**

Laitila, Nieminen, Saksman, and Tylli ask in Problem 4.3(4): "Is there a Banach space E of scalar-valued analytic functions on $\mathbb{D}$ and an analytic map $\varphi : \mathbb{D} \to \mathbb{D}$, for which $C_\varphi \in \mathcal{S}(E) \setminus \mathcal{K}(E)$?" Here $\mathcal{S}(E)$ denotes the strictly singular operators and $\mathcal{K}(E)$ the compact operators.

Answer. Yes. One can take $\varphi(z) = z^2$.

Answer: Answered: yes. **Evidence:** Sketch.

Proof. Write every positive integer uniquely as $2^k j$ with j odd, and represent $f(z) = a_0 + \sum_{j \text{ odd}} a_{0,j} z^j + \sum_{k \ge 1, j \text{ odd}} a_{k,j} z^{2^k j}$. Define

$$\|f\|_E = |a_0| + \sum_{j \text{ odd}} |a_{0,j}| + \left(\sum_{k \ge 1, j \text{ odd}} \frac{|a_{k,j}|^2}{(k!)^2 j^{2(k-1)}} \right)^{1/2}.$$

This is a Banach direct sum of a scalar, an ℓ^1 space, and a weighted ℓ^2 space. If $n = 2^k j$ with $k \ge 1$, then $|a_{k,j}| \le \|f\|_E k! j^{k-1} \le \|f\|_E n^{2k} \le \|f\|_E \exp(2(\log n)^2 / \log 2)$. Hence the power series converges locally uniformly in $\mathbb{D}$, so E is a Banach space of analytic functions; finite coefficient sequences, and therefore polynomials, are dense.

We use that the formal inclusion $J : \ell^1 \to \ell^2$ is strictly singular but not compact. The unit vectors show it is not compact. To see strict singularity, let $M \subset \ell^1$ be infinite-dimensional. For any N and small $\delta > 0$, choose unit vectors $u_1, \dots, u_N \in M$ with successive disjoint coordinate intervals carrying all but δ of each vector's ℓ^1 mass. Their interval restrictions y_i have disjoint supports and $\|u_i - y_i\|_1 < \delta$. For $v = N^{-1} \sum_i u_i$, this gives $\|v\|_1 \ge 1 - 2\delta$ and $\|Jv\|_2 \le N^{-1/2} + \delta$. Letting N grow and then δ decrease shows that J is not bounded below on any infinite-dimensional subspace.

For $\varphi(z) = z^2$, composition fixes the scalar coordinate, maps the $k = 0$ odd layer by J into the $k = 1$ Hilbert layer, and on normalized coordinates $b_{k,j} = a_{k,j}/(k! j^{k-1})$ for $k \ge 1$ acts by

$$b_{k,j} \longmapsto \frac{b_{k,j}}{(k+1)j}.$$

The latter is compact because its diagonal weights tend to zero; the scalar action has finite rank. The remaining layer map is strictly singular: if it were bounded below on an infinite-dimensional subspace of E, its first-layer projection would be infinite-dimensional and J would be bounded below there. Finally, adding the compact pieces preserves strict singularity: approximate the compact part by a finite-rank map, restrict to the kernel of that map, and use strict singularity of the layer map to contradict any proposed lower bound.

The odd monomials z^j have norm one, and their images z^{2j} are separated unit vectors in the $k = 1$ Hilbert layer. Thus C_φ is not compact. This is the requested example. □

17.3. **The BMOA contact-set converse**

Laitila, Nieminen, Saksman, and Tylli ask in Problem 4.3(3): "Is the converse of Proposition 4.1(i) also true?" The proposition states that ℓ^2-singularity of C_φ on BMOA implies $m(E_\varphi) = 0$, where $E_\varphi = \{\zeta \in \mathbb{T} : |\varphi(\zeta)| = 1\}$.

Answer. No.
Answer: Answered: no. **Evidence:** Sketch.

Proof. Let $a_j = 1 - 2^{-j^2}$, let B be the Blaschke product with these zeros, and set $\varphi(z) = (1+z)B(z)/2$. Since B is inner, $|\varphi(\zeta)| = |1+\zeta|/2 < 1$ almost everywhere except at $\zeta = 1$, so $m(E_\varphi) = 0$.

The zeros are thin. For $0 < s < t < 1$, their pseudohyperbolic distance satisfies

$$1 - \rho(s,t) = \frac{(1-t)(1+s)}{1-st} \le \frac{2(1-t)}{1-s}.$$

Thus for $k \neq j$, $1 - \rho(a_j, a_k) \le 2\,2^{-|j^2-k^2|} \le 2\,2^{-j|j-k|}$, and consequently

$$\sum_{k\neq j}(1 - \rho(a_j,a_k)) \le 4\sum_{m\ge 1} 2^{-jm} \le 8\,2^{-j} \longrightarrow 0.$$

It follows that $\delta_j = \prod_{k\neq j}\rho(a_j,a_k) \to 1$. Write $B = \sigma_{a_j} B_j$. Then $\varphi\circ\sigma_{a_j}(z) = z h_j(z)$ with $|h_j(0)| = (1+a_j)\delta_j/2 \to 1$. Schwarz's lemma gives $|h_j| \le 1$; by normality a subsequence converges locally uniformly to a unimodular constant λ. Hence $\varphi\circ\sigma_{a_j} \to \lambda z$ along that subsequence.

For $f \in \mathrm{BMOA}$, the seminorm is $\sup_{a\in\mathbb{D}}\|f\circ\sigma_a - f(a)\|_{H^2}$. Since $\varphi(a_j) = 0$, the corresponding tests for $C_\varphi f$ converge locally uniformly to $f(\lambda z) - f(0)$. Each test norm is at most $\|C_\varphi f\|_{\mathrm{BMOA}}$ by the defining seminorm, and lower semicontinuity of radial H^2 means gives

$$\|C_\varphi f\|_{\mathrm{BMOA}} \ge \|f(\lambda\cdot) - f(0)\|_{H^2} = \|f - f(0)\|_{H^2}.$$

Paley's lacunary theorem identifies the closed span of z^{2^k}, $k \ge 1$, with ℓ^2 in BMOA. The displayed inequality makes C_φ bounded below on this copy, so it is not ℓ^2-singular. This disproves the converse. □

17.4. **Compactness across Hardy exponents**

Brevig and Perfekt state: "$\mathscr{C}_\varphi : \mathscr{H}^2 \to \mathscr{H}^2$ is compact if and only if

$$\lim_{\operatorname{Re} w \to 1/2^+} \frac{\mathscr{M}_\varphi(w)}{\operatorname{Re} w - 1/2} = 0.$$

" The inferred question asks whether this characteristic-zero criterion is independent of finite $p \ge 2$. See Theorem 1.4.
Answer. The condition is necessary for every finite $p \ge 2$ and sufficient whenever the local embedding holds at p. Thus the equivalence is unconditional for even $p \ge 2$ and, conditional on the local-embedding theorem of Chen, Fang, Guo, Hou, Shao, and Zhou, for every finite $p \ge 2$.
Answer: Answered: yes, conditional for non-even p. **Evidence:** Sketch.
Conditional on: arXiv:2609.11560, Theorem 1.1 and Proposition 10.1.

Proof. Assume first that the local embedding holds at this p. Put $\nu = \varphi(+\infty)$ and $T_h f(s) = f(s+h)$. Horizontal translation is contractive on $\mathscr{H}^p$, and $T_h f \to f$ in norm as $h \downarrow 0$, by polynomial density. Since $p \ge 2$, the coefficient and evaluation estimates are

$$|a_n(f)| \le \|f\|_p, \qquad |f(s)| \le \zeta(2\operatorname{Re} s)^{1/2}\|f\|_p.$$

For $g = T_h f$, the Littlewood–Paley formula and Stanton's mean-counting identity give

$$\|C_\varphi g\|_p^p = |g(\nu)|^p + \frac{p^2}{2\pi}\int_{\mathbb{C}_{1/2}} |g(w)|^{p-2}|g'(w)|^2 M_\varphi(w)\,\mathrm{d}A(w). \tag{1}$$

Indeed, apply $\Delta|F|^p = p^2|F|^{p-2}|F'|^2$ to $F = g\circ\varphi$ in the vertical mean formula, and change variables by the mean-counting function. The positive shift makes g, g' bounded on the half-plane of integration, so the truncations in that identity converge.

We need a weighted strip estimate. For $0 < \delta < 1$ and $1/2 < \theta \le (1/2 + \operatorname{Re}\nu)/2$, the local embedding implies

$$\int_{1/2<\operatorname{Re} w<\theta} |f(w)|^{p-2}|f'(w)|^2 \frac{\operatorname{Re} w - 1/2}{|w-\nu|^{1+\delta}}\,\mathrm{d}A(w) \le C\|f\|_p^p. \tag{2}$$

Here is the estimate. The local embedding on the boundary line transfers, by horizontal translation and polynomial approximation, to every line $x > 1/2$. Set $G(s) = f(s)(s + 1/2)^{-(1+\delta)/p}$, using the principal branch, and

$$m(x) = \int_{\mathbb{R}} |G(x+iy)|^p\,\mathrm{d}y.$$

Summing the unit-interval embedding bounds with the weight $|x + 1/2 + iy|^{-1-\delta}$ gives $m(x) \le C\|f\|_p^p$ uniformly for $x > 1/2$. Differentiating under the integral and using the Laplacian identity shows that m is convex and

$$m''(x) = p^2 \int_{\mathbb{R}} |G(x+iy)|^{p-2}|G'(x+iy)|^2, dy.$$

A bounded nonnegative convex function has $m'(x) \uparrow 0$; integrating this identity twice yields

$$\int_{\mathbb{C}_{1/2}} |G(w)|^{p-2}|G'(w)|^2(\operatorname{Re} w - 1/2)\,\mathrm{d}A(w) \le C\|f\|_p^p.$$

Writing $f(w) = G(w)(w + 1/2)^{(1+\delta)/p}$ and differentiating bounds the corresponding integral with denominator $|w + 1/2|^{1+\delta}$. On the indicated strip, $|w + 1/2| \le C_\nu|w - \nu|$, which proves (2).

The compactness condition and the Littlewood-type estimate in the proof of Brevig–Perfekt's Theorem 1.4 imply that for every $\varepsilon > 0$ there is $\theta > 1/2$ such that

$$M_\varphi(w) \le \varepsilon^2 \frac{\operatorname{Re} w - 1/2}{|w-\nu|^{1+\delta}}, \qquad 1/2 < \operatorname{Re} w < \theta. \tag{3}$$

Combining (1)–(3) controls the boundary strip by $C\varepsilon^2\|f\|_p^p$. Away from the strip, the evaluation bound and Cauchy–Schwarz give $|g'(w)| \le C2^{-\operatorname{Re} w}$. The remaining integral is finite because Stanton's formula at $p = 2$ applied to 2^{-s} gives $\int_{\mathbb{C}_{1/2}} |(2^{-w})'|^2 M_\varphi(w)\,\mathrm{d}A(w) < \infty$. This proves a bound for $C_\varphi T_h$ independent of h; letting $h \downarrow 0$ and using weak lower semicontinuity proves that C_φ is bounded on $\mathscr{H}^p$.

Now let (f_j) be a bounded weakly null sequence in $\mathscr{H}^p$. Its coefficients tend to zero, and the coefficient and derivative series show that $f_j, f_j' \to 0$ uniformly on every half-plane strictly to the right of $1/2$. More precisely, after choosing θ in (3), for all sufficiently large j,

$$|f_j(s)| \le \varepsilon, \qquad |f_j'(s)| \le \varepsilon|(2^{-s})'| \quad (\operatorname{Re} s \ge \theta).$$

The same bounds hold for $g = T_h f_j$, uniformly in $h > 0$, and $|g(\nu)| \le \varepsilon$. In (1), the part with $\operatorname{Re} w < \theta$ is at most $C\varepsilon^2$ by (2) –(3). On the rest, the integrand is bounded by $\varepsilon^p|(2^{-w})'|^2 M_\varphi(w)$, whose integral is finite by the preceding $p = 2$ estimate. Thus $\|C_\varphi T_h f_j\|_p^p \le C'\varepsilon^2$ uniformly for large j and all $h > 0$. Let $h \downarrow 0$ and use lower semicontinuity to get $\|C_\varphi f_j\|_p \to 0$. Reflexivity of $\mathscr{H}^p$ now implies compactness.

For necessity, write $\sigma = \operatorname{Re} w$ and use the normalized kernels

$$K_w(s) = \frac{\zeta(s + \overline{w})}{\zeta(2\sigma)^{1/2}}, \qquad G_{w,p}(s) = \frac{\zeta(s + \overline{w})^{2/p}}{\zeta(2\sigma)^{1/p}}.$$

Euler products give $\|K_w\|_2 = \|G_{w,p}\|_p = 1$ and $|K_w|^2 = |G_{w,p}|^p$. As $\sigma \downarrow 1/2$, every fixed coefficient of $G_{w,p}$ tends to zero; reflexivity therefore gives $G_{w,p} \rightharpoonup 0$. If C_φ is compact on $\mathscr{H}^p$, then $\|C_\varphi G_{w,p}\|_p \to 0$. The pointwise identity persists after composition, and the

corresponding Hardy norms satisfy $\|C_\varphi K_w\|_2^2 = \|C_\varphi G_{w,p}\|_p^p$. Brevig–Perfekt's kernel estimate in the necessity proof of Theorem 1.4 gives

$$\frac{M_\varphi(w)}{\sigma - 1/2} \le C\|C_\varphi K_w\|_2^2,$$

so the stated mean-counting condition follows. The local embedding is unconditional for even p by the $p = 2$ case and integer powers; for other finite $p \ge 2$ it is conditional on the cited local-embedding preprint. □

17.5. **Spectrum of a vertical translation**

Bayart and Yao ask: "What is the spectrum of $\mathcal{C}_\varphi$ provided $\varphi(s) = s + i\tau$?" See Question 3.8, p. 15 of their paper on Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series.
Answer. The spectrum is $\{1\}$ for $\tau = 0$ and $\mathbb{T}$ for $\tau \neq 0$.
Answer: Answered: $\{1\}$ if $\tau = 0$, and $\mathbb{T}$ otherwise. **Evidence:** Sketch.

Proof. For $\tau \neq 0$, vertical translation $T_\tau f(s) = f(s + i\tau)$ is an isometry of every seminorm P_ε and has inverse $T_{-\tau}$. The Neumann series therefore puts the spectrum in $\mathbb{T}$. Each n^{-s} is an eigenvector with eigenvalue $n^{-i\tau}$, and these eigenvalues are dense in $\mathbb{T}$: the increasing sequence $|\tau| \log n$ has gaps tending to zero and is unbounded.

Density alone does not imply that every point of the circle belongs to the spectrum on this Fréchet space. Fix $\lambda = e^{i\theta}$. Choose integers n_k nearest to $x_k = \exp((2\pi k - \theta)/\tau)$ (reversing k if necessary), so $|n_k^{-i\tau} - \lambda| = O(1/n_k)$. The series $g(s) = \sum_{k\ge1} k^{-2} n_k^{-s}$ belongs to H_+^∞. If $(T_\tau - \lambda I)f = g$, coefficient comparison gives $|a_{n_k}(f)| \gtrsim n_k/k^2$. But every $f \in H_+^\infty$ satisfies $|a_n(f)| \le n^{1/4} P_{1/4}(f)$, a contradiction. Thus $T_\tau - \lambda I$ is not onto. For $\tau = 0$, the operator is the identity.

The complete proof, including the nonclosed-spectrum point, is in the public note on questions of Bayart and Yao. □

17.6. **Cyclicity and univalence**

Bayart and Yao ask: "The above proof shows that provided C_φ is cyclic, φ is univalent on $\mathbb{C}_{1/2}$. Is it also univalent on $\mathbb{C}_0$?" See Question 4.2, p. 18 of their paper on Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series.
Answer. Yes.
Answer: Answered: yes. **Evidence:** Sketch.

Proof. Suppose that C_φ is cyclic. By Proposition 4.1 of Bayart and Yao, its symbol has characteristic $c_0 = 1$. Now suppose that distinct $a, b \in \ C_0$ satisfy $\varphi(a) = \varphi(b)$. Let φ_N be the finite-prime truncations of the symbol; they converge locally uniformly to φ. Choose a small disc about a that excludes b and on whose boundary $\varphi - \varphi(b)$ has no zero. Hurwitz's theorem gives, for some N, a point $a' \neq b$ with $\varphi_N(a') = \varphi_N(b)$. Since $c_0 = 1$, we have $\varphi_N(s) = s + \psi_N(s)$, where ψ_N depends only on the finite-prime coordinates $z(s)$. If $z(a') = z(b)$, the collision equation would force $a' = b$. Hence $z(a') \neq z(b)$.

By the open mapping theorem, one can choose infinitely many distinct pairs a_k, b_k in small neighborhoods of a', b such that $\varphi_N(a_k) = \varphi_N(b_k)$ and the finite-prime points are distinct. The finite-prime intertwining identity shows that each bounded functional $\ell_{a_k} - \ell_{b_k}$ annihilates the range of C_φ. Interpolation by finite-prime polynomials makes these functionals linearly independent. Thus the orthogonal complement of the range is infinite-dimensional. But for a cyclic operator the closure of the range has codimension at most one, a contradiction. The collision and independence details are in the full proof, linked below.

The full proof appears in the public note on questions of Bayart and Yao. □

17.7. **Minimal commutants of diagonal operators**

Bayart and Yao ask: "Does there exist a diagonal operator associated to an injective sequence (λ_n) which fails to have a minimal commutant? If so, can we give a characterization of diagonal operators with a minimal commutant?" Here $De_n = \lambda_n e_n$ on a Hilbert space with orthonormal basis (e_n) and bounded eigenvalues; see Question 5.6, p. 22.

Answer. A bounded injective diagonal operator has a minimal commutant exactly when there is no nonzero $a \in \ell^1$ such that $\sum_n a_n \lambda_n^k = 0$ for every integer $k \geq 0$. Such injective diagonals without a minimal commutant exist.

Answer: Answered: yes, with a criterion. **Evidence:** Sketch.

Proof. For $a \in \ell^1$, the functional $L_a(x) = \sum_n a_n x_n$ on ℓ^∞ annihilates every polynomial sequence $(p(\lambda_n))$ precisely when all the stated moments vanish. The double commutant description of diagonal operators identifies weak-operator density of polynomials in D with the absence of such a nonzero annihilating functional. To construct one, choose pairwise disjoint discs $B(c_j, \rho_j)$ in a Vitali packing and form $\nu = \sum_j \rho_j^2 \delta_{c_j} - \delta_0$. The Cauchy-transform identity for this packing gives $\int z^k \, \mathrm{d}\nu(z) = 0$ for every $k \geq 0$. Its point masses give an ℓ^1 annihilator; adding distinct nonzero accumulation points if needed makes the diagonal eigenvalues injective without changing the vanishing-moment construction.

The full criterion and construction are written out in the public note on questions of Bayart and Yao. □

17.8. **Removing the zero-free hypothesis**

Under the hypotheses of Corollary 8.5, Bayart and Yao ask: "Can we dispense with the assumption that ψ does not vanish on $\mathbb{D}$ in Corollary 8.5?" Here $\varphi(z) = \lambda z^p \psi(z)$, $p \geq 2$, $\lambda \neq 0$, $\psi(0) = 1$, and some iterate of φ has supremum norm less than one; see Question 8.7, p. 33.

Answer. Yes.

Answer: Answered: yes. **Evidence:** Sketch.

Proof. The Böttcher coordinate B near the attracting fixed point satisfies $B \circ \varphi = B^p$. Choose an iterate $\varphi^{[k]}$ whose image lies in the coordinate neighborhood, and put $v = B \circ \varphi^{[k]}$. Then $v \circ \varphi = v^p$, and v is small on the disc. The map $\widetilde{\varphi} = B^{-1} \circ v^{p+1}$ is a holomorphic self-map commuting with φ, and its zero has order $(p+1)p^k$, which is not a power of p. Testing weak-operator limits of polynomials in C_φ on monomials z^{p^j} and then on $z^{(p+1)p^k}$ shows that $C_{\widetilde{\varphi}}$ is not in that closure. Thus the commutant is not minimal, without a zero-free assumption on ψ.

The complete proof is in the public note on questions of Bayart and Yao. □

17.9. **Lens maps and minimal commutants**

For the lens map $\varphi_\alpha(z) = (C(z)^\alpha - 1)/(C(z)^\alpha + 1)$, where $C(z) = (1+z)/(1-z)$ and $0 < \alpha < 1$, Bayart and Yao ask"Does C_{φ_α} have a minimal commutant?" Here the operator acts on $H^2(\mathbb{D})$; see Question 8.11 and Remark 8.12, p. 36.

Answer. No, for every $0 < \alpha < 1$.

Answer: Answered: no. **Evidence:** Sketch.

Proof. Write $u(z) = \log((1+z)/(1-z))$, so $u \circ \varphi_\alpha = \alpha u$. Reflection $Rf(z) = f(-z)$ commutes with C_{φ_α}. A bounded functional on the commutant algebra, represented by a Cauchy-transform jump across the strip $u(\mathbb{D}) = \mathbb{R} + i(-\pi/2, \pi/2)$, vanishes on every polynomial in C_{φ_α} but not on R. Its density is obtained from a Cauchy transform with zeros at $64i\alpha^{-3k}$ and a nonzero value at $-64i$. Hence R is not in the weak-operator closure of the polynomial algebra, and the commutant is not minimal.

The construction and boundedness verification for the separating functional are in the public note on questions of Bayart and Yao. □

17.10. **Numerical range of a weighted composition operator**

The source asks: "Does 0 lie in the interior of the numerical range $W(C_{\psi,\varphi})$?" Question 2.13, p. 8 of Lahiri, Ojha, Halder, and Birbonshi's paper on numerical ranges and Davis–Wielandt shells of composition operators on the Hardy space of Dirichlet series. The hypotheses are that φ is a nonconstant c_0-symbol, $\psi \in \mathscr{H}^2$, $\psi_1 = 0$, and $C_{\psi,\varphi}$ is bounded.
Answer. No as stated: $\psi = 0$ gives numerical range $\{0\}$. If $\psi \neq 0$, the answer is yes.
Answer: Answered: no as stated; yes for nonzero ψ. **Evidence:** Sketch.

Proof. Let $T = C_{\psi,\varphi}$ and $u = \psi/\|\psi\|_2$. Since $\psi_1 = 0$, 1 and u are orthonormal. Also $T1 = \psi$ and $\langle Tu, 1\rangle = 0$, because φ has positive characteristic and u has zero constant coefficient. Thus the compression of T to $\operatorname{span}\{1, u\}$ has matrix

$$\begin{pmatrix} 0 & 0 \\ \|\psi\|_2 & \lambda \end{pmatrix}, \qquad \lambda = \langle Tu, u\rangle.$$

Its numerical range is an ellipse whose foci are 0 and λ. Its minor and major semiaxes are respectively

$$\frac{\|\psi\|_2}{2} \quad \text{and} \quad \frac{\sqrt{\|\psi\|_2^2 + |\lambda|^2}}{2}.$$

Since 0 is a focus, the ellipse contains the disc centered at 0 of radius

$$\frac{\sqrt{\|\psi\|_2^2 + |\lambda|^2} - |\lambda|}{2} > 0.$$

It therefore contains a disc of positive radius centered at 0. The numerical range of a compression is contained in $W(T)$, proving the claim for nonzero ψ; for $\psi = 0$, $T = 0$ and $W(T) = \{0\}$. □

17.11. **Davis–Wielandt shells of affine composition operators**

Lahiri, Ojha, Halder, and Birbonshi ask in Question 5.4: "Determine the Davis–Wielandt shell $\mathrm{DW}(C_\varphi)$." Here $\varphi(s) = cs + c_1$, c is a nonnegative integer, and boundedness requires $a = \operatorname{Re} c_1 \geq 0$ for $c \geq 1$ and $a > 1/2$ for $c = 0$.
Answer. For $c = 0$ the shell is the solid ellipsoid below. For $c = 1$ it is the ordinary convex hull $\operatorname{conv}\{(n^{-c_1}, n^{-2a}) : n \geq 1\}$. For $c \geq 2$ and $a = 0$ it is $\{(\lambda, 1) : |\lambda| < 1\} \cup \{(1, 1)\}$. For $c \geq 2$ and $a > 0$ it has the support-function description below, with the origin as the only point of the closure that is not attained.
Answer: Answered: exact support-function description. **Evidence:** Sketch.

Proof. When $c = 0$, put $Q = \zeta(2a) > 1$. The operator is rank one: $C_\varphi f = f(c_1)\mathbf{1}$. Its nonzero two-dimensional block is unitarily equivalent to $\left(\begin{smallmatrix} 0 & \sqrt{Q-1} \\ 0 & 1 \end{smallmatrix}\right)$, and it vanishes on an infinite-dimensional orthogonal complement. If a unit vector has coordinates (x, y) on this block, its shell coordinates are $t = Q|y|^2$ and $\lambda = |y|^2 + \sqrt{Q-1}\, y\overline{x}$. Hence

$$Q|\lambda|^2 + t^2 - 2t \operatorname{Re}\lambda - (Q-1)t = Q(Q-1)|y|^2(|x|^2 + |y|^2 - 1) \leq 0.$$

Conversely, for each $0 \leq t \leq Q$, the allowed values of λ form the disk centered at t/Q with radius $\sqrt{(Q-1)(t/Q)(1-t/Q)}$. Choosing $|y|^2 = t/Q$, varying x in the disk $|x| \leq \sqrt{1-|y|^2}$, and placing the remaining norm in the kernel realizes that whole disk. Completing the square shows that the resulting set is exactly

$$\mathrm{DW}(C_\varphi) = \{(\lambda, t) : Q|\lambda|^2 + t^2 - 2t \operatorname{Re}\lambda - (Q-1)t \leq 0\}.$$

For $c = 1$, the monomials $e_n(s) = n^{-s}$ are eigenvectors, with eigenvalues n^{-c_1}. If $f = \sum_n x_n e_n$ is a unit vector, then its shell point is the countable convex combination

$$(\lambda, t) = \sum_{n \geq 1} |x_n|^2 (n^{-c_1}, n^{-2a}).$$

Every such barycenter lies in the ordinary convex hull of the displayed points. Indeed it lies in their closed convex hull; if it is on a relative supporting hyperplane, equality in the

averaged supporting functional forces every point of positive weight onto that hyperplane. Repeating in the lower-dimensional affine span reduces to the relative interior, which belongs to the ordinary convex hull. Carathéodory's theorem then represents it using at most four points. Conversely, every finite convex combination is realized by a unit vector with the corresponding squared coefficient moduli.

Suppose next that $c \geq 2$ and $a = 0$. The monomial map fixes e_1 and is the orthogonal direct sum of copies of the unweighted unilateral shift on the remaining chains. It is an isometry, so every shell point has $t = 1$. The numerical range of the unilateral shift is the open unit disk: Cauchy–Schwarz gives $|\langle Sx, x\rangle| \leq 1$, equality would force a unimodular eigenvector, and the unit vectors $x_j = \sqrt{1-r^2}\, r^j e^{ij\theta}$ give every value $re^{i\theta}$ with $0 \leq r < 1$. Combining the fixed vector with the shift copies gives exactly the stated open disk and its fixed point.

It remains to treat $c \geq 2$ and $a > 0$. Put

$$\mathcal{R}_c = \{n \geq 2 : n \text{ is not a } c\text{-th power}\}.$$

For $0 < r < 1$ and $b \in \mathbb{R}$, define

$$F_{r,b}(z) = \sum_{V,E} (-b)^{|V|} (-1/4)^{|E|} z^{|V|+2|E|} r^{2\sum_{j\in V} c^j + 2\sum_{j\in E} c^j},$$

where V, E run through finite subsets of the nonnegative integers, E has no consecutive elements, and V avoids every endpoint in $E \cup (E+1)$. Define $\tau_r(b)$ to be the first positive real zero of $F_{r,b}$ and set

$$m(b) = \max_{n\in\mathcal{R}_c} \frac{1}{\tau_{n^{-a}}(b)}, \qquad H(\theta, b) = \max\{\cos\theta + b, m(b)\}.$$

Then the shell is

$$\mathrm{DW}(C_\varphi) = \left\{(\lambda, t) : 0 < t \leq 1, \quad \mathrm{Re}(e^{-i\theta}\lambda) + bt \leq H(\theta, b) \text{ for every } \theta, b \in \mathbb{R}\right\}.$$

We justify the scalar series and the support function. On the monomial basis, $C_\varphi e_n = n^{-c_1} e_{n^c}$. Every $n \geq 2$ has a unique representation $n = q^{c^j}$ with $q \in \mathcal{R}_c$ and $j \geq 0$. A diagonal unitary removes the phases of the weights along each chain, giving

$$C_\varphi \simeq 1 \oplus \bigoplus_{q\in\mathcal{R}_c} S_{q^{-a}}, \qquad S_r e_j = r^{c^j} e_{j+1}.$$

For each chain and $b \in \mathbb{R}$, let $J_r(b) = \mathrm{Re}\, S_r + bS_r^* S_r$. Its diagonal entries are br^{2c^j} and its off-diagonal entries are $r^{c^j}/2$. Their absolute sums are finite, so $J_r(b)$ is compact and self-adjoint. For large j, the compression to $\mathrm{span}\{e_j, e_{j+1}\}$ has determinant

$$r^{2c^j}\left(b^2 r^{2c^{j+1}} - 1/4\right) < 0.$$

It therefore has a positive eigenvalue, and the compact operator $J_r(b)$ has a largest eigenvalue $h_r(b) > 0$. Also $h_r(b) \leq r + |b|r^2$.

Let $J_{r,K}$ be the compression to $e_0, \ldots, e_{K-1}$, and let $P_K(z) = \det(zI - J_{r,K})$. Expanding along the last row gives

$$P_0(z) = 1, \quad P_1(z) = z - br^2, \quad P_{j+1}(z) = (z - br^{2c^j})P_j(z) - \frac{r^{2c^{j-1}}}{4} P_{j-1}(z).$$

Extending $J_{r,K}$ by zero,

$$\|J_r(b) - (J_{r,K} \oplus 0)\| \leq r^{c^{K-1}} + |b| r^{2c^K}.$$

Thus the largest eigenvalues of these compressions converge to $h_r(b)$.

The determinant $\det(I - zJ_{r,K})$ expands into exactly the finite choices of V, E above: a nonzero permutation term of a tridiagonal matrix consists of fixed points and disjoint adjacent transpositions. The unrestricted absolute sum is bounded, uniformly on compact z-sets, by

$$\prod_{j\geq 0} (1 + |bz| r^{2c^j}) \prod_{j\geq 0} (1 + |z|^2 r^{2c^j}/4) < \infty.$$

Consequently these finite determinants converge locally uniformly to the entire function $F_{r,b}$. Since the top eigenvalues converge to $h_r(b)$, $F_{r,b}(1/h_r(b)) = 0$. There is no earlier positive zero: for $0 < z < 1/h_r(b)$, every positive eigenvalue η of a finite compression satisfies $z\eta < 1$, while the sum of the absolute eigenvalues is at most

$$B = |b| \sum_{j\ge 0} r^{2c^j} + \sum_{j\ge 0} r^{c^j} < \infty.$$

The negative eigenvalues contribute factors at least one, and $\log(1 - z\eta) \ge -z\eta/(1 - zh_r(b))$ for positive eigenvalues. Therefore

$$\det(I - zJ_{r,K}) \ge \exp\left(-\frac{zB}{1 - zh_r(b)}\right) > 0.$$

Passing to the limit proves $F_{r,b}(z) > 0$ before $1/h_r(b)$, so $\tau_r(b) = 1/h_r(b)$.

For the support direction (θ, b), the self-adjoint operator $\mathrm{Re}(e^{-i\theta}C_\varphi) + bC_\varphi^*C_\varphi$ has top spectral value

$$\max\{\cos\theta + b,\ \sup_{q\in\mathcal{R}_c} h_{q^{-a}}(b)\}.$$

Rotating the basis on each chain removes θ from its shift term. Since $h_{n^{-a}}(b) > 0$ and $h_{n^{-a}}(b) \le n^{-a} + |b|n^{-2a} \to 0$, the supremum is the maximum $m(b)$. This gives $H(\theta, b)$. The vertical support is $0 \le t \le 1$, since C_φ is a contraction with $\|C_\varphi e_1\| = 1$. The Davis–Wielandt shell is convex in dimension at least three; hence these support inequalities determine its closure.

For completeness, the series also gives certified finite approximations. Let $\ell_{n,K}(b)$ be the largest real zero of the above P_K with $r = n^{-a}$, and put

$$L_{M,K}(b) = \max\{0,\ \max_{n\in\mathcal{R}_c,\ n\le M} \ell_{n,K}(b)\}.$$

Then

$$0 \le m(b) - L_{M,K}(b) \le \max\{2^{-ac^{K-1}} + |b|2^{-2ac^K},\, (M+1)^{-a} + |b|(M+1)^{-2a}\}.$$

The shell is compactly generated because C_φ is compact: its images of the monomial basis are orthogonal and have lengths $n^{-a} \to 0$. It is also injective. The inequalities force $\lambda = 0$ when $t = 0$. Indeed, for every unit vector x on a chain and $u > 0$,

$$\langle \mathrm{Re}\, S_r x, x\rangle - u\|S_r x\|^2 \le \|S_r x\| - u\|S_r x\|^2 \le \frac{1}{4u}.$$

Thus $m(-u) \le 1/(4u)$ and $H(\theta, -u) \le 1/(4u)$ for $u \ge 1$. If $(\lambda, 0)$ lies in the closure, its support inequalities give $\mathrm{Re}(e^{-i\theta}\lambda) \le 1/(4u)$ for every $u \ge 1$; letting $u \to \infty$ and varying θ forces $\lambda = 0$.

Every nonzero boundary point of the closure is attained. A supporting functional with nonzero horizontal part has positive support value because $H(\theta, b) \ge m(b) > 0$. A vertical support at the bottom is only the origin; the top support has value $1 > 0$. Let A be the associated compact self-adjoint operator and $h > 0$ its largest eigenvalue. Choose unit vectors whose shell points tend to the boundary point and pass to a weakly convergent subsequence with limit f. Compactness of C_φ and $C_\varphi^*C_\varphi$ gives both the limiting shell coordinates and $\langle Af, f\rangle = h$. Since $\langle Af, f\rangle \le h\|f\|^2$ and $\|f\| \le 1$, we have $\|f\| = 1$, so the point is attained. The relative interior of a closed convex set belongs to the convex set whose closure it is, and the shell is convex; therefore all interior points are attained as well. The origin is not attained because C_φ is injective, but the shell points $(0, n^{-2a})$ for $n > 1$ converge to it. Hence the origin alone is omitted. □

17.12. **Which boundedness sets can occur?**

After stating Theorem 1.1, Andersson writes: "We are unaware of whether this completely characterizes all possible sets A, and find it an interesting question to determine precisely what sets A can occur." See §1 of "Exotic operators on the Hardy spaces of the infinite-dimensional torus." The inference is to characterize the exact boundedness sets among the theorem's common-domain operators.

Answer. Exactly the sets $A \subseteq [1,\infty]$ for which $A \setminus \{\infty\}$ is F_σ in $[1,\infty)$.

Answer: Answered: exact characterization. **Evidence:** Sketch.

Proof. Let D be the common dense domain, with D and $T(D)$ contained in every finite H^p. For each nonzero $f \in D$, the ratio $r_f(p) = \|Tf\|_p/\|f\|_p$ is continuous: on a compact interval $a \le p \le b$, $|g|^p \le |g|^a + |g|^b$ gives dominated convergence for each $g \in \cap_{q<\infty} H^q$. Thus $N(p) = \sup_{f\neq 0} r_f(p)$ is lower semicontinuous. Boundedness is equivalent to $N(p) < \infty$, so the finite part of A is $\bigcup_{k\ge 1}\{N \le k\}$, an F_σ set.

Conversely, write $A \cap [1,\infty) = \bigcup_k F_k$ with F_k increasing and closed, and put $U_k = [1,\infty)\setminus F_k$. Cover each U_k by compact intervals inside relatively open bounded intervals with overlap at most three. Andersson's rank-one block lemma supplies, for each compact interval, an operator R_ν of norm at most one outside its open interval and norm at least $1+\rho_\nu$ on a neighborhood of the compact interval. Taking an ℓ_ν-fold tensor block on fresh variables gives norm $\|R_\nu\|_p^{\ell_\nu}$. Choose ℓ_ν so $2^{-\nu}(1+\rho_\nu)^{\ell_\nu} \ge k(\nu)$ and sum $T = \sum_\nu 2^{-\nu} R_\nu^{(\ell_\nu)}$ on analytic polynomials. Each polynomial uses only finitely many blocks. At $p \in A$, only finitely many blocks can have norm above $2^{-\nu}$; for $p \notin A$, a block at every level forces the polynomial-domain operator norm to be arbitrarily large. If $\infty \notin A$, add Andersson's one-variable Volterra operator, bounded at every finite exponent and unbounded on H^∞, acting in a fresh variable. This realizes the prescribed finite part and the chosen behavior at infinity. □

17.13. **Approximation numbers of composition operators**

Queffélec and Queffélec conclude: "The result of 1. is optimal, and that of 2. as well, up to the logarithmic factor." This sentence follows Theorem 8.9.3, p. 272, of *Diophantine Approximation and Dirichlet Series*. The inferred questions ask whether the logarithm can be removed from the positive-characteristic lower bound in Theorem 8.8.1 and what the exact order is in the critical finite-dimensional case of Theorem 8.9.3.

Answer. In the positive-characteristic case, $a_N(C_\varphi) \ge (2N)^{-\operatorname{Re} c_1}$, and this power is sharp. In the critical finite independent-frequency case of dimension $d \ge 2$, $a_n(C_\varphi) \asymp n^{-(d-1)/2}$.

Answer: Answered: yes in (a), and $a_n \asymp n^{-(d-1)/2}$ in (b). **Evidence:** Sketch.

Proof. For the first claim, let $e_j(s) = j^{-s}$ and $I_N = \{N+1,\dots,2N\}$. In the expansion of $C_\varphi e_j$, the leading term is $j^{-c_1} e_{jc_0}$ and every other frequency is $j^{c_0} m$ with $m \ge 2$. For distinct $j,k \in I_N$, no such latter frequency equals k^{c_0}: if $(k/j)^{c_0}$ is an integer while $1/2 < k/j < 2$, then $j = k$. Projection onto $\operatorname{span}\{e_{jc_0} : j \in I_N\}$ therefore gives $\|C_\varphi f\| \ge (2N)^{-\operatorname{Re} c_1}\|f\|$ on an N-dimensional subspace. Every rank-$(N-1)$ approximation has a unit kernel vector there, proving the bound. For $\varphi(s) = c_0 s + c_1$, the operator is diagonal up to an orthonormal change of basis and its approximation numbers equal $N^{-\operatorname{Re} c_1}$.

For the critical case, the Bohr lift identifies the norm with an embedding of the Hardy space into $L^2(\mu)$, where μ is the image of Haar measure under $W(z) = \sum_{j=1}^d |c_j|(1-z_j)$. Replacing the zeta kernel by the half-plane Laplace kernel changes the squared embedding operator by an analytic-kernel operator whose approximation numbers decay geometrically. The Laplace embedding is dominated by the model measure $x^{(d-3)/2} e^{-y^2/x}\,\mathrm{d}x\,\mathrm{d}y$ on a bounded half-plane region. Dyadic decomposition in x and Gaussian localization in y give an upper bound $O(n^{-(d-1)/2})$ for the model embedding; the lower bound is the published one in Theorem 8.9.3. Together these prove the asserted order.

The full model-measure and approximation-number argument is in the sharpness note. □

18. Dirichlet series and frequencies

18.1. **Maximal convergence abscissas for general frequencies**

In arXiv:1902.02073v1, Defant and Schoolmann ask: "Given a frequency λ, give reasonable upper and lower estimates for $\sup_{D\in H_p(\lambda)}\sigma_c(D)$ and $\sup_{D\in H_p(\lambda)}\sigma_c(D)$, for example in terms of $L(\lambda)$." This is Problem 4.17, p. 21; the published Problem 4.21 corrects the second σ_c to σ_a.

Answer. For every frequency, with $L = L(\lambda)$,

$$L/2 \le C_p \le S_p \le L/p \quad (1 \le p \le 2), \qquad L/p \le C_p \le S_p = L/2 \quad (2 \le p < \infty).$$

These universal bounds are sharp.

Answer: Answered: sharp bounds in terms of $L(\lambda)$ and p. **Evidence:** Sketch.

Proof. For finite L, choose L' with $L < L' < \tau$. Since $\limsup_{n\to\infty}(\log n)/\lambda_n = L$, we have $\lambda_n > (\log n)/L'$ eventually, and hence

$$\sum_n e^{-\tau\lambda_n} < \infty.$$

If $L = \infty$, the upper bounds in the theorem are automatic. For $D = \sum a_n e^{-\lambda_n s} \in \mathcal{H}_p(\lambda)$, the coefficient map satisfies $\|a\|_{\ell^\infty} \le \|D\|_1$, and orthogonality gives $\|a\|_{\ell^2} \le \|D\|_2$. For $1 \le p \le 2$, interpolation yields $\|a\|_{\ell^{p'}} \le \|D\|_p$; for $p \ge 2$, monotonicity of the L^p norms gives $\|a\|_{\ell^2} \le \|D\|_p$. Hölder's inequality and Cauchy–Schwarz now give $S_p \le L/p$ when $p \le 2$ and $S_p \le L/2$ when $p \ge 2$. Since convergence implies absolute-convergence abscissa is no smaller, $C_p \le S_p$.

If $L = 0$, the series $D_0(s) = \sum_n n^{-2}e^{-\lambda_n s}$ converges in $\mathcal{H}_p(\lambda)$ because each character has norm one. Since $\lambda_n/\log n \to \infty$, for every real $\sigma < 0$ its terms at σ tend to infinity. Thus $C_p \ge 0$, giving the lower bound in this case.

Suppose $L > 0$ and fix $0 < M < L$. Choose N_k recursively so that $N_k \ge e^k$, $N_k \ge 2N_{k-1}$, and $0 < \lambda_{N_k} < (\log N_k)/M$. Such indices exist by the definition of L. The disjoint blocks $B_k = \{n : N_k/2 < n \le N_k\}$ have $|B_k| \ge N_k/2$ and $N_k \le 2|B_k|$. Monotonicity of λ gives, for $\sigma, b \ge 0$,

$$\sum_{n\in B_k} be^{-\sigma\lambda_n} \ge b|B_k|e^{-\sigma\lambda_{N_k}} \ge b|B_k|^{1-\sigma/M}2^{-\sigma/M}.$$

Put $\theta = \max\{1/2, 1 - 1/p\}$ and $Q_k = \sum_{n\in B_k} h_{\lambda_n}$. Orthogonality and the triangle inequality give $\|Q_k\|_2 = |B_k|^{1/2}$ and $\|Q_k\|_\infty \le |B_k|$, so $\|Q_k\|_p \le |B_k|^\theta$: use $\|Q_k\|_p \le \|Q_k\|_2$ for $p \le 2$, and $\|Q_k\|_p^p \le \|Q_k\|_\infty^{p-2}\|Q_k\|_2^2$ for $p \ge 2$. Therefore

$$D_1(s) = \sum_k k^{-2}|B_k|^{-\theta} \sum_{n\in B_k} e^{-\lambda_n s}$$

converges in $\mathcal{H}_p(\lambda)$. If $\sigma < (1-\theta)M$, put $\varepsilon = 1 - \theta - \sigma/M > 0$. The sum of the terms of $D_1(\sigma)$ over B_k is at least $c_\sigma k^{-2}|B_k|^\varepsilon$, which tends to infinity because $|B_k| \ge e^k/2$. These are differences of partial sums, so D_1 diverges at σ. Letting $M \uparrow L$ proves $C_p \ge (1-\theta)L = \min\{L/2, L/p\}$.

For the absolute-convergence lower bound, Khintchine's inequality and Fubini's theorem give signs ε_n, $n \in B_k$, such that $\|\sum_{n\in B_k}\varepsilon_n h_{\lambda_n}\|_p \le K_p|B_k|^{1/2}$. Hence

$$D_2(s) = \sum_k k^{-2}|B_k|^{-1/2} \sum_{n\in B_k} \varepsilon_n e^{-\lambda_n s}$$

converges in $\mathcal{H}_p(\lambda)$. If $\sigma < M/2$, the absolute sum over B_k is at least $c_\sigma k^{-2}|B_k|^{1/2-\sigma/M}$, which diverges as $k \to \infty$. Thus $S_p \ge M/2$, and then $S_p \ge L/2$ by letting $M \uparrow L$.

For sharpness, fix $A, B > 0$. Let $M_j = \lceil e^{Aj} \rceil$. Choose $\delta_j > 0$ so that $M_j\delta_j < 1/4$ and $\{1, \delta_1, \delta_2, \ldots\}$ is linearly independent over $\mathbb{Q}$. The frequency

$$\mu_{j,k} = j + k\delta_j, \qquad j \ge 1, \quad 0 \le k < M_j,$$

is increasing when ordered by blocks; the blocks are disjoint and contained in $[j, j+1/4)$. Their cumulative counting function gives $L(\mu) = A$.

The corresponding characters can be written $z_0^j z_j^k$ on a product torus, so a polynomial in $\mathcal{H}_p(\mu)$ has the form $F = \sum_j z_0^j P_j(z_j)$, with $\deg P_j < M_j$. Conditional expectation onto the coordinates z_0, z_j, followed by extraction of the jth Fourier coefficient in z_0, is contractive on L^p and gives

$$\|P_j\|_{L^p(\mathbb{T})} \le \|F\|_{\mathcal{H}_p(\mu)}.$$

This remains true by density for every $F \in \mathcal{H}_p(\mu)$.

Let $D_m(z) = 1 + \cdots + z^{m-1}$. The geometric-sum bound $|D_m(e^{it})| \le \min(m, C/|t|)$ gives $\|D_m\|_1 \le C\log(m+1)$ and $\|D_m\|_p \le C_p m^{1-1/p}$ for $p > 1$. If Q is any initial segment of the coefficients of a polynomial P of degree less than m, convolution with a Dirichlet kernel and Young's inequality give $\|Q\|_p \le C\log(m+1)\|P\|_p$. Nikolskii's inequality then yields $\|Q\|_\infty \le C_p m^{1/p}\log(m+1)\|P\|_p$ for $p > 1$. At $p = 1$, Hausdorff–Young bounds each coefficient of P by $\|P\|_1$, so $\|Q\|_\infty \le m\|P\|_1$. Consequently every partial sum in block j is bounded on $\operatorname{Re} s \ge \sigma$ by

$$C_p e^{-j\sigma} M_j^{1/p}\log(M_j+1)\|F\|_{\mathcal{H}_p(\mu)}$$

for $p > 1$, and by $e^{-j\sigma}M_j\|F\|_{\mathcal{H}_1(\mu)}$ for $p = 1$. Summing over j proves uniform convergence on every such half-plane when $\sigma > A/p$, hence $C_p(\mu) \le A/p$.

For the reverse inequality set $K_{p,j} = \|D_{M_j}\|_{L^p(\mathbb{T})}$ and take

$$F_p(s) = \sum_{j\ge 1} j^{-2} K_{p,j}^{-1} \sum_{k=0}^{M_j-1} e^{-(j+k\delta_j)s}.$$

The jth block has $\mathcal{H}_p(\mu)$ norm j^{-2}, so the series converges in $\mathcal{H}_p(\mu)$. Its coefficients are positive. At real $0 \le \sigma < A/p$, its jth block value is at least

$$j^{-2}K_{p,j}^{-1}M_j e^{-(j+M_j\delta_j)\sigma} \ge c_\sigma j^{-2}K_{p,j}^{-1}M_j e^{-j\sigma}.$$

For $p > 1$, $\|D_{M_j}\|_p \le C_p M_j^{1-1/p}$, so this lower bound grows exponentially like $e^{j(A/p-\sigma)}$ up to a polynomial factor. For $p = 1$, $\|D_{M_j}\|_1 \le C\log(M_j+1)$, and it grows like $e^{j(A-\sigma)}$ up to a polynomial factor. Thus the partial sums fail to converge for every $\sigma < A/p$, proving $C_p(\mu) = A/p$.

Next choose a positive increasing sequence ν_n in small disjoint intervals about $\log(n+1)/B$, avoiding the rational span of $\{1, \delta_1, \delta_2, \ldots\}$ and all previously chosen terms. Then ν_n are rationally independent, $\operatorname{span}_{\mathbb{Q}}\{\nu_n\} \cap \operatorname{span}_{\mathbb{Q}}\{1, \delta_j : j \ge 1\} = \{0\}$, and $L(\nu) = B$. Its characters are independent uniform circle variables. For a finite first-chaos sum $X = \sum_n a_n z_n$,

$$\|X\|_2^2 = \sum_n |a_n|^2, \qquad \|X\|_4^4 = 2\Big(\sum_n |a_n|^2\Big)^2 - \sum_n |a_n|^4 \le 2\Big(\sum_n |a_n|^2\Big)^2.$$

Interpolation gives $\|X\|_1 \ge \|X\|_2^3/\|X\|_4^2 \ge 2^{-1/2}\|a\|_2$. For each integer $r \ge 1$, expansion of the $2r$th moment also gives $\|X\|_{2r}^{2r} \le r!\|a\|_2^{2r}$. Thus the $\mathcal{H}_p(\nu)$ norm is equivalent to the ℓ^2 norm of the coefficients for every finite p. Cauchy–Schwarz gives $C_p(\nu) \le B/2$, and the coefficient sequence $a_n = ((n+1)^{1/2}\log(n+1))^{-1}$ belongs to ℓ^2 and has convergence abscissa $B/2$, so $C_p(\nu) = B/2$.

Let λ be the sorted union of μ and ν. Its counting function is the sum of the two component counting functions, so $L(\lambda) = \max\{A, B\}$. The direct-sum condition realizes $\mathcal{H}_p(\lambda)$ on independent product factors. Conditional expectation onto either factor extracts the corresponding component contractively. If $\sigma > \max\{A/p, B/2\}$, the μ component converges uniformly by the block estimate above and the ν component converges absolutely by Cauchy–Schwarz. A partial sum in the sorted union consists of prefixes of these two components, so $C_p(\lambda) \le \max\{A/p, B/2\}$. The positive witnesses in the two factors give the reverse inequality.

For $1 \le p \le 2$, the general upper estimate for $S_p(\mu)$ gives $\sigma_a(F_\mu) \le A/p$ for every μ component, while the ν component has absolute convergence abscissa at most $B/2$. Hence $S_p(\lambda) \le \max\{A/p, B/2\} = C_p(\lambda)$, and equality follows from $C_p \le S_p$. For $1 \le p \le 2$, choosing $B = L$ and $A = pc$ realizes $C_p = S_p = c$ for every $L/2 \le c \le L/p$. For $2 \le p < \infty$, choosing $A = L$ and $B = 2c$ realizes $C_p = c$ for every $L/p \le c \le L/2$; the general theorem gives $S_p = L/2$. Thus all the bounds are attained. □

18.2. **Cyclicity when the omitted curve has unbounded argument**

Dan and Guo ask: "In Theorem 7.1, can the assumption that γ has a bounded argument function be dropped?" See Question 7.6, p. 42 of their paper on ordinary Dirichlet series and dilation systems. The theorem concerns a zero-free $F \in H^2(\mathbb{T}^\infty)$ whose range on $X = \mathbb{D}^\infty \cap \ell^2$ avoids a simple curve from zero to infinity.

Answer. Yes: every such F is cyclic, without a bounded-argument assumption on the omitted curve.

Answer: Answered: yes. **Evidence:** Sketch.

Proof. We first show that if the range of a zero-free $F \in H^2(\mathbb{T}^\infty)$ on X lies in a simply connected domain $\Omega \subset \mathbb{C} \setminus \{0\}$, then F is cyclic. Let $\psi : \mathbb{D} \to \Omega$ be conformal and write $F = \psi \circ h$, where $h = \psi^{-1} \circ F$ maps X to $\mathbb{D}$. A univalent function omitting zero and its reciprocal belong to $H^q(\mathbb{D})$ for every $q < 1/2$. Consequently $u = \log|\psi^*| \in L^2(\mathbb{T})$ and the outer factorization has the form $\psi = e^{i\alpha}\exp(Hu)$, where H is the analytic Poisson extension. Truncate u from below by $u_\varepsilon = \max(u, \log\varepsilon)$ and set $b_\varepsilon = e^{-i\alpha}\exp(-Hu_\varepsilon)$. Then b_ε is bounded analytic, $\|b_\varepsilon\|_\infty \le \varepsilon^{-1}$, $|\psi b_\varepsilon| \le 1$, and $\psi b_\varepsilon \to 1$ locally uniformly. The bounded analytic multiplier $b_\varepsilon \circ h$ shows $A_\varepsilon = (\psi b_\varepsilon) \circ h$ belongs to the cyclic subspace generated by F. Also $\|A_\varepsilon\|_2 \le 1$ and $A_\varepsilon(0) \to 1$, so $\|A_\varepsilon - 1\|_2^2 \le 2 - 2\operatorname{Re} A_\varepsilon(0) \to 0$. Thus 1 is in that subspace.

For a simple curve γ joining 0 to infinity, its closure in the Riemann sphere is a Jordan arc. Its complement is a simply connected domain avoiding zero. The image $F(X)$ is connected and avoids γ, so it lies in this domain. The preceding argument applies and proves cyclicity. This removes only the bounded-argument restriction; zero-freeness and the range-avoidance hypothesis remain. □

18.3. **Bass stable rank of a Dirichlet-series algebra**

Sasane's Conjecture 6.1 asks: "Does $R(S) = \infty$ imply that the Bass stable rank of $\mathscr{H}^\infty_S$ is infinite?" Here $S \subseteq \mathbb{N}$ is a multiplicative subsemigroup containing 1, $R(S)$ is the supremum of the sizes of independent subsets, and independence means that no member lies in the subsemigroup generated by the others.

Answer. No. For $S_2 = \{2^k : k \ge 0\}$, $R(S_2) = \infty$ but $\operatorname{bsr}(\mathscr{H}^\infty_{S_2}) = 1$.

Answer: Answered: no. **Evidence:** Sketch.

Proof. More generally, if $a > 1$ lies in S, then for each $M \ge 2$ the set $E_M = \{a^k : M \le k < 2M\}$ is independent. A product of two or more of its elements has exponent at least $2M$, and a product of one other element cannot equal the omitted power. Thus independent sets have arbitrarily large finite size.

For S_2, the map $Ug(s) = g(2^{-s})$ is an isometric algebra isomorphism from $H^\infty(\mathbb{D})$ onto $\mathscr{H}^\infty_{S_2}$: the half-plane map $s \mapsto 2^{-s}$ has image $\mathbb{D} \setminus \{0\}$, and a bounded series supported on powers of two is a Taylor series in 2^{-s}. Treil's theorem gives $\operatorname{bsr}(H^\infty(\mathbb{D})) = 1$. Bass stable rank is invariant under unital algebra isomorphism, proving the counterexample. The source's additional unique-product generator condition is therefore essential to its positive theorem. □

18.4. **The translated partial-sum inequality**

In the Oberwolfach report, Bayart asks: "Is it the case that $\sigma_N \to 0$ as $N \to \infty$?" Here σ_N is the supremum over $f \in H^\infty(\mathbb{C}_0)$ of the infimum of $\sigma > 0$ for which $\|S_N f\|_{H^\infty(\mathbb{C}_\sigma)} \le \|f\|_{H^\infty(\mathbb{C}_\sigma)}$.
Answer. No. In fact, $\sigma_N = +\infty$ for every $N \ge 2$.
Answer: Answered: no. **Evidence:** Sketch.

Proof. Let $k = \lfloor \log_2 N \rfloor$ and $m = 2^k$, so $m \le N < m^2$. Set $f_N(s) = 1 + m^{-s} - m^{-2s}/16$. Its partial sum is $S_N f_N = 1 + m^{-s}$. At height $\sigma > 0$, put $r = m^{-\sigma} \in (0,1)$. The two supremum norms are the suprema on $|z| \le r$ of $P(z) = 1 + z - z^2/16$ and $1 + z$, respectively. The latter is $1 + r$.

It remains to show $|P(re^{it})| < 1 + r$. Write $x = \cos t$. If $x \le 3/4$, then

$$1 + r - |1 + re^{it}| \ge \frac{r(1-x)}{1+r} \ge \frac{r}{8},$$

whereas the quadratic term has modulus $r^2/16 < r/8$. If $x \ge 3/4$, direct expansion gives

$$|P(re^{it})|^2 - |1 + re^{it}|^2 = \frac{r^4}{256} - \frac{r^2}{8}(\cos 2t + rx) \le \frac{r^4}{256} - \frac{r^2}{64} < 0,$$

since $\cos 2t = 2x^2 - 1 \ge 1/8$. The maximum-modulus principle now gives $\|f_N\|_{H^\infty(\mathbb{C}_\sigma)} < \|S_N f_N\|_{H^\infty(\mathbb{C}_\sigma)}$ for every $\sigma > 0$. The defining set of admissible σ is empty, proving the claim. □

19. Vertical limits

19.1. **Counting functions in the completed Nevanlinna class**

Andersson asks: "Let $f \in \mathscr{N}$. Does there exist a set $E \subseteq \mathbb{T}^\infty$ of full measure such that if $\chi \in E$, then f_χ has an analytic continuation to $\mathbb{C}_0$ for which

$$\lim_{\sigma_0 \downarrow 0} \lim_{T \to \infty} \frac{\pi}{T} \sum_{\substack{s \in \mathscr{Z}(f_\chi - \zeta) \\ |\operatorname{Im} s| < T \\ \operatorname{Re} s > \sigma_0}} (\operatorname{Re} s - \sigma_0)$$

exists and is finite for all $\zeta \in \mathbb{C} \setminus \{f(+\infty)\}$?" Question 3, §6, pp. 26–30; zeros are counted with multiplicity.
Answer. Yes. On one full-measure set the value is

$$J_\zeta(0+) - \log|f(+\infty) - \zeta|, \qquad J_\zeta(x) = \int_{\mathbb{T}^\infty} \log|f_\chi(x) - \zeta| \, \mathrm{d}m(\chi).$$

Answer: Answered: yes. **Evidence:** Sketch.
Conditional on: arXiv:2606.20293v2, Lemma 3.1.

Proof. Write $d_0(q,r) = \|q - r\|_0$ for the metric defining $\mathscr{N}$, and let $\theta_t\chi$ denote the prime Kronecker flow. For $q \in \mathscr{N}_u$, the Bohr mean identity gives

$$\int_{\mathbb{T}^\infty} \log(1 + |q_\chi(s)|) \, \mathrm{d}m(\chi) \le \|q\|_0.$$

The disk submean inequality for the nonnegative subharmonic function $\log(1 + |q_\chi|)$ then gives, for every compact $K \Subset \mathbb{C}_0$,

$$\int \sup_{s \in K} \log(1 + |q_\chi(s)|) \, \mathrm{d}m(\chi) \le C_K \|q\|_0.$$

Apply this estimate to differences of a rapidly Cauchy sequence $q_j \in \mathscr{N}_u$ converging to f. Tonelli's theorem on a countable compact exhaustion shows that, almost surely, $q_{j,\chi}$ converges locally uniformly to an analytic representative F_χ of f_χ.

We also need control at infinity. Fix $u > 0$. If $x - u > 1$, the Bohr lift of $q(u + \cdot)$ and its analytic Poisson evaluation give

$$\sup_{\chi,t} \log(1 + |q_\chi(x + it)|) \le \prod_p \frac{1 + p^{-(x-u)}}{1 - p^{-(x-u)}} \, \|q\|_0.$$

The product is finite. Applying this to $q_j - q_k$ shows that the approximants are uniformly Cauchy on a sufficiently far-right half-plane, uniformly in χ and t. Their constant coefficients are Cauchy as well, since $\log(1 + |a_1(q_j - q_k)|) \le d_0(q_j, q_k)$. Since each fixed q_j tends to its constant coefficient there, we obtain the uniform anchor

$$\lim_{x\to\infty} \sup_{\chi,t} |F_\chi(x + it) - a_1| = 0, \qquad a_1 = f(+\infty).$$

In particular, for any compact target set $A \Subset \mathbb{C} \setminus \{a_1\}$, all F_χ avoid A to the right of one common vertical line.

Fix such an A and a positive cutoff σ_*. Jensen's disk inequality and the compact-sup estimate above give an integrable random bound for the number of ζ-points, uniformly over $\zeta \in A$, in any fixed-width rectangle contained in $\{\operatorname{Re} s \ge \sigma_*\}$. For a continuous compactly supported weight h in that half-plane, the divisor sum $H_\zeta(\chi) = \sum_{F_\chi(s)=\zeta} h(s)$ is continuous in ζ and has this integrable envelope. Apply the ergodic theorem for θ_t to finite target nets and to the moduli of continuity of H_ζ; this yields uniform convergence of the time averages over $\zeta \in A$. Choose h to smooth the vertical cutoff and the factor $(\operatorname{Re} s - \sigma_0)_+$. The smoothed average differs from the sharp count only in bounded-width blocks at the two vertical ends. The integrable Jensen bound and Borel–Cantelli make that error $o(1)$ after division by T, uniformly on A. Taking countable compact exhaustions of $\mathbb{C} \setminus \{a_1\}$ and rational $\sigma_0 > 0$ gives one full-measure set where the inner limit exists for every such target and cutoff. Rational cutoffs on either side and continuity extend the assertion to every real $\sigma_0 > 0$.

To identify the value, fix $\zeta \ne a_1$. Jensen's inequality for the Dirichlet polynomials, followed by approximation in d_0, gives $J_\zeta(x) \ge \log|a_1 - \zeta|$; the gauge bound gives a finite upper bound. Thus J_ζ is finite and convex. Poincaré–Lelong's formula, averaged over characters, says that the expected zero measure per unit vertical length is $J_\zeta''(dx)/(2\pi)$. The uniform anchor implies that F_χ has no ζ-points beyond a common far-right line and that $J_\zeta(x) \to \log|a_1 - \zeta|$ there; convexity then makes J_ζ constant past that line. Therefore integration by parts gives, for each $\sigma_0 > 0$,

$$2\pi \int_{x>\sigma_0} (x - \sigma_0) \frac{J_\zeta''(dx)}{2\pi} = J_\zeta(\sigma_0) - \log|a_1 - \zeta|.$$

The ergodic limit of the sharp counts equals this expectation. Finally, the finite upper bound for J_ζ makes $J_\zeta(0+)$ finite; letting $\sigma_0 \downarrow 0$ proves the stated formula in the required order of limits. If f is constant, all counts for $\zeta \ne a_1$ vanish, so the claim is immediate. □

19.2. **The optimal H^p order-to-mean abscissa for $1 \le p \le 2$**

Brevig and Kouroupis state: "Fix $\beta > 0$. If an element f of $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ such that $\mu_f(\sigma) \le \beta$ for every $\sigma > 0$, then $\sigma_{\mathrm{m}}(f) \le \beta/(1+2\beta)$." Theorem 1, p. 2. Their Theorem 4 proves sharpness in $\mathscr{H}^2$. The inferred question asks whether the same optimal bound holds for $1 \le p \le 2$.

Answer. Yes. For every $1 \le p \le 2$, the sharp bound is $\beta/(1 + 2\beta)$.
Answer: Answered: $\beta/(1 + 2\beta)$, sharply. **Evidence:** Sketch.
Conditional on: arXiv:2609.35496, Theorems 1 and 4.

Proof. Fix $\sigma > \beta/(1 + 2\beta)$ and choose $\varepsilon > 0$ so small that $\sigma - \varepsilon$ is still above this threshold. By Helson's inequality,

$$\sum_{n\ge1} \frac{|a_n|^2}{d(n)} \le \|f\|^2_{\mathscr{H}^1} \le \|f\|^2_{\mathscr{H}^p}.$$

Since $d(n)n^{-2\varepsilon}$ is bounded, $g(s) = f(s+\varepsilon)$ lies in $\mathscr{H}^2$. Approximate g in $\mathscr{H}^2$ by Dirichlet polynomials P_j. The differences $g - P_j$ continue with finite order and have order at most β; Brevig–Kouroupis' Theorem 1 therefore gives their H^2 mean identity at $\sigma - \varepsilon$.

Let $A_T(u)$ be the normalized L^p norm of $u(\sigma - \varepsilon + it)$ on $[-T, T]$. Minkowski's inequality and $p \le 2$ give

$$|A_T(g) - A_T(P_j)| \le \left(\frac{1}{2T} \int_{-T}^{T} |g - P_j|^2 (\sigma - \varepsilon + it)\, \mathrm{d}t \right)^{1/2}.$$

The right side tends, as $T \to \infty$, to at most $\|g - P_j\|_{\mathscr{H}^2}$. The vertical mean of each polynomial is its $\mathscr{H}^p$ norm after translation, and these translated polynomial norms converge to that of g. Letting $j \to \infty$ proves the upper bound for the p-mean abscissa.

For sharpness, the entire extremal f_β in Theorem 4 belongs to $\mathscr{H}^2 \subset \mathscr{H}^p$ and has convergence abscissa $\beta/(1+2\beta)$. A bounded first vertical mean on any line, together with finite order, forces convergence of the Dirichlet series to its right; the truncated-Perron proof of this Titchmarsh-type implication is included in the linked book chapter. Thus its p-mean abscissa cannot be smaller than its convergence abscissa, proving equality. □

The order-to-mean abscissa argument

19.3. **The order-zero mean-value formula in H^p**

Brevig and Kouroupis state: "If f in $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ of order 0, then

$$\|f\|_{\mathscr{H}^2}^2 = \lim_{\sigma \to 0^+} \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |f(\sigma + it)|^2 \, \mathrm{d}t.$$

" Corollary 2, §2. The inferred question asks whether this ordered formula extends to every $1 \le p < \infty$.

Answer. Yes, conditional on the source's H^2 mean-value theorem and the hypercontractivity theorem for ordinary Dirichlet series.

Answer: Answered: yes. **Evidence:** Sketch.

Conditional on: arXiv:2609.35496, Theorem 1, and arXiv:2401.09996, Corollary 1.2.

Proof. Let $k = \lceil p/2 \rceil$. First suppose $g \in \mathscr{H}^{2k}$ has an order-zero continuation. Choose Dirichlet polynomials $P_j \to g$ in $\mathscr{H}^{2k}$. The function $(g - P_j)^k$ belongs to $\mathscr{H}^2$ and has order zero. Brevig–Kouroupis' Corollary 2 therefore gives, for every $\sigma > 0$,

$$\lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |g(\sigma + it) - P_j(\sigma + it)|^{2k} \, \mathrm{d}t = \|(g - P_j)(\sigma + \cdot)\|_{\mathscr{H}^{2k}}^{2k}.$$

Minkowski's inequality and $L^p \le L^{2k}$ on the normalized interval compare the p-mean of g with that of P_j. The latter converges to $\|P_j(\sigma + \cdot)\|_{\mathscr{H}^p}$ by the mean theorem for polynomials; letting $j \to \infty$ proves

$$\lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |g(\sigma + it)|^p \, \mathrm{d}t = \|g(\sigma + \cdot)\|_{\mathscr{H}^p}^p.$$

For a general $f \in \mathscr{H}^p$, fix $\sigma > 0$ and choose $0 < \varepsilon < \sigma$. Ordinary-frequency hypercontractivity implies $f_\varepsilon(s) = f(s + \varepsilon) \in \mathscr{H}^{2k}$; its continuation still has order zero. Apply the preceding result to f_ε at $\sigma - \varepsilon$. This gives the inner mean identity for f at every $\sigma > 0$. Finally, horizontal translations are contractive and $f_\sigma \to f$ in $\mathscr{H}^p$ as $\sigma \downarrow 0$, first for polynomials and then by density. Taking the outer limit proves the ordered formula. □

20. Optimal polynomial approximants and cyclicity

20.1. **Pairing spurious zeros with electrostatic partners**

Let $\{\xi_{n,k}\}$ be the set of spurious roots of φ_n and $\{s_{n,\ell}\}$ the set of the roots of the electrostatic partner S_n which are inside the disk and do not accumulate

on $\mathbb{T}$. Then both sets has the same amount of elements and we can reorganize all of them in pairs $(\xi_{n,k}, s_{n,k})$ in such a way that

$$\xi_{n,k} = s_{n,k} - \frac{s_{n,k}}{n} + o(1/n),$$

for any of these pairs.

Orive–Sánchez-Lara–Seco, *Electrostatic interpretation of the zeros of optimal polynomial approximants*, arXiv:2507.15488, Conjecture 1, p. 20.
Answer: Answered: no. **Evidence:** Sketch.

Proof outline. The counterexample uses one fixed real cubic

$$f_\theta(z) = (1-z)(z-e^{i\theta})(z-e^{-i\theta}), \qquad 0 < \theta < \pi.$$

For the orthogonal polynomials of $|f_\theta|^2\,\mathrm{d}m$, the low-degree numerator is asymptotically quadratic. A nested phase construction fixes a single θ and a subsequence n_j for which this quadratic has an almost double root $\eta_j \to \xi$, with $0 < \xi < 1/4$. On the scale $u = n_j(z-\eta_j)$, the orthogonal-polynomial numerator tends to a nonzero multiple of u^2, while the electrostatic-partner numerator tends to a nonzero multiple of $u(u-2\xi)$. Hurwitz's theorem therefore gives

$$n_j(\xi_{j,1}-\eta_j),\ n_j(\xi_{j,2}-\eta_j) \to 0, \qquad n_j(s_{j,1}-\eta_j) \to 0, \quad n_j(s_{j,2}-\eta_j) \to 2\xi.$$

The conjecture would require both partner displacements to be ξ, so either pairing has a nonzero limiting error. The cardinalities are equal in this example; the displacement assertion fails. The phase construction, numerator estimates and zero counts are proved in *Remarks on the zeros of optimal polynomial approximants*, Theorem 3 and Section 4. □

20.2. **Weak limits of approximant zeros**

It is conjectured that the same should hold for the OPA $\{p_n\}$ themselves.

Orive–Sánchez-Lara–Seco, *Electrostatic interpretation of the zeros of optimal polynomial approximants*, arXiv:2507.15488, unnumbered conjecture, p. 3.
Answer: Answered: Haar measure. **Evidence:** Sketch.

Proof outline. For a critical polynomial or a positive power of one, write

$$F(z) = g(z)\prod_{k=1}^{m}(1-\overline{a_k}z)^{\beta_k}, \quad a_k \in \mathbb{T}, \quad \beta_k > 0,$$

where g is analytic and zero-free near $\overline{\mathbb{D}}$. Let φ_n be the monic orthogonal polynomial for $|F|^2\,\mathrm{d}m$. Martínez-Finkelshtein–McLaughlin–Saff, *IMRN* (2006), article 91426, Corollary 1 and Theorem 1(iii), show that at most $m-1$ zeros stay in each compact subset of $\mathbb{D}$ and that $z^{-n}\varphi_n(z)$ is locally uniformly bounded for $|z| > 1$. Cauchy estimates and Newton's identities bound every fixed power sum of the zeros. Hence the normalized zero measures converge to Haar measure on $\mathbb{T}$: their limits live on $\mathbb{T}$ and all nonzero Fourier moments vanish.

The normal equations give $p_n = \overline{F(0)}\varphi_n^*/\|\varphi_n^*\|^2_{|F|^2}$. Thus finite zeros reverse by $z \mapsto 1/\overline{z}$, and $\deg p_n \geq n-m+1$. At most $m-1$ reversed zeros can escape to infinity; their normalized mass tends to zero. Reflection changes the remaining zeros by arbitrarily little once they are near $\mathbb{T}$, so the approximants have the same Haar limit. The full argument, including degree loss and escape control, is in *Remarks on the zeros of optimal polynomial approximants*, Theorem 4 and Section 5. □

20.3. **Irrational local maxima of the autocorrelation**

Does there exist a positive irrational where A has a strict local maximum?

Dirichlet Series and Function Theory in Polydiscs, Oberwolfach Report 06/2014, Balazard's talk, Question 2, p. 350.
Answer: Answered: no. **Evidence:** Sketch.

Proof outline. Write $A(\lambda) = \int_0^\infty \{t\}\{\lambda t\} t^{-2}\,\mathrm{d}t$ and let p_k/q_k be the continued-fraction convergents of an irrational $\lambda > 0$. The argument proves that, for infinitely many k, with $h_k = |\lambda - p_k/q_k|$,

$$A(\lambda + h_k) + A(\lambda - h_k) - 2A(\lambda) > 0.$$

The key step is to apply the exact rational-neighbourhood functional equation of Báez-Duarte–Balazard–Landreau–Saias to $\varphi_2(x) = \sum_{j\geq 1} B_2(jx)/j^2$. Its remainder is invariant under a constant shift of φ_2; centring its range improves the remainder estimate enough for a positive symmetric second difference along a subsequence of convergents. Since $h_k \to 0$, one of the two neighbouring values is larger than $A(\lambda)$. Thus λ is not a local maximum. The rational points are strict local maxima by the cited authors' published result. The full estimates and continued-fraction subsequence argument are in *The multiplicative autocorrelation of the fractional part has no local maximum at irrational points*, Theorem 1 and its proof. □